

Real Analysis

Self-Study Lecture Note (Analysis I & II)

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Note: This lecture note was created for the purpose of self-study. Part I is based on the content of Professor Kim Doyoon's Analysis 1 class, and Part II is based on the content of Professor Koo Hyungwoon's Analysis 2 class. Redistribution is strictly prohibited as it contains portions directly taken from these lectures. Please feel free to comment on wrong explanations or typos via e-mail.

Key Reference: Walter Rudin et al. *Principles of mathematical analysis*, volume 3. McGraw-hill New York, 1964

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Part I — Analysis I

The Real Field, Topology, Sequences, and Continuity

The real and Complex number systems

Def : Order

Let S be a set. An ordered on S is a relation, denoted by $<$, with the following two properties:

1. If $x \in S$ and $y \in S$ then one and only one of statements $x < y, x = y, y < x$
2. $x < y, y < z \Rightarrow x < z$

Def : Ordered set

Ordered set is a set S in which and order is defined. For example, $\mathbb{Q}$ is an ordered set if $r < s$ is defined to mean that $s - r$ is a positive rational number.

Def : bounded above

Suppose S is an ordered set, and $E \subset S$. If there exists ($= \exists$) a $\beta \in S$ such that ($=$ s.t) $x \leq \beta$ for every ($= \forall$) $x \in E$, then E is called **bounded above**, and β is called **upper bound** of E . On the other hands, If $\exists \alpha \in S$ s.t $x \geq \alpha \forall x \in E$ then E is called **bounded below**, and α is called **Lower bound** of E

Def : Supremum and Infimum

Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

1. α is an upper bound of E
2. If $\gamma < \alpha$ is not an upper bound of $E \Leftrightarrow$ For γ s.t $\gamma < \alpha$ where $\exists a \in E$ such that $\gamma < a \leq \alpha$

On the other hands, there exists an $\beta \in S$ such that :

1. β is an upper bound of E
2. If $\beta < \gamma$ is not a Lower bound of $E \Leftrightarrow$ for γ s.t $a < \gamma$ where $\exists a \in E$ such that $\beta \leq a < \gamma$

And ordinarily denote that $\alpha = \sup E, \beta = \inf E$

Thm: $\sup E$ is unique ★

▷ **proof** : Suppose α, β both are $\sup E$ but those are not same and S is ordered set , $E \in S$. then $e \leq \alpha \forall e \in E$ is true and also for any $\epsilon > 0$ there exists $e \in E$ such that $\alpha - \epsilon < e \leq \alpha$ (by Def : supremum). On the other hand, let choose $\epsilon = \alpha - \beta > 0$ because $\alpha \neq \beta$ and then in terms of $\sup E = \beta$ there exists $e \in E$ such that $\beta - \epsilon = \alpha < e \leq \beta$ this indicate α is not upper bound of E and this is contradiction ! ■

Def : least-upper-bound property

If S is ordered set and $E \subset S$, E is not empty and E is bounded above then $\exists \sup E \in S$ then we call that ordered set S have the least-upper bound property

Thm : supremum and Infimum in least-upper-bound property ★

Suppose S is an ordered set with least-upper-bound property and $B \subset S$, B is not empty, B is bounded below. and let L is the set of all lower bound of B then $\alpha = \sup L$ exists in S , and $\alpha = \inf B$ exists in S .

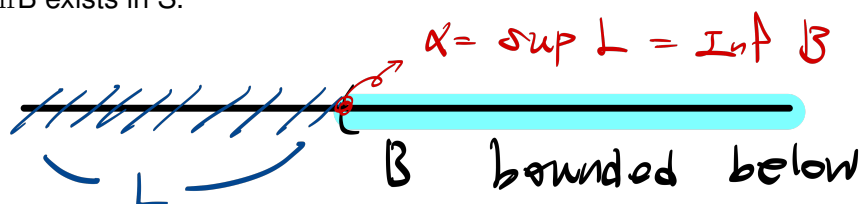


Figure 1: Thm : supremum and Infimum in least-upper-bound property

▷ **proof**

Proof sketch

step1 : Does L exist $\rightarrow$ Yes !

step2 : Show L is bounded above

step3 : $\sup L$ exists because S is least-upper-bound property step4 : $\sup L = \sup B$?

By assumption $L = \{y \in S, y \leq x \forall x \in B\}$ so, $L \neq \emptyset$ (step 1) and by Def: bounded above ($x \leq \beta$ for every $(= \forall) x \in E$) L is bounded above (step 2) $\Rightarrow \sup L = \alpha$ because set has least-upper-bound property there exists Supremum and which denoted to α (step 3) only thing we need to show is $\alpha = \inf B$. that is, we need to show that

1. α is lower bound of B i.e. $\alpha \leq x \forall x \in B$
2. If $\alpha < \beta$ then β is not a lower bound of $B \Leftrightarrow$ for β s.t $y < \beta$ where $\exists y \in B$ such that $\alpha \leq y < \beta$

To get a contradiction, let's assume there exists $x \in B$ such that $x < \alpha$ (counter-assumption of 1) and by definition of least upper bound of L there exists $y \in L$ such that $x < y \leq \alpha$. And because $y \in L$, it is lower bound of B this indicate $y \leq x \forall x \in B$ this is contradiction ! (We show that " α is lower bound of B i.e. $\alpha \leq x \forall x \in B$ " is true) and If $\alpha < \beta$ then $\beta \notin L$ this means β is not a lower bound of B ■

Field

Def : field axioms

A field is a set $\mathcal{F}$ (= field) with **two operation**, called *addition* and *multiplication*, which satisfy the following axiom (let all component in $\mathcal{F}$)

1. Addition

- (a) $x \in \mathcal{F}, y \in \mathcal{F} \Rightarrow x + y \in \mathcal{F}$
- (b) Commutative : $x + y = y + x$
- (c) Associative : $(x + y) + z = x + (y + z)$
- (d) $0 + x = x$
- (e) $x + (-x) = 0$

2. Multiplication

- (a) $xy \in \mathcal{F}$
- (b) $xy = yx$
- (c) $(xy)z = x(yz)$
- (d) $\exists 1 \neq 0$ such that $1x = x$
- (e) $x \neq 0, 1/x \in \mathcal{F}$ such that $x \cdot (1/x) = 1$

3. Distribution law

- (a) $x(y + z) = xy + xz$

Prop: Addition

1. $x + y = y + z \Rightarrow y = z$

▷ **Proof** : $y = 0 + y = (-x + x) + y$ [1-e] $= -x + (x + y)$ [1-c] $= -x + (x + z)$ [by assumption] $= -(x + x) + z$ [1-c] $= 0 + z$ [1-e] $= z$
 $\therefore y = z$ ■

2. $x + y = x \Rightarrow y = 0$

▷ **Proof** : $y = 0 + y = (-x + x) + y$ [1-e] $= -x + (x + y)$ [1-c] $= -x + x$ [by assumption] $= 0$
 $\therefore y = 0$ ■

3. $x + y = 0 \Rightarrow y = -x$

▷ **proof** : $y = 0 + y = (-x + x) + y$ [1-e] $= -x + (x + y)$ [1-c] $= -x + 0$ [by assumption] $= -x$
 $\therefore y = -x$ ■

4. $-(-x) = x$

▷ **proof** : with [3.] replace x to $-x$ than $-x + y = 0 \Rightarrow y = -(-x) \Rightarrow x - x + y = x$ [1.] $= y = x \Rightarrow -(-x) = x$ ■

Prop : multiplication

1. $xy = xz \Rightarrow y = z$

▷ **proof** : $y = x(1/x)y$ [2-e] $= 1/x(xy)$ [2-b] $= 1/x(xz)$ [by assumption] $= z$
 $\therefore y = z$ ■

2. $xy = x \Rightarrow y = 1$

▷ **proof** : $y = 1y$ [2-d] $= x \cdot (1/x)y$ [2-e] $= (1/x) \cdot xy$ [2-b] $= (1/x)x$ [by assumption] $= 1$ [2-e]
 $\therefore y = 1$ ■

3. $xy = 1 \Rightarrow y = 1/x$

▷ **proof** : $y = (1/x)xy$ [2-e,2-b] $= 1/x(xy)$ [2-c] $= 1/x(1)$ [by assumption] $= 1/x$ [2-d]
 $\therefore y = 1/x$ ■

4. $1/(1/x) = x$

▷ **proof** : with [3.] replace x to $1/x$ than $(1/x)y = 1$ then $y = 1/(1/x)$ and by [2-e] $y = x$
 $\therefore x = 1/(1/x)$ ■

Prop : Furthermore Statement

1. $0x = 0$

▷ **proof** : $0x + 0x = (0 + 0)x = 0x \Rightarrow 0x + 0x - 0x = 0x - 0x \Rightarrow 0x = 0$

2. $x \neq 0, y \neq 0 \Rightarrow xy \neq 0$

▷ **proof** : let $x \neq 0, y \neq 0$ but $xy = 0$, then $1 = (1/x)x(1/y)y = (1/x)(1/y)xy = (1/x)(1/y)0 = 0, 1 = 0$ contradiction ! ■

3. $(-x)y = -(xy) = x(-y)$

▷ **proof** : $-x + x = 0 \Rightarrow (-x + x)y = 0y \Rightarrow -xy + xy = 0 \Rightarrow xy - xy - xy = -xy \Rightarrow (-x)y = -(xy)$ on the other hand, $-y + y = 0 \Rightarrow x(-y + y) = x0 \Rightarrow x(-y) + xy = 0 \Rightarrow x(-y) + xy - xy = -xy \Rightarrow x(-y) = -xy$ ■

4. $(-x)(-y) = xy$ ★

▷ **proof** : $-y + y = 0 \Rightarrow -x(-y + y) = -x0 \Rightarrow (-x)(-y) + (-x)(y) = 0 \Rightarrow (-x)(-y) + (-x)(y) + -(-x)(y) = 0 + -(-x)(y) \Rightarrow (-x)(-y) = -(-x)(y) \Rightarrow (-x)(-y) = (x)(y)$ [by Prop: Addition 4.] ■

Def : Ordered field

An ordered field is a field $\mathcal{F}$ which is also an ordered set , such that

- $x + y < x + z \text{ if } y < z$

▷ **proof** : let $y < z \Rightarrow z - y > 0, z - y = (x + z) - (x + y)$ then $(x + z) - (x + y) > 0 \Rightarrow (x + z) - (x + y) + (x + y) > 0 + (x + y) \Rightarrow (x + z) > (x + y)$ ■

- $xy > 0$ if $x > 0$ and $y > 0$

▷ **proof** let $x > 0$ and $y > 0$ but $xy \leq 0$

case 1. if $xy = 0$ then $x = 0$ or $y = 0$ by Prop : $0x = x$, this is contradiction!

case 2. if $xy < 0$ two positive numbers multiplication should be positive this is also contradiction ! ■

Prop : Ordered field

1. $x > 0 \Rightarrow -x < 0, x < 0 \Rightarrow -x > 0$

▷ **proof** : $x > 0 \Rightarrow -x + x > -x + 0 \Rightarrow -x < 0,$
 $x < 0 \Rightarrow 0 = -x + x < -x + 0 \Rightarrow -x > 0$ ■

2. $x > 0, y < z \Rightarrow xy < xz$

▷ **proof** : $z > y \Rightarrow z - y > y - y = 0 \Rightarrow x(z - y) > x0 = 0(x > 0) \Rightarrow xz - xy + xy > 0 + xy \Rightarrow xz > xy$ ■

3. $x < 0, y < z \Rightarrow xy > xz$

▷ **proof** : $z > y \Rightarrow z - y > y - y = 0 \Rightarrow -x(z - y) > -x0 = 0(x < 0 \Rightarrow -x > 0) \Rightarrow -x(z - y) = (-x)(z - y) > 0 \Rightarrow x(z - y) < 0$ [by Prop : Ordered field 1.] $\Rightarrow xz - xy < 0 \Rightarrow xz < xy$ ■

4. $x \neq 0 \Rightarrow x^2 > 0$

▷ **proof** : If $x > 0$ [by Def : Ordered field Second] $\Rightarrow xx = x^2 > 0$

If $x < 0$ [by Prop : Ordered field 1.] $-x > 0 \Rightarrow (-x)(-x) > 0$ [by Prop : Furthermore

Statement 4.] $(-x)(-x) = xx$, thus $(-x)(-x) > 0 \Rightarrow xx = x^2 > 0$ ■

This implies that since $1 \cdot 1 = 1^2 = 1 > 0$

5. $0 < x < y \Rightarrow 0 < 1/y < 1/x$

▷ **proof** : First of all we need to know that if $x > 0 \Rightarrow 1/x > 0$, let $x > 0$ and then $x > 0 \Rightarrow x(1/x) = 1 > 0(1/x) = 0 \Rightarrow 1 > 0$, [by Def : Ordered field Second] $1/x > 0$ Likewise, $1/y > 0$. Now $x < y \Rightarrow (1/x)(1/y)x < (1/x)(1/y)y \Rightarrow 1/y < 1/x$ and we now $x, y > 0$ then $1/x, 1/y > 0$ thus we get $0 < 1/y < 1/x$ ■

The real field

Thm : archimedean property ★

If $x, y \in \mathbb{R}, x > 0 \Rightarrow \exists n \in \mathbb{Z}^+$ such that $nx > y$

▷ **proof** : Assume there exists y s.t $nx \leq y$ for all positive integer. $A = \{nx : x \in \mathbb{Z}^+\}$ y is upper bound of nx and A is bounded above and let $\sup A = \alpha$ then by Def:Supremum $\alpha - x < nx \leq \alpha$ where $nx \in A$ then we can get $\alpha - x < nx \Rightarrow \alpha < (n+1)x$ and $(n+1)x \in A$ ($n \in \mathbb{Z}$ then $n+1 \in \mathbb{Z}$). Since α is not upper bound of A So, α no more Supremum of A this is contradiction ! ■

Thm : $\mathbb{Q}$ is dense in $\mathbb{R}$ ★

If $x, y \in \mathbb{R}, x < y$ then there exists $p \in \mathbb{Q}$ such that $x < p < y$

▷ **Proof** : $x > y \Rightarrow y - x > 0$ and by the Thm: Archimedean Property, $n(y - x) > z$ where $z \in \mathbb{R}$. Then we can choose z as 1. Thus, we can get $n(y - x) > 1 \Rightarrow ny - nx > 1 \Rightarrow nx + 1 < ny$. Hold it, and again by the Thm: Archimedean Property, we can have these

two inequalities: $m_1 > nx_1, m_2 > nx_2$ where $m_1, m_2 \in \mathbb{Z}^+$. With these, we can have nx such that $-m_2 < nx < m_1$. Hence, there exists m such that $m - 1 \leq nx < m$ With this,

$$\underbrace{m - 1 \leq nx < m}_{\text{(i)}}$$

we know $nx < m$ (i), $m \leq 1 + nx$ (ii), and $nx + 1 < ny$ (iii) $\Rightarrow nx < m < ny \Rightarrow x < \frac{m}{n} < y$ which $\frac{m}{n} \in \mathbb{Q}$ because of our assumption we can know $n \neq 0$ ■

Thm : Irrational case ★

For any $x, y \in \mathbb{R}, x < y$ there exists irrational number $q \in \mathbb{R} \setminus \mathbb{Q}$ (Note $A \setminus B = A \cdot B, A \cap B^c$) such that $x < q < y$

▷ **proof** : Following assumption, $y - x > 0$ as we did above by the Thm: Archimedean Property we can get $n(y - x) > \sqrt{2} \Rightarrow \sqrt{2} + nx < ny$ and let $m_1 > nx, m_2 > -nx$ then we can get nx such that $-m_2 < nx < m_1$ there exists m such that $m \leq nx < m + \sqrt{2}$ with this one might easily know that $nx < m + \sqrt{2} < nx + \sqrt{2} < ny \Rightarrow nx < m + \sqrt{2} < ny \Rightarrow x < \frac{m+\sqrt{2}}{n} < y$ ■

This implies that between any two real numbers, there exists at least one irrational number. In other words, irrational numbers are also dense in $\mathbb{R}$. Note that the sum of a rational number and an irrational number is irrational

Uniqueness of Positive n-th Roots

For every real $x > 0$, for every integer $n > 0$ there is one and only one positive real y such that $y^n = x$ and this number y is written $\sqrt[n]{x}$ or $x^{1/n}$

▷ **proof** : That is clear that [by Prop:Ordered field 3.] for $0 < y_1 < y_2$ this implies $y_1^n < y_2^n$ (finish the rest later if time permits)

The complex field

Euclidean Spaces

Def: Vector Space

For each positive integer k , let $\mathbb{R}^k$ be the set of all ordered k -tuples:

$$\mathbf{x} = (x_1, x_2, \dots, x_k),$$

where $x_1, \dots, x_k$ are real numbers, called the *coordinates* of $\mathbf{x}$. The elements of $\mathbb{R}^k$ are called points or vectors, especially when $k > 1$. Let $\mathbf{y} = (y_1, \dots, y_k) \in \mathbb{R}^k$ and $\alpha \in \mathbb{R}$. We define:

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_k + y_k),$$

$$\alpha \mathbf{x} = (\alpha x_1, \dots, \alpha x_k),$$

where $\mathbf{x} + \mathbf{y} \in \mathbb{R}^k$ and $\alpha \mathbf{x} \in \mathbb{R}^k$. These operations satisfy the commutative, associative, and distributive laws, making $\mathbb{R}^k$ a *vector space* over the real field $\mathbb{R}$. The zero element of $\mathbb{R}^k$ (also called the *origin* or *null vector*) is the point $\mathbf{0} = (0, 0, \dots, 0)$.

We also define the *inner product* of $\mathbf{x}$ and $\mathbf{y}$ by:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i,$$

and the *norm* of $\mathbf{x}$ by:

$$|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

Thm : Norm Properties

let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k, \alpha \in \mathbb{R}$ following are true :

1. $|\mathbf{x}| \geq 0$
2. $|\mathbf{x}| = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$
3. $|\alpha \mathbf{x}| = |\alpha| |\mathbf{x}|$
4. $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|$
5. $|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$
6. $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$

Basic Topology

Def : Function

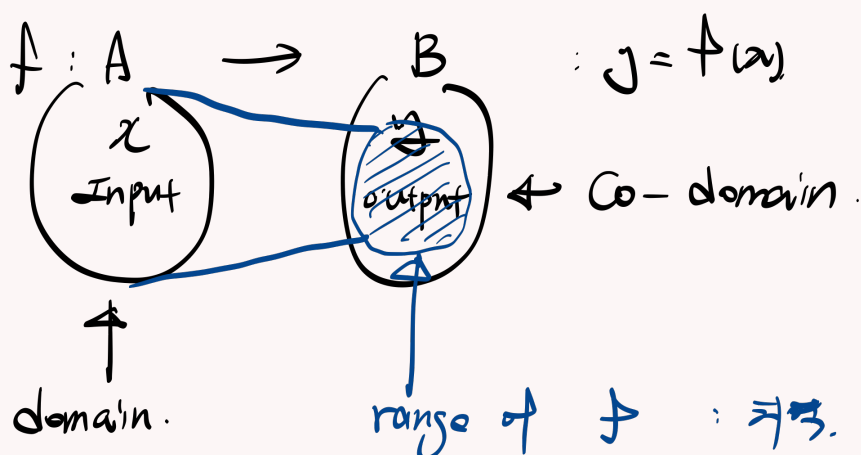


Figure 2: Function

Consider two sets A and B whose elements can be any objects. We say that a function f is defined from set A (the **domain**) to set B if for every element $x \in A$, there is an associated element in B , denoted as $f(x)$. The values of f are the corresponding elements in B that arise from the elements of A .

In summary, we express this relationship as follows:

$$f: A \rightarrow B \quad (\text{where } f \text{ is a function that maps elements from } A \text{ into } B.)$$

Alternatively, we can express this as:

$$x \mapsto y \quad (x \text{ is mapped to } y.)$$

Def : One to one and Inverse function

Again let A and B there are two sets and let f be a mapping A into B . If $E \subset A$, $f(E)$ is defined to be the set of all elements $f(x)$, for $x \in E$ we call this :

$$\text{Range of } f: f(A) = \{f(x) : x \in A\}$$

$$\text{Image of } f: f(E) = \{f(x) : x \in E\}$$

and it is clear that $f(A) \subset B$ and $f(A) = B \Rightarrow f$ maps A **onto** B

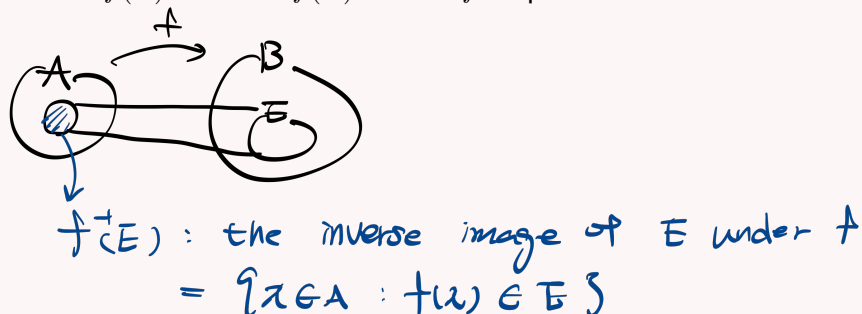


Figure 3: Inverse Function

And if $E \in B$, $f^{-1}(E)$ denotes the set of all $x \in A$ such that $f(x) \in E \Leftrightarrow f^{-1} = \{x \in A : f(x) \in E\}$

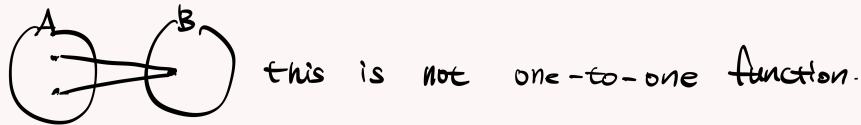


Figure 4: One to one Function

If $y \in B$, $f^{-1}(y)$ is the set of all $x \in A$ such that $f(x) = y$ (= onto) $\Leftrightarrow f^{-1}(y) = \{x \in A : f(x) = y\}$. If, for each $y \in B$, $f^{-1}(y)$ consists of at most one element of A then f is said to be a **one to one**, 1-1 mapping if A into B we state this way:

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \text{ where } x_1, x_2 \in A$$

Def : 1-1 correspondence

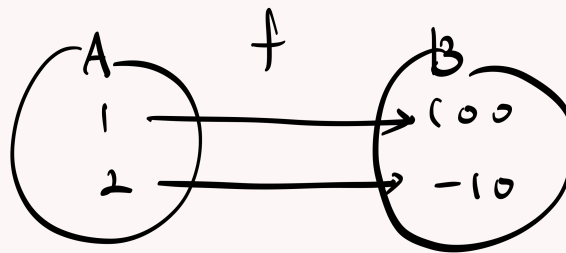


Figure 5: One to one correspondence

If there exists a one to one function from A onto B (1-1, onto) $\Rightarrow$ "one to one correspondence" = "same cardinal number" = " A and B are equivalent". And properties of 1-1 & onto as following :

1. Reflexive : $A \sim A$
2. Symmetric : $A \sim B \Rightarrow B \sim A$
3. Transitive : $A \sim B, B \sim C \Rightarrow A \sim C$

Def : Countable and Infinite ★

For $n \in \mathbb{Z}^+$ let $J_n = \{1, 2, \dots, n\}$ and $J = \{1, 2, \dots\}$ For any set A , following statements defined :

1. Finite : $A \sim J_n$
2. Infinite : Not finite, **The set equivalent to one of its proper subsets ★**
3. Countable : $A \sim J$
4. Uncountable : Neither finite nor countable
5. Finite or countable $\Rightarrow$ at most countable

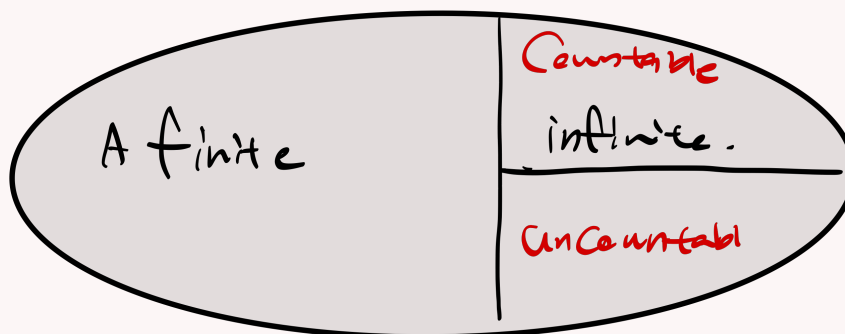


Figure 6: Finite, Countable, Uncountable

Let have example of [Def : Countable and Infinite 2.,3.] If we define set A :

$$A = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

then , $A \sim J$ so A is infinite still countable. and another example is this. Consider $\mathbb{N} = \{1, 2, 3, 4, \dots\}$, and $S = \{2, 3, 4, 5, \dots\}$ where $S \subset \mathbb{N}$ then, we can easily know that $\mathbb{N} \sim S$ If we define the function as $f(n) = n + 1$ for $n \in \mathbb{N}$.

Def : Sequence

If f mapping J onto A and $f(n) = x_n$ then we call it "sequence". Since every countable set is the range of a 1-1 function defined on J this means by definition any set A which is countable $A \sim J$ so we can say that the elements of any "Countable set can be arranged in a sequence"

Thm : Infinite subset of Countable set

Every infinite subset of a countable set A is countable.

▷**Proof:** If $E \subset A$ and A is a countable set, we construct a sequence $\{n_k\}$ to index the elements of E in A . The sequence begins with $n_1 = \min\{n \in \mathbb{N} : x_n \in E\}$ and for $k \geq 2$, it is defined as $n_k = \min\{n > n_{k-1} : x_n \in E\}$. This process yields a sequence $\{x_{n_k}\}$ that includes all elements from E , demonstrating that E is countable. Therefore, every infinite subset of a countable set A is countable. ■

Def : Union and Intersection

let A and Ω be set and $\alpha \in A$, $E_\alpha \subset \Omega$ and , $\{E_\alpha\}_{\alpha \in A}$ is called collection of sets or family of sets then, following can be defined :

1. Union set : The union sets E_α is defined to be the set S such that $x \in S \Leftrightarrow x \in E_\alpha$ **for at least one** $\alpha \in A \Rightarrow S = \bigcup_{\alpha \in A} E_\alpha$
2. Intersection sets : Set P such that $x \in P \Leftrightarrow x \in E_\alpha$ **for every** $\alpha \in A \Rightarrow P = \bigcap_{\alpha \in A} E_\alpha$

Ex : Union and Intersection

let A be the set of real numbers x such that $0 < x \leq 1 \Rightarrow A = (0, 1]$. For every $x \in A$, let E_x be the set of all real numbers y such that $0 < y < x \Rightarrow E_x = (0, x)$ then we can say :

- $E_x \subset E_z \Leftrightarrow 0 < x \leq z \leq 1$. Because we can rewrite $(0, x) \subset (0, z)$ and this also means z always bigger than x and this is defined from 0 to 1.
- $\bigcup_{x \in A} E_x = E_1$. Because we defined that x from 0 to 1 and then E_x cover all range from 0 to 1. This indicate itself denote E_1
- $\bigcap_{x \in A} E_x$ is **empty** ★. Because we can rewrite this statement as $\bigcap_{x \in [0, 1]} (0, x)$. And a value y that belongs to the intersection must satisfy $y \in E_x$ for all $x \in A$, which means $0 < y < x$. However, since $A = (0, 1]$ and x can be arbitrarily close to 0, y must also satisfy $y < x$ for arbitrarily small x . Therefore, no $y > 0$ can satisfy this condition for all

$x \in A$, and thus, such a y does not exist.

Thm : Union of Countable sets

Union of countable sets still countable.

▷ **proof** : Let there is sequence $E_n, (n = 1, 2, 3, \dots)$ and $S = \bigcup_{n=1}^{\infty} E_n$ and here each E_n is countable. Then we need to show $S \sim J$ 1-1 and onto function. Image that each E_n are defined following :

$$E_1 = \{x_{11}^1, x_{12}^3, x_{13}^6, \dots\}$$

$$E_2 = \{x_{21}^2, x_{22}^5, x_{23}^9, \dots\}$$

$$E_3 = \{x_{31}^4, x_{32}^8, x_{33}^{13}, \dots\}$$

$$E_4 = \{x_{41}^7, x_{42}^{12}, x_{43}^{18}, \dots\}$$

⋮

And array the union of $E_n = S$ as $S = \{(1, 1), (2, 1), (1, 2), \dots\}$ and then lets correspond this with J then we can define the function f mapping S to $J : f(x, y) = y + \sum_{i=0}^{x+y-2} i$ then we can get 1-1 and onto function this means S has same cardinal with J by definition S is countable ■.

Cor : Union of at most countable set

Union of at mos countable sets still at most countable.

▷ **proof** : we proved that union of countable sets still countable. So, it is sufficient that to prove in finite case, since at most countable means finite or countable. let A is at most countable, and , for every $\alpha \in A$ B_α is at most countable. we can denote $T = \bigcup_{\alpha \in A} B_\alpha$ then B_α can be defined following :

$$B_1 = \{b_{11}, b_{12}, b_{13}, \dots b_{1m}\}$$

$$B_2 = \{b_{21}, b_{22}, b_{23}, \dots b_{2m}\}$$

$$B_3 = \{b_{31}, b_{32}, b_{33}, \dots b_{3m}\}$$

$$B_4 = \{b_{41}, b_{42}, b_{43}, \dots b_{5m}\}$$

⋮

$$B_n = \{b_{n1}, b_{n2}, b_{n3}, \dots b_{nm}\}$$

And array the union of B_α, T appropriately and lets correspond this with J_n then we can define function f mapping T to J as $f(b_{\alpha_i, j}) = n \in \mathbb{N}$ then we can know $T \sim J_n$ so, T is finite ■

Thm : Countable of n-tuples

Let A be a countable set, B_n be the set of all n -tuples $(a_1, a_2, \dots, a_n)$, where $a_k \in A$, $(k = 1, 2, 3, \dots)$. Then B_n is also countable. $\triangleright$ **proof** : B_1 is countable because $B_1 = A$, and suppose B_{n-1} is countable ($n = 2, 3, \dots$) the elements of B_n can be of the form (b, a) , $b \in B_{n-1}$, $a \in A$ the set of (b, a) has same cardinal with A . Thus B_n is the union of a countable set. by Thm : Union of countable set, B_n is countable set ■

Cor : Rational is countable

The sets of all rational numbers is countable.

$\triangleright$ **proof** : There is rational number sets defined as $E_n = \{\frac{n}{m} : m \in \mathbb{N}\}$, $(n = 1, 2, 3, \dots)$ and array sets with the number of m then we can write :

$$\begin{aligned} E_1 &= \{\frac{1}{1}, \frac{1}{2}, \dots\} \\ E_2 &= \{\frac{2}{1}, \frac{2}{2}, \dots\} \\ E_3 &= \{\frac{3}{1}, \frac{3}{2}, \dots\} \\ E_4 &= \{\frac{4}{1}, \frac{4}{2}, \dots\} \\ &\vdots \end{aligned}$$

Then, with the same manner which we've shown in Thm : Union of Countable sets we can get $J \sim S$ ■

Metric Spaces

Def : Distance function

If there is a set of called metric spaces, then this set contains components. These components are referred to as points. Let d, p be points in a metric space X . An associated real number $d(p, q)$ is defined, which represents the distance between points p and q . If this distance satisfies the following three properties, it is classified as a distance function

1. $p \neq q \Rightarrow d(p, q) > 0$
2. $d(p, q) = d(q, p)$
3. $d(p, q) \leq d(p, r) + d(r, q) \forall r \in X$

Def : Segment, Interval , Open ball

There is useful definition defined in metric space :

- interval $[a, b] \Rightarrow a \leq x \leq b$
- Segment $(a, b) \Rightarrow a < x < b$
- k -cell : For every integer $(i = 1, 2, \dots)$, let $a_i, b_i \in \mathbb{R}$ such that $a_i < b_i$ The set of all points $x = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ is a k -cell

- Open ball $\Rightarrow B = \{y \in \mathbb{R} : |y - x| < r\}$
- Closed ball $\Rightarrow B = \{y \in \mathbb{R} : |y - x| \leq r\}$
- Convex : If $x, y \in E$ and there is $0 \leq \lambda \leq 1$, and If the set is convex the following property should be satisfied : $\lambda x + (1 - \lambda)y \in E$

Thm : Open ball is convex

every open ball is convex

▷ **proof** : Open ball $B_r(x) = \{y \in \mathbb{R} : |x - y| < r\}$, and suppose there is $z, y \in B_r(x)$, and $w = \lambda y + (1 - \lambda)z$. Then let's check whether w is in $B_r(x)$ or not. which means $|x - w| = |w - x| < r$.

$$\begin{aligned}
 |w - x| &= |\lambda y + (1 - \lambda)z - x| \\
 &= |\lambda(y - x) + (1 - \lambda)(z - x)| \\
 &\leq \lambda|y - x| + (1 - \lambda)|z - x| \text{ (by triangular inequality)} \\
 &< \lambda r + (1 - \lambda)r \text{ where } |y - x| < r, |z - x| < r \\
 &= r \\
 \therefore |w - x| &< r \blacksquare
 \end{aligned}$$

Thm : K-cells convex

K-cells is convex ▷ **proof** : Suppose there is $x, y \in C$ and C is K-cells set, then we can know that all points in the set x, y is $x = (x_1, x_2, \dots, x_k), y = (y_1, y_2, \dots, y_k) \in \mathbb{R}^k$ for $a_i \leq x_i \leq b_i, a_i \leq y_i \leq b_i$ such that $a_i < b_i$ and both are in real number. Then let $z = \lambda x + (1 - \lambda)y$ then :

$$\begin{aligned}
 z &= \lambda x_1 + (1 - \lambda)y_1 \\
 &= \lambda x_2 + (1 - \lambda)y_2 \\
 &= \lambda x_1 + (1 - \lambda)y_2 \\
 &\vdots \\
 &= \lambda x_k + (1 - \lambda)y_k
 \end{aligned}$$

And by the condition that $a_i \leq x_i, a_i \leq y_i$ we can know $z_i = \lambda x_i + (1 - \lambda)y_i \geq \lambda a_i + (1 - \lambda)a_i = a_i \therefore z_i \geq a_i$ and also condition that $b_i \geq x_i, b_i \geq y_i$ we can derive that $z_i = \lambda x_i + (1 - \lambda)y_i \leq \lambda b_i + (1 - \lambda)b_i = b_i \therefore b_i \geq z_i = b_i \therefore z_i \leq b_i$ and combination those two then we can say $a_i \leq z_i \leq b_i$. Thus $z \in C$ and this indicate convex ■

Def : Various subset of metric Space ★

let X be a metric space. and there are various subset ($= E$) in metric space X

1. **Neighborhood** : $N_r(p) = \{q \in X : d(p, q) < r\}$, for here r is radius of $N_r(p)$

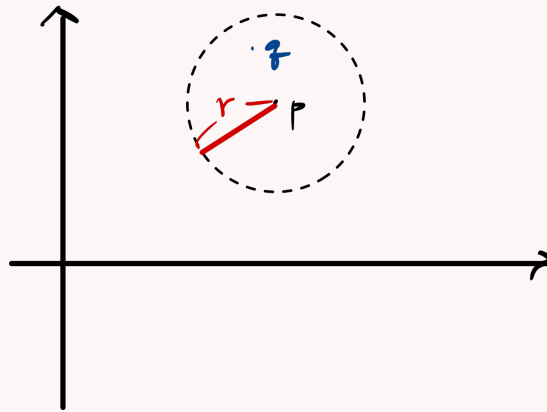


Figure 7: Neighborhood

2. **Limit point** : The point p whose every neighborhood(=nhd) of p contains a point q such that $q(\neq p) \in E$
3. **Isolated point** : $p \in E$ and p is not a limit point of E this means, there is no $q \in N_r(p)$ such that $q \in E$
4. **closed** : if every limit point of E is a point of E
5. **interior point** : there exists of a nhd of N of p such that $N \subset E$
6. **Open** : every point of set E is interior point

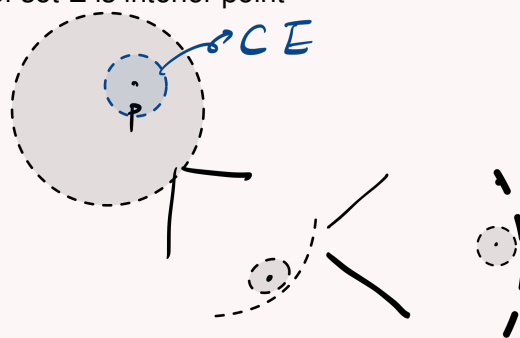


Figure 8: Open

7. **Complement** : E^c
8. **Perfect** : E is closed and every point of E is a limit point of E
9. **Bounded** : $\exists M \in \mathbb{R}$ and a point $q \in X$ such that $d(p, q) < M$ for all $p \in E$

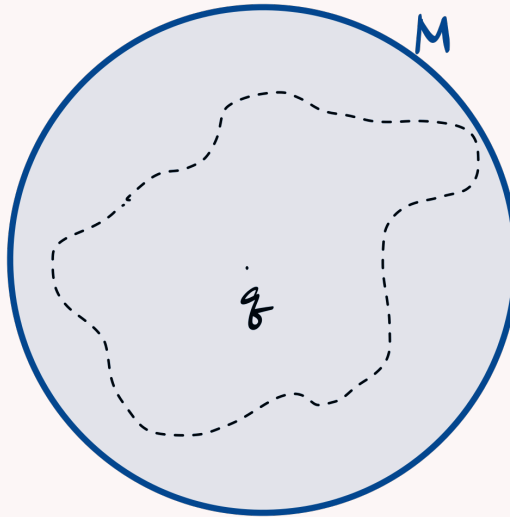


Figure 9: Bounded

10. **Dense:** X if every point of X is a limit point of E , or a point of E .

With the property of **Dense**, we can know rational number $\mathbb{Q}$ is dense in $\mathbb{R}$

▷ **proof** : The statement that the set of rational numbers $\mathbb{Q}$ is dense in the set of real numbers $\mathbb{R}$ means that for any point $x \in \mathbb{R}$ and for any $r > 0$, there exists a rational number $q \in \mathbb{Q}$ such that q lies within the nhd of $x \in X$, $q \in N_r(x)$. This implies that every real number is either a limit point of the set of rational numbers or a rational number itself, thus demonstrating that $\mathbb{Q}$ is dense in $\mathbb{R}$.

Thm : Neighborhood is Open set

Every neighborhood is an open set.

▷ **proof** : Remind that $N_r(p) = \{q \in X : d(p, q) < r\}$. we need to check every point of $q \in N_r(p)$ is an interior point. let q be any point of E $d(p, q) = r - h$ here $h > 0$ then we can write $N_r(p, q) = \{s \in X : d(q, s) < h\}$. And we have to check whether $N_r(p, q) \subset E$ or not. Choose any point in $N_r(p, q)$ let s . then $d(p, s)$ following is true : $d(p, s) \leq d(p, q) + d(q, s)$ by triangular inequality. and this means $d(p, s) < r - h + h = r$. So that $s \in E$ thus q is an interior point of E . ■

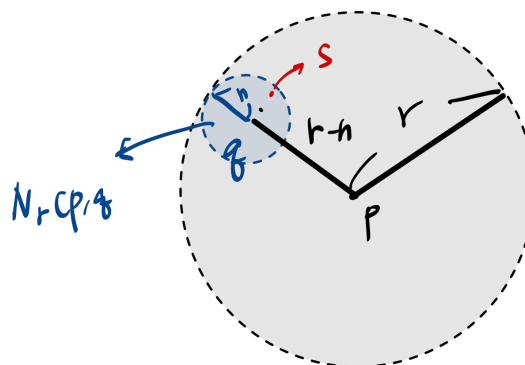


Figure 10: Neighborhood is Open set

Thm : Neighborhood of limit point

If p is a limit point of set E , then every nhd of p contains infinitely many points of E

▷ **proof** : To get contradiction, Assume that $\exists N = N_r(p)$ such that contains finite number of points of E and let those $q_1 \cdots q_n \in N \cap E$ and select r such that $r = \min d(p, q_m)$ where $1 \leq m \leq n$ then $N_r(p) \cap E = \emptyset$. Because we select radius which smallest one near point p , this is contradiction of definition of limit point ■

Thm: De Morgan's Laws

Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α . Then we can say

$$\left(\bigcup_{\alpha} E_{\alpha} \right)^c = \bigcap_{\alpha} (E_{\alpha}^c)$$

▷ **Proof**: First, let us consider the case where $(\subseteq)$, assume $x \in \left(\bigcup_{\alpha} E_{\alpha} \right)^c$. This means $x \notin \bigcup_{\alpha} E_{\alpha}$, which implies that $x \notin E_{\alpha}$ for all α . Therefore, x belongs to E_{α}^c for all α . Thus, $x \in \bigcap_{\alpha} E_{\alpha}^c$. Hence,

$$\left(\bigcup_{\alpha} E_{\alpha} \right)^c \subseteq \bigcap_{\alpha} (E_{\alpha}^c).$$

Now, consider the case $(\supseteq)$ where $x \in \bigcap_{\alpha} (E_{\alpha}^c)$. This means $x \notin E_{\alpha}$ for all α . This implies that $x \notin \bigcup_{\alpha} E_{\alpha}$, hence $x \in \left(\bigcup_{\alpha} E_{\alpha} \right)^c$. Therefore,

$$\left(\bigcup_{\alpha} E_{\alpha} \right)^c \supseteq \bigcap_{\alpha} (E_{\alpha}^c).$$

Thus, we conclude that

$$\left(\bigcup_{\alpha} E_{\alpha} \right)^c = \bigcap_{\alpha} (E_{\alpha}^c).$$

■

Thm : Open Set and Complement Closedness

A set $E \subseteq X$ (where X is a metric space) is open if and only if E^c is closed.

▷ **Proof:(1)** $(\Leftarrow)$ Assume that E^c is closed. We need to prove that E is open, which means every point of E is an interior point of E . Let $x \in E$. We wish to show that x is an interior point of E . Since $x \in E$, it follows that $x \notin E^c$. Now, because E^c is closed, x is not a limit point of E^c . Therefore, there exists some neighborhood $N_r(x)$ of x such that $N_r(x) \cap E^c = \emptyset$, meaning $N_r(x) \subset E$. Hence, x is an interior point of E . Since $x \in E$ was arbitrary, we conclude that E is open.

(2) $(\Rightarrow)$ Assume E is open. We need to prove that E^c is closed, meaning that every limit point of E^c belongs to E^c . Let y be a limit point of E^c . We want to show that $y \in E^c$. Since y is a limit point of E^c , every neighborhood $N_r(y)$ of y contains a point of E^c other than y itself. Now, assume $y \notin E^c$. This means $y \in E$, but since E is open, there exists a neighborhood $N_r(y) \subset E$, which contradicts the assumption that y is a limit point of E^c . Therefore, $y \in E^c$, and thus every limit point of E^c belongs to E^c , proving that E^c is closed. Thus, we conclude that E is open if and only if E^c is closed. ■

Cor : Open Set and Complement Closedness ★

A set F is closed if and only if F^c is open

▷ **proof** : (1) ($\Rightarrow$) Assume that F is closed then by the Thm : Open Set and Complement Closedness F^c is open .

(2) ($\Leftarrow$) Assume that F^c is open then, $(F^c)^c = F$ is closed ■

Thm : Properties of Unions and Intersections of Open and Closed Sets

1. For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.

▷ **proof** : Let's assume we have a collection of open sets $\{G_\alpha\}$. By the definition of open sets, for every point $x \in \{G_\alpha\}$, there exists a neighborhood $N_r(x) \subset \{G_\alpha\}$. Now, let's consider an arbitrary point $x \in \bigcup_\alpha G_\alpha \Rightarrow x \in G_\alpha$, so $N_r(x) \subset \bigcup_\alpha G_\alpha$ also holds. Therefore, we conclude that $\bigcup_\alpha G_\alpha$ is an open set. ■

2. For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.

▷ **proof** Let's assume that F_α is a collection of closed sets. If F_α is closed, then F_α^c is open. By De Morgan's laws, we have $\left(\bigcap_\alpha F_\alpha\right)^c = \bigcup_\alpha F_\alpha^c$. Since F_α^c is open, $\bigcup_\alpha F_\alpha^c$ is also open. ■

3. For any finite collection $\{G_i\}_1^n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.

▷ **proof** : Let's assume $G_1, \dots, G_n$ are open sets. According to the definition, for every point $x \in G_i$, there exists a neighborhood $N_r(x) \subset G_i$. Now, if we consider $x \in \bigcap_{i=1}^n G_i$, then x belongs to each G_i , and since $G_1, \dots, G_n$ are open sets, $N_r(x) \subset G_i$ for every i . Therefore, $N_r(x)$ is also completely contained in $\bigcap_{i=1}^n G_i$. ■

4. For any finite collection $\{F_i\}_1^n$ of open closed sets, $\bigcup_{i=1}^n F_\alpha$ is closed.

▷ **proof** : Let's say $F_1, \dots, F_n$ are closed sets. Using a similar method as in point 2, we know that the complement of any closed set is an open set. Specifically, $\left(\bigcup_{i=1}^n F_\alpha\right)^c = \bigcap_{i=1}^n F_\alpha^c$, and since each F_α^c is an open set and we've already proven that the intersection of open sets is also an open set, it follows that $\bigcap_{i=1}^n F_\alpha^c$ is an open set.

Therefore, since $\left(\bigcup_{i=1}^n F_\alpha\right)^c$ is an open set, we conclude that $\bigcup_{i=1}^n F_\alpha$ is a closed set. ■

Def : Closure

If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X , then the closure of E is the set $\bar{E} = E \cup E'$.

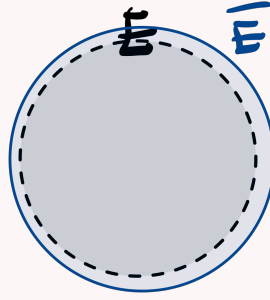


Figure 11: $\bar{E} = E \cup E'$

Thm : Properties of Closure

1. $\bar{E}$ is closed

▷ **proof** : What we prove is that $\bar{E}^c$ is open, thus for $x \in \bar{E}^c$ x is an interior point of $\bar{E}^c$. Because x is not a limit point of E , $N_r(x) \cap E = \emptyset$ also $N_r(x) \cap E' = \emptyset$. Assume, for contradiction, that $N_r(x) \cap E' \neq \emptyset$, meaning there exists a limit point of E within the neighborhood $N_r(x)$. By the definition of a limit point, this would imply that x must be a limit point of E , as every neighborhood of a limit point contains points of E . However, this contradicts the assumption that $x \in \bar{E}^c$, meaning x is not a limit point of E . Therefore, our assumption is false, and we conclude that $N_r(x) \cap E' = \emptyset$ thus, $N_r(x) \cap (E \cup E') = \emptyset \Rightarrow N_r(x) \subset \bar{E}^c$ ■

2. $E = \bar{E} \Leftrightarrow E$ is closed

▷ **proof** :

(1)($\Rightarrow$) If $E = \bar{E}$, Because $\bar{E}$ is closed (we proved), E is closed

(2)($\Leftarrow$) If E is closed then $E' \subset E$. By definition of closed set, every limit point (E') in set E . And $\bar{E} = E \cup E'$ this $\bar{E} = E$. ■

3. $\bar{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

▷ **proof** : Since F is closed set $F \supset F'$, and thus it is trivial that $F \supset E \Rightarrow F' \supset E'$ means if $x \in E'$ then $x \in F'$. Combine this two then we can get $F \supset E'$ because $F \supset F'$ thus, $F \supset E \cup E' = \bar{E}$ ■

Thm : Closed set contains Supremum

Let E be a nonempty set of real numbers which is bounded above. Suppose $y = \sup E$. Then $y \in \bar{E}$. Hence E is closed $\Rightarrow y \in E$ ▷ **proof** : If $y \in E$ then $y \in \bar{E}$ is very trivial. So we focus on the case that $y \notin E$. For every $h > 0$ there exists a point $x \in E$ such that $y - h < x < y$ and $x \in N_h(y)$ such that $x \in E$ this means y is a limit point of E so, $y \in E'$ but $y \notin E$. Hence $y \in \bar{E}$ ■

Remark : Closed set contains Supremum

Thm : Relative openness★

If $E \subset Y \subset X$, where X is metric space then E is **open relative to Y** $\Leftrightarrow E = Y \cap G$ for some open subset G of X

▷ **proof** :

(1)($\Rightarrow$) Since E is open relative to Y , for every $p \in E$, there exists a neighborhood $N_r(p) \subset Y$ such that $N_r(p) \subset E$. Here, $N_r(p) = \{q \in Y : d(p, q) < r_p\}$, where r_p is a positive radius ensuring that p is an interior point of E in Y . Now, define a neighborhood

$V_p = \{q \in X : d(p, q) < r_p\}$. Since this is a neighborhood of p in X , $V_p \subset X$ is open in X . Let $G = \bigcup_{p \in E} V_p$. Clearly, G is an open set in X , as it is a union of open sets V_p . For each $p \in E$, we have $p \in Y$ and $p \in V_p$, so $p \in Y \cap G$. Therefore, $E \subseteq Y \cap G$. On the other hand, because $N_r(p) \subset E$, we also have $V_p \cap Y \subset E$ for each $p \in E$. Thus, $G \cap Y \subset E$, which gives $E = Y \cap G$.

(2)($\Leftarrow$) Assume $E = Y \cap G$, where G is an open set in X . We will show that E is open relative to Y . For any $p \in E$, since $p \in G$ and G is open in X , there exists a neighborhood $N_r(p) \subset G$ in X , where $N_r(p) = \{q \in X : d(p, q) < r\} \subset G$. Now, define $N_1 = N_r(p) \cap Y$. This is a neighborhood of p in Y , since $N_1 = \{q \in Y : d(p, q) < r\} \subset G \cap Y = E$. Therefore, p is an interior point of E in Y , meaning E is open relative to Y . ■

Cor : Relative closedness ★

Let, $F \subset Y \subset X$ and F is closed relative to $Y \Leftrightarrow F = Y \cap C$ for some closed subset C of X ▸ **proof :**

(1)($\Rightarrow$) Assume F is closed relative to Y . We will show that $F = Y \cap C$ for some closed set $C \subset X$. Since F is closed relative to Y , we know that $F = Y \cap G^c$ for some open set $G \subset Y$, where G^c is the complement of F in Y . Now, by the theorem of relative openness, since G is open in Y , we can write $G = Y \cap G'$ for some open set $G' \subset X$. Taking complements, we get $F = Y \cap (G')^c$, where $(G')^c$ is closed in X . Let $C = (G')^c$.

(2)($\Leftarrow$) Assume $F = Y \cap C$, where $C \subset X$ is closed. We will show that F is closed relative to Y . Since C is closed in X , for any point $x \in C$, every limit point of C lies in C . For any sequence $\{x_n\} \subset F$ that converges to a point $x \in Y$, since $F = Y \cap C$, the limit point x must lie in C . Therefore, $x \in F$, meaning F contains all its limit points in Y . This shows that F is closed in Y , or closed relative to Y . ■

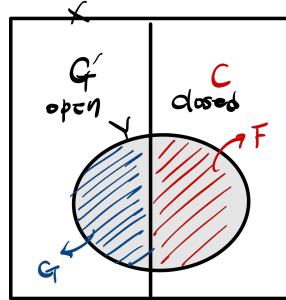


Figure 12: Relative closedness

Compact Set

Def : Open Cover

By an Open cover of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \bigcup_{\alpha} G_\alpha$

For Instance, $X = \mathbb{R}$ and E is interval $(0, 1)$ and let $G = (0, \frac{1}{n})$, for $n = 1, 2, 3 \dots$ and then $\bigcup_{n=1}^{\infty} G_n = (0, 1)$ this indicate that $\{G_\alpha\}$ is an open cover of E

Def : Compact

A subset k of a metric space X is said to be **Compact** if every open cover of k contains a finite sub cover. More explicitly, if $\{G_\alpha\}$ is an open cover of k , then there exists finitely many indices $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $k \subset G_{\alpha_1} \cup G_{\alpha_2} \dots \cup G_{\alpha_n}$

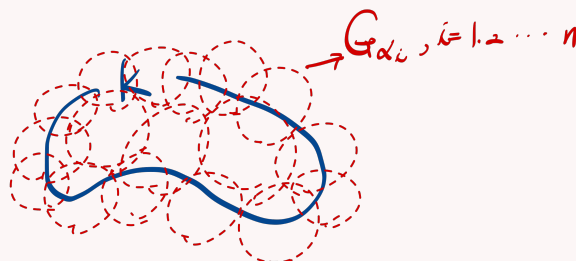


Figure 13: Compact

For example, $E = (1,0)$ is not compact

▷ **proof** : a set E to be compact, every given "finite" open cover must be able to cover E . To explore a set that cannot be covered by a finite open cover but can be covered by an infinite one, consider the sets defined by $G_n = (0, 1 - \frac{1}{n})$. We have $\bigcup_{n=1}^{\infty} G_n = (0, 1)$. However, for any finite collection of indices n , we have $G_{n_1} \cup G_{n_2} \cup \dots \cup G_{n_R} = (0, 1 - \frac{1}{R}) \not\supset E$. Thus, we conclude that the set E is not compact. ■

Thm : Relative compact

★ Suppose $K \subset Y \subset X$. Then K is **compact relative to X** if and only if K is **compact relative to Y**

▷ **proof** : (1) ($\Rightarrow$) Let there exist a finite open cover $\{G_\alpha\}$ such that $\bigcup_\alpha G_\alpha \supset K$ and each G_α is open in X . We need to show that $\{G_\alpha\}$ is an open cover of Y , which means that G_α is open in Y . By the theorem on relative openness, we know that $G_\alpha = Y \cap V_\alpha$, where V_α is some open set in X . If we collect these sets $\{V_\alpha\}$ and take their union, we have $\bigcup_\alpha V_\alpha \supset \bigcup_\alpha G_\alpha \supset K \Rightarrow \bigcup_\alpha V_\alpha \supset K$. This means that $\{V_\alpha\}$ is an open cover of K in X . Since K is compact in X ,

$$\begin{aligned} V_{\alpha_1} \cup V_{\alpha_2} \dots \cup V_{\alpha_n} &\supset K \\ \Rightarrow G_{\alpha_1} \cup G_{\alpha_2} \dots \cup G_{\alpha_n} &\supset K \\ &= (Y \cap V_{\alpha_1}) \cup (Y \cap V_{\alpha_2}) \cup (Y \cap V_{\alpha_3}) \dots \cup (Y \cap V_{\alpha_n}) \\ &= Y \cup (V_{\alpha_1} \cup V_{\alpha_2} \dots \cup V_{\alpha_n}) \supset K \end{aligned}$$

This means that K is compact in Y .

(2) ($\Leftarrow$) Assume that K is compact in Y and we aim to show that K is also compact in X . Let $\{V_\alpha\}$ be an open cover of K in X , meaning $\bigcup_\alpha V_\alpha$ covers K , where each V_α is an open set in X . Since $V_\alpha \cap Y$ is an open set, the collection $\{V_\alpha \cap Y\}$ serves as an open cover for $K \subset Y$. Because K is compact in Y , there exists a finite subcover from $\{V_\alpha \cap Y\}$. In other words, we can write $K \subset \bigcup_{i=1}^n (V_{\alpha_i} \cap Y)$. This implies that K is covered by finitely many sets from the original open cover in X , demonstrating that K is compact in X . ■

Thm : Compact subsets are closed

Compact subsets of metric spaces are **closed**. For example $E=(0,1)$ is open and E is not compact

▷ **proof** :let X : metric space and $K \subset X$, then K is compact. We will prove K^c is open and then we finally can know that K is closed. Choose point in K^c , $p \in K^c$ then lets chech whether it is an interior point of K^c , which means there exists open nhd of p in K^c . For $d(p, q) > 0$ and let $\frac{d(p, q)}{2} = r$ then $p \notin N_r(q)$ is true. So we can denote the open cover of K as following : $\{N_{r_q}(q) : q \in K\}$. For here r may differ depending where q is in K thus,
 $\bigcup_{q \in K} N_{r_q}(q) \supset K$.

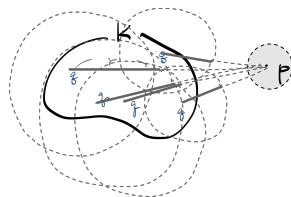


Figure 14: Open cover of K

And now, Set $r = \min\{r_{q_1}, r_{q_2}, \dots, r_{q_n}\}$ then $N_r(p) \cap K = \emptyset$ thus, $N_r(p) \subset K^c$ so K^c is open so K is closed ■

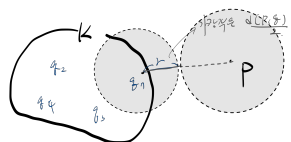


Figure 15: Empty set of $N_r(p) \cap K$

Def : Closed subsets of compact are compact

Closed subsets of Compact sets are compact Suppose F is closed and X is compact such that $K \subseteq X$. Let $\{V_\alpha\}$ be an open cover of F . Consider the collection $\{V_\alpha, F^c\}$. Then we have:

$$V_{\alpha_1} \cup V_{\alpha_2} \cup \dots \cup V_{\alpha_n} \cup F^c \supset K$$

Since $K = F \cup (F^c \cap K)$, we can conclude that F is compact. ■

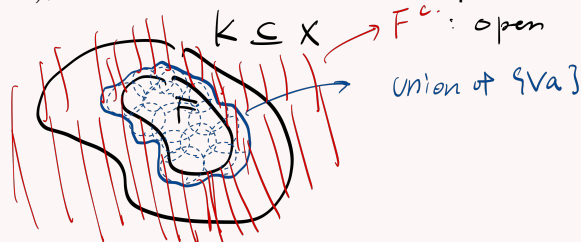


Figure 16: Closed subsets of Compact sets are compact.

Cor : Intersection of closed and compact set

Intersection of closed and compact set is compact. ▷ **proof** : let F is closed set and K is compact set then $F \cap K = E$ is closed by Thm Compact subsets are closed. Thus E is compact by Thm : Closed subsets of compact are compact $E \subset K$ also compact. ■

Thm : collection of compact

If $\{K_\alpha\}$ is collection of compact subsets of a metric space X such that the intersection of every finite sub collection of $\{K_\alpha\}$ is nonempty, then $\cap K_\alpha$ is nonempty.

▷ **proof** : To get contradiction let $\cap K_\alpha = \emptyset$. And fix K_1 for collection of compact set $\{K_\alpha\}$. Suppose G_α is the complement of K_α then one might easily know that G_α is open. We assumed that $\cap K_\alpha = \emptyset$ thus, G_α is open cover of K_1 . Because K_1 is compact, by definition there exists finite open cover of K_1 , this denote that $G_{\alpha_1}, G_{\alpha_2}, \dots, G_{\alpha_n} \Rightarrow K_{\alpha_1}^c, K_{\alpha_2}^c, \dots, K_{\alpha_n}^c \Rightarrow K_{\alpha_1}^c \cup K_{\alpha_2}^c \cup \dots \cup K_{\alpha_n}^c$

AFTER MIDTERM

Perfect set

Thm : Heine-Borell Theorem

If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two :

1. E is closed and bounded.
2. E is compact.
3. Every infinite subset of E has a limit point in E .

▷ **proof** : Skip

Def : Perfect sets

E is perfect if E is **closed** and **every point of E is limit point of E**

For example, Suppose $X = \mathbb{R}$ and $E = \{0\} \cup \{\frac{1}{n} : n = 1, 2, \dots\}$ Since 0 is limit

point of $E(q \in N_r(p), \text{ s.t } q \in \{\frac{1}{n} : n = 1, 2, \dots\})$ is not perfect because, there is no problem at 0 but every point of $A = \{\frac{1}{n} : n = 1, 2, \dots\}$ is not a limit point rather isolated point. We can find neighborhood of every point of A has at least one of them(=nhd) does not contain point of E .

Thm : perfect set is uncountable

Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is **uncountable**.

▷ **proof** : Suppose P is a nonempty perfect set in $\mathbb{R}^k$ and is countable, so we can enumerate its elements as $P = \{x_1, x_2, x_3, \dots\}$. By definition, a perfect set is closed and every point is a limit point. For any point x_1 , an open neighborhood $N_r(x_1)$ must contain infinitely many other points from P . The same applies to x_2, x_3 , and so forth, implying that between any two points in P , there must be infinitely many points of P . This contradicts the assumption that P is countable, as it requires P to have isolated points, which a perfect set cannot have. Therefore, P must be uncountable. ■

The Cantor set

The set which we are now going to construct shows that there exist perfect sets in $\mathbb{R}^1$ which contain no segment. Suppose there is the set $E = \{E_1, E_2, \dots\}$ which has following property:

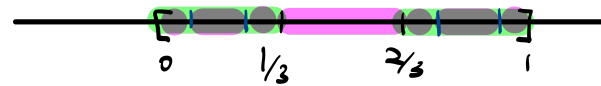
$$\begin{aligned} E_1 &= [0, 1] \\ E_2 &= \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] \\ E_3 &= \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{3}{9}\right] \cup \left[\frac{6}{9}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] \\ &\vdots \end{aligned}$$


Figure 17: The Cantor set

Each set E_i in the construction of the Cantor set is closed and, by the Heine-Borel theorem, also compact. The Cantor set is defined by the property $E_{i+1} \subset E_i$, and thus, the intersection $P = \bigcap_{i=1}^{\infty} E_i$ is nonempty and compact. To show that P is a perfect set, we need to demonstrate that P is closed and that every point in P is a limit point of P .

Given any $x \in P$, it follows that $x \in E_n$ for all n . By the construction of the Cantor set, segments of the form $(\frac{3k+1}{3^m}, \frac{3k+2}{3^m})$ (where k and m are positive integers) are removed at each stage. The inequality $3^{-m} < \frac{\beta-\alpha}{6}$ indicates that such segments do not contain points from P , so $(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}) \not\subset P$ for any $(\alpha, \beta) \subset (\frac{3k+1}{3^m}, \frac{3k+2}{3^m})$.

To prove that P is perfect, we must show that every $x \in P$ is a limit point of P . Since $x \in P$, it lies in E_n for every n . Assume there exists an interval $I_n \subseteq E_n$ such that $x \in I_n$. Let x_n be an endpoint of I_n with $x_n \neq x$ and $x_n \in E_n$. By construction, as n becomes sufficiently large, the length of I_n approaches 0, and $x_n \rightarrow x$. This demonstrates that x is a limit point of P , proving that P has no isolated points and is, therefore, perfect. Hence, the intersection defining the Cantor set is a perfect set.

Connected Sets

Def : Separated

Two Subsets A and B of a metric X are said to be **Separated** if both $A \cup \overline{B}$ and $\overline{A} \cup B$ are empty

For example, $A = [0,1)$ and $B = (1,2]$. For here $\overline{A} \cap B = [0,1] \cap (1,2] = \emptyset$ Where $\overline{A} = [0,1]$ On the other hand, $\overline{B} \cap A = [0,1) \cap [1,2] = \emptyset$ thus, A and B are separated.

A set $E \in X$ is said to be **Connected** if E is not a **union of two nonempty separated sets**.

For example, $E = \underbrace{[0,1)}_A \cup \underbrace{(1,2]}_B = [0,2] \setminus \{1\}$ this set is not Connected. Since E is union of Two nonempty separated sets.

Remark

Separated sets are of course disjoint, but disjoint sets need not be separated. For example, the interval $[0,1]$ and the segment $(1,2)$ are not separated, since 1 is a limit point of $(1,2)$. However, the segments $(0,1)$ and $(1,2)$ are separated. The connected subsets of the line have a particularly simple structure.

For example, $A = [0,1], B = (1,2)$, $A \cap B = \emptyset$ But A and B are not separated since $\overline{A} \cup B = \emptyset$ but, $\overline{B} \cup A = [0,1] \cap [1,2] \neq \emptyset$ this example indicates that $A \cap B = \emptyset$ does not mean A and B is separated.

Thm : property of Connected sets in $\mathbb{R}^1$

A subset E of the real line $\mathbb{R}^1$ is connected (A) if and only if it has the following property:
If $x \in E, y \in E$, and $x < z < y$, then $z \in E$ (B).

Note : $A \Rightarrow B \Leftrightarrow \sim B \Rightarrow \sim A$

▷ **Proof**:

($\Rightarrow$) Suppose there exist $x, y \in E$ such that $z \notin (x, y) \cap B$. We need to show that E is not connected ($\sim A$), meaning we need to show that E can be separated into two nonempty, disjoint sets. Let $A = E \cap (-\infty, z)$ and $B = E \cap (z, \infty)$. This implies:

$$A \cup B = E \quad \text{and} \quad z \notin E.$$

Without loss of generality, assume $x < z < y$. We know $x, y \in E$, so $x \in A$ and $y \in B$. Thus, $A \neq \emptyset$ and $B \neq \emptyset$. However, $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Since E is the union of A and B , and A and B are nonempty and separated, by the definition of connectedness, E is not connected. Therefore, if E is connected, then for any $x, y \in E$ and any z such that $x < z < y$, we must have $z \in E$.

($\Leftarrow$) Wish to show if suppose E is not connected ($\sim A$). Then there exist x, y, z such that $x, y \in E$, $x < z < y$, and $z \notin E$. For $A, B \neq \emptyset$, by assumption that E is separated $\overline{A} \cap B = A \cap \overline{B} = \emptyset, E = A \cup B$. Assume $x \in A$ and $y \in B$ and $x < y$ let $z = \sup(A \cap [x, y])$ then $z \in \overline{A}$ by theorem that the set is bounded above hence $z \notin B$. In particular $x \leq z < y$, if $z \notin A$ then $x < z < y$ and $z \in E$ otherwise If $z \in A$ then $z \notin \overline{B}$, hence there exists z_1 such that $z < z_1 < y$ and $z_1 \notin B$ and then $x < z_1 < y$ and $z_1 \notin E$. ■

Convergent Sequences

Def : Converge

A sequence $\{p_n\}$ in a metric space X is said to **Converge** if there exists a point in X with the following property:

$\forall \epsilon > 0 \exists N$ such that $d(p_n, p) < \epsilon$ whenever $n > N$, where N is integer. ★

For example, $X = \mathbb{R}$, $\{p_n\} = \frac{1}{n} \rightarrow p = 0$ as $n \rightarrow \infty$. Since, $\exists N \in \mathbb{N}$ such that $|\frac{1}{n} - 0| < \epsilon, \forall \epsilon > 0, n > N$. Because by archimedean Theorem that $\exists$ integer n such that $nx > y$ if $x = 1$ and $y = \frac{1}{\epsilon} \in \mathbb{R} \Rightarrow N \cdot 1 > \frac{1}{\epsilon} \Rightarrow \epsilon > \frac{1}{N}$ thus we can say that $\frac{1}{n} = 0 = \frac{1}{N} < \epsilon$

Another example, let $\{p_n\} = \{1, 1, \dots\}$ then we know $\forall \epsilon > 0, \exists N$ such that $|p_n - 1| < \epsilon$ whenever $n \leq N$ which N is 1 or 2, or 3.. whatever.

One more, Suppose $\{p_n\} = \frac{(-1)^n}{n} + 1$ then we can state that, $\forall \epsilon > 0, |p_n - 1| < \epsilon \forall n \geq N$. Then we can find N by : $|\frac{(-1)^n}{n} + 1 - 1| < \epsilon \Rightarrow |\frac{(-1)^n}{n}| < \epsilon \Rightarrow |\frac{1}{n}| < \epsilon \Rightarrow \frac{1}{n} < \frac{1}{N} < \epsilon$

And also, the sequence $\{p_n\}$ is said to be **bounded if its range is bounded**

Thm : Various properties of Sequence

Let $\{p_n\}$ be a sequence in a metric space X .

1. $\lim_{n \rightarrow \infty} p_n = p \Leftrightarrow$ every nhd of p contains $\{p_n\}$ all but finitely many of the n .

▷ **proof:**

($\Rightarrow$) Let V be a nhd of p and wish to show that $\forall \{p_n\}$ except finitely many of them in V . there exists $\epsilon > 0$ such that $N_\epsilon(p) \in V$ and $N_\epsilon(p) = \{q \in X : d(p, q) < \epsilon\}$ there exists N such that $d(p_n, p) < \epsilon \forall n \geq N$ since $\lim_{n \rightarrow \infty} p_n = p$ thus $\{p_N, p_{N+1}, \dots\} \subset V$ thus nhd of p contain $\{p_n\}$ such that "finitely many n "

($\Leftarrow$) Suppose every nhd of p contains $\{p_n\}$ all but finitely many of the n . then wish to show that $\forall \epsilon > 0, \exists N$ such that $d(p_n, p) < \epsilon \forall n \geq N$. Thus, If we choose biggest n outside of nhd of V as N then we can say that $n \geq N$ implies $d(p_n, p) < \epsilon, \forall \epsilon > 0$ thus $\lim_{n \rightarrow \infty} p_n = p$

2. If $p \in X, p' \in X$ and if $\lim_{n \rightarrow \infty} p_n = p, p'$ then $p = p'$

▷ **proof :** Suppose there exists other integer N, N' such that $n \geq N \Rightarrow |p_n - p| < \epsilon$ and $n \geq N' \Rightarrow |p_n - p'| < \epsilon$. This is true because we assumed $\lim_{n \rightarrow \infty} p_n = p, p'$ and lets choose $n \geq \max(N, N')$ then we have by triangular inequality $|p_n - p'| \leq |p_n - p| + |p_n - p'| < 2\epsilon = \epsilon$ since this statement should be satisfied for any $\epsilon > 0$ thus $|p - p'| = 0$ which means $p = p'$ ■

3. $\lim_{n \rightarrow \infty} p_n = p$ then $\{p_n\}$ is bounded

▷ **proof :** Since $\lim_{n \rightarrow \infty} p_n = p$, $\exists N$ such that $|p_n - p| < \epsilon \forall n > N$. Suppose $\epsilon = 1$ then $\exists |p_1 - p|, |p_2 - p|, |p_3 - p|, \dots, |p_N - p|$ And if we choose $r = \max\{1, |p_1 - p|, |p_2 - p|, |p_3 - p|, \dots, |p_N - p|\}$ then $|p_n - p| < r$ we can know $\{p_n\}$ is bounded. ■

4. If $E \subset X$ and if p is a limit point of E , then there is a sequence $\{p_n\}$ in E such that $\lim_{n \rightarrow \infty} p_n = p$

▷ **proof** : Since p is a limit point of E , every neighborhood of p contains at least one point of E distinct from p . For each $n \in \mathbb{N}$, consider the open neighborhood $N_{1/n}(p)$ of radius $\frac{1}{n}$ centered at p . By the definition of a limit point, there exists a point $p_n \in E$ such that $p_n \neq p$ and $d(p_n, p) < \frac{1}{n}$. Now, given any $\epsilon > 0$, choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$. For all $n > N$, we have $d(p_n, p) < \frac{1}{n} < \epsilon$. This implies:

$$|p_n - p| < \epsilon \quad \text{for all } n > N.$$

Therefore, $\lim_{n \rightarrow \infty} p_n = p$. This shows that there is a sequence $\{p_n\}$ in E converging to p .

Thm : Complex Sequences ★

Suppose s_n, t_n are complex sequences, and $\lim_{n \rightarrow \infty} s_n = s, \lim_{n \rightarrow \infty} t_n = t$ then,

1. $\lim_{n \rightarrow \infty} (s_n + t_n) = s + t$

▷ **proof** : Given $\epsilon > 0$, $\exists N_1$ and N_2 such that $n \geq N_1 \Rightarrow |s_n - s| < \epsilon, n \geq N_2 \Rightarrow |t_n - t| < \epsilon$ and if we choose $N = \max(N_1, N_2)$ then $n \geq N$ implies $|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < 2\epsilon$. For here ϵ is arbitrary number thus we can know $2\epsilon \Rightarrow \epsilon$. Finally we can know the limit point of $\lim_{n \rightarrow \infty} (s_n + t_n)$ is $s + t$. ■

2. $\lim_{n \rightarrow \infty} cs_n = cs, \lim_{n \rightarrow \infty} (c + s_n) = c + s, \forall c$

▷ **proof** : We can consider this with two case, If $c = 0$ then there exists N , for $|cs_n - cs| < \epsilon$ such that $n \geq N$. This is trivial since $cs_n = 0$ and cs also 0. However in case $c \neq 0$ then, we can consider ϵ as $\frac{\epsilon}{|c|}$ since ϵ be arbitrary number bigger than 0. then we have N such that $|cs_n - cs| = |c||s_n - s| < |c| \cdot \frac{\epsilon}{|c|} = \epsilon$ if whenever $n > N$. thus $cs_n \rightarrow cs$ as $n \rightarrow \infty$. On the other hand, By the assumption that there exists N for $\forall \epsilon > 0 |s_n - s| < \epsilon \forall n > N$. We can know there exists N such that $|(c + s_n) - (c + s)| = |s_n - s| < \epsilon$. for $n > N$. ■

3. $\lim_{n \rightarrow \infty} s_n t_n = st$

▷ **proof** : We use the identity that $s_n t_n - st = (s_n - s)(t_n - t) + s(t_n - t) + t(s_n - s)$. Suppose $\epsilon > 0 \exists N_1$ and N_2 such that $n \geq N_1 \Rightarrow |s_n - s| < \epsilon, n \geq N_2 \Rightarrow |t_n - t| < \epsilon$. If we take $N = \max(N_1, N_2)$ then $|(s_n - s)(t_n - t) - 0| < \epsilon^2 = \epsilon$. This indicate that $\lim_{n \rightarrow \infty} (s_n - s)(t_n - t) = 0$ be continue that $\lim_{n \rightarrow \infty} (s_n t_n - s_n t - t s_n + st) = 0$ and by we if we use the theorem that (a) and (b) then we can say $\lim_{n \rightarrow \infty} (s_n t_n) - st = 0 \Rightarrow \lim_{n \rightarrow \infty} s_n t_n = st$ ■

4. $\lim_{n \rightarrow \infty} \frac{1}{s_n} = \frac{1}{s} \Rightarrow s_n \neq 0 (n = 1, 2, \dots), s \neq 0$

▷ **proof** : Choosing m such that $n \geq m$ and suppose ϵ as $\frac{|s|}{2}$. Then, we have $|s_n - s| < \frac{|s|}{2}$. By the triangular inequality: $|s_n| = |s_n - s + s| \geq ||s| - |s_n - s||$. Substituting this, we get: $|s_n| \geq |s| - |s_n - s| > |s| - \frac{|s|}{2} = \frac{|s|}{2}$. Thus, $|s_n| > \frac{|s|}{2}$. Now, let's consider the limit: $\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s - s_n}{s s_n} \right|$. Since we proved that $|s_n| > \frac{|s|}{2}$, it follows that: $|s s_n| > |s| \cdot \frac{|s|}{2} = \frac{|s|^2}{2}$. Therefore, by our assumption: $\left| \frac{s - s_n}{s s_n} \right| < \frac{\epsilon}{\frac{|s|^2}{2}}$. Substituting the result above, we obtain: $\left| \frac{s - s_n}{s s_n} \right| < \frac{\epsilon}{\frac{|s|^2}{2}} = \frac{2\epsilon}{|s|^2}$. As a result, we can find N such that for $n > N$, the following holds: $\left| \frac{1}{s_n} - \frac{1}{s} \right| < \epsilon$. ■

Thm : Convergent Sequences in $\mathbb{R}^k$

1. Suppose $x_n \in \mathbb{R}^k (n = 1, 2, 3, \dots)$ and $x_n = (a_{1,n}, a_{2,n}, \dots, a_{k,n})$. Then x_n Converges to $x = (a_1, \dots, a_k) \Leftrightarrow \lim_{n \rightarrow \infty} a_{j,n} = a_j (1 \leq j \leq k)$

▷ **proof :**

$(\Rightarrow) \lim_{n \rightarrow \infty} x_n = x$ this indicate that by the definition for $\epsilon > 0$ there exists N satisfy $n > N$ and this implies $\|x_n - x\| < \epsilon$. $\|\cdot\|$ means Euclid norm. And by the definition of Euclid norm.

$\|x_n - x\| = \sqrt{(a_{1,n} - a_1)^2 + (a_{2,n} - a_2)^2 + \dots + (a_{k,n} - a_k)^2} < \epsilon$. this can be expand. And to be x_n converges to x , the norm should be

$\|x_n - x\| = \sqrt{\sum_{j=1}^k (a_{j,n} - a_j)^2} < \sqrt{k \cdot \left(\frac{\epsilon}{\sqrt{k}}\right)^2} = \sqrt{k \cdot \frac{\epsilon^2}{k}} = \epsilon$. If the difference of each component is less than $\frac{\epsilon}{\sqrt{k}}$, it can be guaranteed that the overall difference of the vector x_n is less than ϵ . thus for $|a_{j,n} - a_j| < \frac{\epsilon}{\sqrt{k}}$ there exists N such that $n > N$. Therefore $\lim_{n \rightarrow \infty} a_{j,n} = a_j$.

$(\Leftarrow)$ suppose $\lim_{n \rightarrow \infty} a_{j,n} = a_j$, then In other words, for any $\epsilon > 0$, there exists a sufficiently large N_j such that if $n > N_j$, then $|a_{j,n} - a_j| < \frac{\epsilon}{\sqrt{k}}$. And suppose we denote $N = \max(N_1, N_2, \dots, N_k)$, then the following condition for convergence holds:

$\|x_n - x\| = \sqrt{(a_{1,n} - a_1)^2 + (a_{2,n} - a_2)^2 + \dots + (a_{k,n} - a_k)^2} < \epsilon$. Thus, $x_n \rightarrow x$. ■

2. Suppose $\{x_n\}, \{y_n\} \subset \mathbb{R}^k, \{\beta_n\}$ is a sequence of real numbers, and $\lim_{n \rightarrow \infty} x_n =$

$x, \lim_{n \rightarrow \infty} y_n = y, \lim_{n \rightarrow \infty} \beta_n = \beta$ then,

- (a) $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$,

▷ **proof :** since $\{x_n\}, \{y_n\}$ converge to x and y $\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} (a_{1,n} + b_{1,n}, \dots, a_{k,n} + b_{k,n}) = (\lim_{n \rightarrow \infty} a_{1,n} + \lim_{n \rightarrow \infty} b_{1,n}, \dots, \lim_{n \rightarrow \infty} a_{k,n} + \lim_{n \rightarrow \infty} b_{k,n})$. and We proved that in Thm complex sequences #1 this converges to $(a_1 + b_1, \dots, a_k + b_k)$. And this is $x + y$ thus, $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$ ■

- (b) $\lim_{n \rightarrow \infty} x_n \cdot y_n = x \cdot y$

▷ **proof :** The inner product of $x_n = (a_{1,n}, \dots, a_{k,n})$ and $y_n = (b_{1,n}, \dots, b_{k,n})$ is this : $x_n \cdot y_n = \sum_{j=1}^k a_{j,n} b_{j,n}$. by the thm Complex sequences #3 $\lim_{n \rightarrow \infty} (x_n \cdot$

$y_n) = \lim_{n \rightarrow \infty} \sum_{j=1}^k a_{j,n} b_{j,n} = \sum_{j=1}^k \lim_{n \rightarrow \infty} (a_{j,n} b_{j,n})$. And from 1. we know that

$\sum_{j=1}^k \lim_{n \rightarrow \infty} (a_{j,n} b_{j,n}) = \sum_{j=1}^k (a_j b_j) = x \cdot y$. ■

- (c) $\lim_{n \rightarrow \infty} \beta_n x_n = \beta x$

▷ **proof :** Note that β defined in $\mathbb{E}$ but x_n defined in vector space. Thus we can expand the with this two sequences, $\lim_{n \rightarrow \infty} (\beta_n x_n) = \lim_{n \rightarrow \infty} (\beta_n a_{1,n}, \dots, \beta_n a_{k,n}) = (\lim_{n \rightarrow \infty} (\beta_n a_{1,n}), \dots, \lim_{n \rightarrow \infty} (\beta_n a_{k,n}))$. $\beta_n \rightarrow \beta$ and $a_{j,n} \Rightarrow a_j$, so the limit of each component is: $\lim_{n \rightarrow \infty} (\beta_n a_{j,n}) = \beta a_j$. thus, $(\beta a_1, \dots, \beta a_k) = \beta x$. ■

Subsequences

Def : subsequence and subsequential limit

Given a sequence $\{p_n\}$, consider a sequence $\{n_k\}$ of positive integers, such that $n_1 < n_2 < n_3 < \dots$. Then the sequence $\{p_{n_k}\}$ is called a subsequence of $\{p_n\}$. If $\{p_{n_k}\}$ converges to its limit, this is called a subsequential limit of $\{p_n\}$.

For example, let's think sequence $p_n = (-1)^n$, $n_k = 2k - 1$, then $\{p_{n_k}\} = \{p_1, p_3, p_5, \dots\} \rightarrow \{-1\}$ as $n \rightarrow \infty$. Thus, we can know even if sequence doesn't converge to something, subsequence could converge.

Thm : convergent sequence and subsequence

$\{p_n\} \rightarrow p \Leftrightarrow$ every subsequence of $\{p_n\} \rightarrow p$

▷ **proof :**

($\Rightarrow$) Suppose, $\{p_n\} \rightarrow p$ then for $\epsilon > 0 \exists N$ such that $n > N$ implies $|p_n - p| < \epsilon$. Now consider the subsequence of $\{p_n\}$ which is $\{p_{n_k}\}$, this is trivial that for $n_k > N$ $|p_{n_k} - p| < \epsilon$ holds thus $\{p_{n_k}\} \rightarrow p$.

($\Leftarrow$) Suppose subsequence $\{p_{n_k}\} \rightarrow p$ and for get contradiction $\{p_n\} \rightarrow p$ then by the definition for $\epsilon > 0 \nexists N$ thus even though $n > N$, $|p_n - p| \geq \epsilon$ could hold, and we can make sequence with $\{p_n\}$ such that $|p_n - p| \geq \epsilon$ and let it $\{p_{n_k}\}$ then this holds $|p_{n_k} - p| \geq \epsilon$ does not converge to p this is contradiction thus $\{p_n\} \rightarrow p$ ■

Thm : Compact and subsequence

1. if $\{p_n\}$ is a sequence in a compact metric space X , then some subsequence of $\{p_n\}$ converges to a point of X .

▷ **proof :**

(a) In Finite case,

Let E be the range of the sequence $\{p_n\}$. Then, the sequence $\{p_n\}$ must contain repeating points. For example, we can create a subsequence $\{p_{n_k}\}$ consisting of points where $p \in E$ with indices n_k such that $n_1 < n_2 < \dots$, like: $p_{n_1} = p_{n_2} = \dots = p$. In this case, since $\{p_{n_k}\}$ is a subsequence consisting of the same point p , it trivially converges to p .

(b) In Infinite case,

By the Bolzano-Weierstrass theorem which indicates that infinite subset of compact set has a limit point. thus, $\{p_n\}$ has a limit point p in X ■

2. Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

▷ **proof :** Every bounded sequence of $\mathbb{R}^k$ lies in a compact subset of $\mathbb{R}^k$ since, by Heine-Borell Theorem which states that bounded subset is compact subset and by 1, sequence in compact subset is convergent. ■

Thm : closed subset and subsequential limits

The subsequential limits of a sequence $\{p_n\}$ in a metric space X form a closed subset of X

▷ **proof :** Let E be the set of all subsequential limits of $\{p_n\}$ to show that if E is closed then we need to show that if q is a limit point of E then $q \in E$. First, let's choose $q \neq p_{n_1}$. If there were no such p_{n_1} , then E would consist only of q , which would be a very trivial result. We define $\delta = d(q, p_{n_1})$ and assume we have chosen $n_1 \dots n_{i-1}$. Since q is a limit point of E , there exists an x in the neighborhood of q such that $d(x, q) < \frac{\delta}{2^i}$ satisfying these conditions.

Furthermore, since $x \in E$, there exists a subsequence n_i such that $n_i > n_{i-1}$ and satisfies $d(x, p_{n_i}) < \frac{\delta}{2^i}$. Now we have $d(q, p_{n_i}) \leq d(q, x) + d(x, p_{n_i}) < 2^{-i}\delta + 2^{-i}\delta = 2 \cdot 2^{-i}\delta = 2^{1-i}\delta$. We can show that $d(q, p_{n_i}) \rightarrow 0$. This means that $\{p_{n_i}\}$ converges to q , and the limit point q is in E . ■

Cauchy Sequences

Def : Cauchy sequence

A sequence $\{p_n\}$ in a metric space X is said to be a Cauchy sequence if for every $\epsilon > 0$ there is an integer N such that $d(\{p_n\}, \{p_m\}) < \epsilon$ if $n \geq N$ and $m \geq N$

Def : Cauchy sequence in Geometric concept

Let E be a nonempty subset of a metric space X , and let S be the set of all real number of the form $d(p, q)$, with $p \in E$ and $q \in E$. The sup of S is called the diameter of E . If $\{p_n\}$ is a sequence in X and if $\{E_n\}$ consists of points $p_N, p_{N+1}, \dots$. It is clear from the two preceding definitions that $\{p_n\}$ is a Cauchy sequence $\Leftrightarrow \lim_{N \rightarrow \infty} \text{diam } E_n = 0$

Thm : Closure and Diameter

1. If $\overline{E}$ is the closure of a set E in a metric space X , then $\text{diam } \overline{E} = \text{diam } E$.
 ▷ **proof** : Lets show that $\text{diam } \overline{E} \leq \text{diam } E$ holds and $\text{diam } \overline{E} \geq \text{diam } E$ also holds.
 $\text{diam } \overline{E} \geq \text{diam } E$ this is trivial since, $\text{diam } E \subseteq \text{diam } \overline{E}$
 then lets see $\text{diam } \overline{E} \leq \text{diam } E$, For any $\epsilon > 0$, suppose $\exists p, q \in \overline{E}$ then by the definition of closure there are points p' and q' in E such that $d(p, p') < \epsilon$ and $d(q, q') < \epsilon$. then by triangular inequality we have, $d(p, q) \leq d(p, p') + d(p', q') + d(q', q) < 2\epsilon + \text{diam } E$. this indicate, For any $p, q \in \overline{E}$, $\overline{E} \leq \text{diam } E + 2\epsilon$ thus, $\text{diam } \overline{E} \leq \text{diam } E$ since choose ϵ which so small that close to 0. ■
2. If $\{k_n\}$ is a sequence of compact sets in X such that $k_n \supset k_{n+1}$ $\lim_{n \rightarrow \infty} \text{diam } k_n = 0$, then $\bigcap_{n=1}^{\infty} k_n$ consists of exactly one point.
 ▷ **proof** : by the theorem that intersection of compact sets is nonempty, $K = \bigcap_{n=1}^{\infty} k_n \neq \emptyset$.
 And If K consists of one more point, then $\text{diam } K > 0$. But for each n , $k_n \supset K$, so that $\text{diam } k_n \leq \text{diam } K$. This contradicts the assumption that $\text{diam } k_n \rightarrow 0$ ■

Thm : various properties of Cauchy Sequence

1. If any metric space X , every convergent sequence is a Cauchy sequence.
 ▷ **proof** : Suppose there is convergent sequence that $\{p_n\} \rightarrow p$, then by the definition, $\exists N$ such that $d(p_n, p) < \epsilon$ for whenever $n \geq N$. then choose m satisfying this statement, then we can get $d(p_n, p_m) \leq d(p, p_m) + d(p, p_n) < 2\epsilon = \epsilon$ and this is definition of Cauchy sequence. ■
2. If X is a compact metric space and if $\{p_n\}$ is a Cauchy sequence in X , then $\{p_n\}$ converges to some point of X .

▷ **proof** : Suppose X is compact and $\{p_n\}$ is a Cauchy sequence in X , thus there $\exists E_N = \{p_N, p_{N+1}, p_{N+2}, \dots\}$ and by the definition of Cauchy sequence $\text{diam}(E_N) \rightarrow 0$ and also by theorem $\text{diam}(\overline{E_N}) \rightarrow 0$. Now since, $\overline{E_N}$ is closed subset of compact set X and also it is compact by theorem. Also $E_N \supseteq E_{N+1}$ is hold, Because $E_N = \{p_N, p_{N+1}, p_{N+2}, \dots\} \supseteq E_{N+1} = \{p_{N+1}, p_{N+2}, p_{N+3}, \dots\}$ So that $\overline{E_N} \supseteq \overline{E_{N+1}}$. thus exists $p \in X$ which lies in every $\overline{E_N}$. And let $\epsilon > 0$ be given, thus, by the definition of convergent $\text{diam}(\overline{E_N}) \rightarrow 0$ could be stated by $\exists N_0$ such that $|\text{diam}(\overline{E_N}) - 0| < \epsilon$ whenever $N \geq N_0$. Since $p \in \overline{E_N}$, it follows that $d(p, q) < \epsilon$ for every $q \in \overline{E_N}$, hence for every $q \in E_N$. This indicate particularly, $p_n \in E_N$ indicate $d(p, p_n) < \epsilon$. Thus, If $n \geq N_0$ then $d(p, p_n) < \epsilon$ so, $\{p_n\} \rightarrow p$ ■

3. In $\mathbb{R}^k$, every Cauchy sequence converges.

▷ **proof** : Let $\{x_n\}$ be a Cauchy sequence in $\mathbb{E}^k$.

Define $E_N = \{x_N, x_{N+1}, x_{N+2}, \dots\}$. Fix $\epsilon = 1$ and choose N satisfy $\text{diam} E_N < 1$ this indicate E_N is bounded, on the other hand, $\{x_n\} = \{x_1, x_2, \dots, x_{N-1}\} \cup E_N$. and $\{x_1, x_2, \dots, x_{N-1}\}$ is also bounded. By theorem that bounded set union bounded set is bounded $\Rightarrow \{x_n\}$ is bounded, since every bounded subset of $\mathbb{E}^k$ has compact closure in $\mathbb{R}^k$ by the upper theorem of 2 the Cauchy sequence in $\mathbb{R}^k$ convergent. ■

Def : Complete of Cauchy sequence

A metric space in which every Cauchy sequence converges in said to be *complete*.

By the thm : various properties of Cauchy Sequence that " If X is a compact metric space and if $\{p_n\}$ is a Cauchy sequence in $X \Rightarrow \{p_n\} \rightarrow p \in X$ " and "In $\mathbb{R}^k$, every Cauchy sequence converges.", all compact $\Rightarrow$ Complete, all Euclidean spaces $\Rightarrow$ Complete."

For example, $\mathbb{Q}$ (Rational number) is not a complete. Consider one of Cauchy sequence likes $\{1, 1.4, 1.41, 1.414, \dots\}$ in $\mathbb{Q}$. this converge $\sqrt{2} \notin \mathbb{Q}$ rather $\in \mathbb{R}$. Thus this is not complete.

Def : Monotonically in, decreasing

A sequence $\{s_n\}$ of real numbers is said to be :

1. monotonically increasing $\Rightarrow s_n \leq s_{n+1} (n = 1, 2, 3, \dots)$
2. monotonically decreasing $\Rightarrow s_n \geq s_{n+1} (n = 1, 2, 3, \dots)$

$\{s_n\}$ is monotone $\Rightarrow$ Should be monotonically increasing, decreasing.

Thm : Monotonic convergent and bounded

Suppose $\{s_n\}$ is monotonic. Then $\{s_n\}$ converge $\Leftrightarrow$ it is bounded. ▷ **proof** :

($\Rightarrow$) By the thm: Various properties of Sequence one of which that <if $\{p_n\} \rightarrow p$ then $\{p_n\}$ is bounded>.

($\Leftarrow$) Suppose $\{s_n\}$ is monotonically increasing thus, $s_n \leq s_{n+1}$ (the proof is analogous in the other case), And $\{s_n\}$ is bounded above. then we do have supremum and let $S = \sup\{s_n\}$. Now we prove that $\lim_{n \rightarrow \infty} s_n = s$ for any $\epsilon > 0$, $\exists N$ such taht $|s_n - s| < \epsilon, \forall n \geq N$.

Now since S is supremum this is hold : $S - \epsilon < s_N \leq S$. And because of monotocically increasing property $n \geq N \rightarrow s_n \geq s_N$ thus we can rewrite $S - \epsilon < s_n \leq S \Rightarrow |s_n - s| < \epsilon$. Thus, $s_n \rightarrow s$ ■.

Upper and Lower Limits

Def : Positive and Negative infinity

Let $\{s_n\}$ be a sequence of real numbers with the following property: For every real M , integer $\exists N$ such that $n \leq N$ implies $s_n \geq M$. We then write $s_n \rightarrow \infty$. Similarly, if for every real M , integer $\exists N$ such that $n \geq N$ implies $s_n \leq M$, $s_n \rightarrow -\infty$.

Def : lim sup, lim inf

Let $\{s_n\}$ be a sequence of real numbers. Let E be the set of number x such that $s_{n_k} \rightarrow x$ for some subsequence $\{s_{n_k}\}$. $s^* = \sup E$, $s_* = \inf E$. The numbers s^*, s_* are called the upper and lower limits of $\{s_n\}$; we use the notation, $\limsup_{n \rightarrow \infty} s_n = s^*$, $\liminf_{n \rightarrow \infty} s_n = s_*$. For example, $s_n = (-1)^n = \{-1, 1, -1, 1, \dots\}$. Thus $s_{2n} \rightarrow 1$ and $s_{2n+1} \rightarrow -1$. Then $\limsup_{n \rightarrow \infty} s_n = 1$, and $\liminf_{n \rightarrow \infty} s_n = -1$.

Thm : monotonicity of liminf and limsup

If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then

$$\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n$$

$$\limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n$$

▷ **proof** : For any arbitrary $n \geq N$, since $s_n \leq t_n$, it follows that for all $k \geq n$, $s_k \leq t_k$. Then, each inf satisfies the following property: $\inf_{n \geq k} s_n \leq \inf_{n \geq k} t_n \forall k \geq 1$. Taking the Supremum on both sides will yield the following: $\sup_{k \geq 1} \inf_{n \geq k} s_n \leq \sup_{k \geq 1} \inf_{n \geq k} t_n \Rightarrow \liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n$ vice versa ■

Convergence to Negative Infinity of Limsup and Limit ★

$$\limsup_{n \rightarrow \infty} s_n = -\infty \Leftrightarrow \exists \lim_{n \rightarrow \infty} s_n = -\infty$$

▷ **proof** :

Suppose $\lim_{n \rightarrow \infty} s_n = -\infty$ this indicates that $\forall M \in \mathbb{R}, \exists N \in \mathbb{N}$ such that $n \geq N$ implies $s_n < M$

thus, $\sup_{n \geq k} s_n < M (\forall k \geq N)$ also holds. By definition, $\limsup_{n \rightarrow \infty} s_n = \inf_{k \geq 1} \left(\sup_{n \geq k} s_n \right)$. Here,

$\sup_{n \geq k} s_n < M$ always holds under the given condition $k \geq N$. Therefore, $\inf_{k \geq 1} \left(\sup_{n \geq k} s_n \right) < M$ and

accordingly, $\limsup_{n \rightarrow \infty} s_n < M \Rightarrow \limsup_{n \rightarrow \infty} s_n = -\infty$.

($\Leftarrow$) By definition $\limsup_{n \rightarrow \infty} s_n = -\infty$ indicate that $\sup_{n \geq k} s_n < M$. So $\forall M \in \mathbb{R}, \exists N \in \mathbb{N}$ such that $n \geq N$ implies $\sup_{n \geq k} s_n < M$ And by property of supremum, $s_n \leq \sup_{n \geq k} s_n \Rightarrow s_n < M, \forall k \geq n$ holds. And this means $\lim_{n \rightarrow \infty} s_n = -\infty$ ■

Various properties of limsup and liminf ★

$$\limsup(s_n + t_n) \leq \limsup(s_n) + \limsup(t_n)$$

▷ **proof** : $\limsup(s_n + t_n)$ indicates $\limsup_{n \rightarrow \infty}(s_n + t_n) = \inf_{k \geq 1} \left(\sup_{n \geq k}(s_n + t_n) \right)$. The following inequality holds for the sum of s_n and t_n : $\sup_{n \geq k}(s_n + t_n) \leq \sup_{n \geq k}s_n + \sup_{n \geq k}t_n$. Because, $s_n < \sup s_n$, $t_n < \sup t_n$ so, $s_n + t_n < \sup s_n + \sup t_n \Rightarrow \sup_{n \geq k}(s_n + t_n) \leq \sup_{n \geq k}s_n + \sup_{n \geq k}t_n$ thus we can conclude that $\inf_{k \geq 1} \sup_{n \geq k}(s_n + t_n) \leq \inf_{k \geq 1} \left(\sup_{n \geq k}s_n + \sup_{n \geq k}t_n \right)$. And this indicates $\limsup_{n \rightarrow \infty}(s_n + t_n) \leq \limsup_{n \rightarrow \infty}s_n + \limsup_{n \rightarrow \infty}t_n$.

Series

$$s_n = \sum_{n=1}^{\infty} a_n$$

What we have done in the above formula is to answer the question of whether a_n converges. But we also have question that sum of sequences also convergent?. If $\{s_n\}$ converges to s , we say that the series converges, and write $\sum_{n=1}^{\infty} a_n = s$.

Continuity

Def : Limits of Functions

Let X and Y be metric spaces ; suppose $E \subset X$, f maps E into Y and p is a limit point of E . We write $f(x) \rightarrow q$ as $x \rightarrow p$ or $\lim_{x \rightarrow p} f(x) = q$. If there is a point $q \in Y$ with following property: For every $\epsilon > 0$ there exists a $\delta > 0$ such that $d_Y(f(x), q) < \epsilon$ for all points $x \in E$ for which $0 < d_X(x, p) < \delta$. Note that $p \in X$ must hold but $p \in E$ doesn't need to hold. Moreover, even if $p \in E$, we may very well have $f(p) \neq \lim_{x \rightarrow p} f(x)$.

For example, to show that $\lim_{x \rightarrow 1} f(x) = 1$ for $f(x) = x^2$, let $\epsilon > 0$. We need to find $\delta > 0$ such that $|x - 1| < \delta$ implies $|f(x) - 1| < \epsilon$. Note that $|f(x) - 1| = |x^2 - 1| = |(x - 1)(x + 1)|$. If $|x - 1| < \delta$, then $-\delta < x - 1 < \delta$, which gives $-\delta + 2 < x + 1 < \delta + 2$. To bound $|x + 1|$, we choose $\delta = \min\left(\frac{\epsilon}{5}, \frac{1}{2}\right)$, ensuring $|x + 1| < \frac{5}{2}$. Thus, $|(x - 1)(x + 1)| \leq |x - 1||x + 1| < \delta \cdot \frac{5}{2} = \frac{\epsilon}{5} \cdot \frac{5}{2} = \epsilon$. Hence, $|f(x) - 1| < \epsilon$ whenever $|x - 1| < \delta$, proving that $\lim_{x \rightarrow 1} f(x) = 1$.

Thm : Sequential Characterization of Limits ★

Let X, Y, E, f and p be as in upper Definition. then,

$$\lim_{x \rightarrow p} f(x) = q \Leftrightarrow \lim_{n \rightarrow \infty} f(p_n) = q$$

For every sequence $\{p_n\}$ in E such that $p_n \neq p$, $\lim_{n \rightarrow \infty} p_n = p$

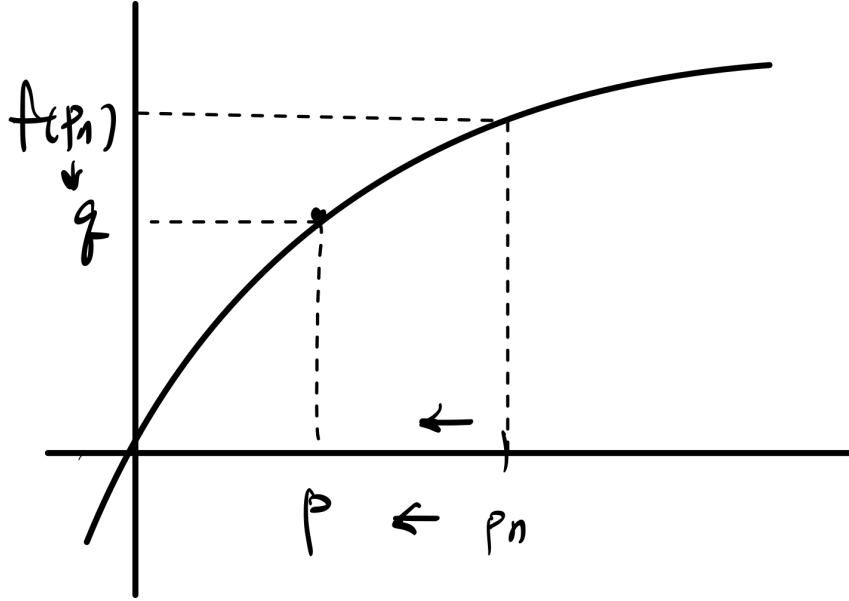


Figure 18: Converge of function

For example, there is a function that $f(x) = \sin\left(\frac{1}{x}\right)$ and $x \rightarrow 0$. Consider the subsequence which construct x, is p_{1n} and p_{2n} ,

$$p_{1n} = \frac{1}{2n\pi} \Rightarrow f(p_{1n}) = \sin(2n\pi) \rightarrow 0, \quad \text{as } n \rightarrow \infty$$

$$p_{2n} = \frac{1}{2n\pi + \pi/2} \Rightarrow f(p_{2n}) = \sin(2n\pi + \pi/2) \rightarrow 1, \quad \text{as } n \rightarrow \infty$$

there is two limit points of $f(x)$, In this case, limit doesn't exist.

▷ **proof :**

($\Rightarrow$) Assume that $\lim_{x \rightarrow p} f(x) = q$ thus, by definition, $\exists \delta$ such that $d_Y(f(x), q) < \epsilon$ whenever $0 < d_X(x, p) < \delta$. Choose $\epsilon = \delta$ then $d_X(p_n, p) < \epsilon$ whenever $n \geq N$ since, $p_n \rightarrow p$ by assumption. For this N $d_X(p_n, p) < \delta$ holds and $d_Y(f(p_n), q) < \epsilon$ also holds. Thus, $\exists \delta$ such that $d_Y(f(p_n), q) < \epsilon$ whenever $0 < d_X(p_n, p) < \delta \Rightarrow \lim_{n \rightarrow \infty} f(p_n) = q$.

($\Leftarrow$) **To prove $B \Rightarrow A$ I will use $\sim A \Rightarrow \sim B$.** Suppose this statement

$$A : \lim_{x \rightarrow p} f(x) = q \Leftrightarrow \forall \epsilon > 0 \quad \exists \delta > 0 \text{ satisfying } C$$

$$C : \underbrace{0 < d_X(x, p) < \delta}_P \Rightarrow \underbrace{0 < d_Y(f(x), q) < \epsilon}_Q$$

$$\text{Note : } P \Rightarrow Q = \sim P \vee Q, \sim C = \sim (\sim P \vee Q) = P \vee \sim Q$$

$$\sim C : \exists x \in E \text{ such that } 0 < d_X(x, p) < \delta \text{ but } d_Y(f(x), q) \geq \epsilon_0$$

$$\sim A : \text{there exists at least one } \epsilon_0 > 0, \text{ for any } \delta > 0 \text{ we have } \sim C$$

Thus, If choose $\delta_n = \frac{1}{n}$ ($n = 1, 2, 3, \dots$) we thus find a sequence in E satisfying $p_n \neq p$, $\lim_{n \rightarrow \infty} p_n = p$ for which $\lim_{n \rightarrow \infty} f(p_n) \neq q$.

Cor : Uniqueness of limits of functions

If f has a limit at p , then the limit is unique

$$\lim_{x \rightarrow p} f(x) = q_1, \lim_{x \rightarrow p} f(x) = q_2 \Rightarrow q_1 = q_2$$

▷ **proof** : Let X and Y are metric spaces and $E \subset X, f : E \rightarrow Y$ and p be a limit point of E . And assume that $\lim_{x \rightarrow p} f(x) = q_1, \lim_{x \rightarrow p} f(x) = q_2$. From $\lim_{x \rightarrow p} f(x) = q_1$ We can know $\forall \epsilon > 0, \exists \delta_1 > 0$ such that $d_Y(f(x), q_1) < \epsilon, \forall x \in E$ with $0 < d_X(x, p) < \delta_1$. On the other hand, From $\lim_{x \rightarrow p} f(x) = q_2$ We can know $\forall \epsilon > 0, \exists \delta_2 > 0$ such that $d_Y(f(x), q_2) < \epsilon, \forall x \in E$ with $0 < d_X(x, p) < \delta_2$.

To get contradiction let suppose $q_1 \neq q_2$ thus $d(q_1, q_2) > 0$. Let $\epsilon_0 = \frac{1}{2}d_Y(q_1, q_2)$ and we choose $\delta = \min(\delta_1, \delta_2)$. Then for all $x \in E$ with $0 < d_X(x, p) < \delta$ we have $d_Y(f(x), q_1) < \epsilon_0$ and $d_Y(f(x), q_2) < \epsilon_0$. With triangular inequality $d_Y(q_1, q_2) \leq d_Y(f(x), q_1) + d_Y(f(x), q_2) < \epsilon_0 + \epsilon_0 = 2\epsilon_0$ However, by our choice of ϵ_0 $d_Y(q_1, q_2) < d_Y(q_1, q_2)$, and this is contradiction. ■

Def : Operations on Functions

Suppose we have complex function f and g , both defined on E . By $f + g$ we mean the function which assigns to each point x of E the number $f(x) + g(x)$. Similarly we define the difference $f - g$, the product fg , and the quotient f/g of the two functions, with the understanding that the quotient is defined only at those points x of E at which $g(x) \neq 0$. If f assigns to each point x of E the same number c , then f is said to be a constant function, or simply a constant, and we write $f = c$. If f and g are real functions, and if $f(x) \geq g(x)$ for every $x \in E$, we shall sometimes write $f \geq g$, for brevity. Similarly, if f and g map E into $\mathbb{R}^k$, we define $f + g$ and $f \cdot g$ by $(f + g)(x) = f(x) + g(x)$, $(f \cdot g)(x) = f(x) \cdot g(x)$ and if λ is a real number $(\lambda f)(x) = \lambda f(x)$

Thm : Properties of Limits of Complex Functions

Suppose $E \subset X$, a metric space, p is a limit point of E , f and g are complex functions on E , and $\lim_{x \rightarrow p} f(x) = A, \lim_{x \rightarrow p} g(x) = B$. then following properties are hold.

- $\lim_{x \rightarrow p} (f + g)(x) = A + B$
- $\lim_{x \rightarrow p} (fg)(x) = AB$
- $\lim_{x \rightarrow p} \left(\frac{f}{g}\right)(x) = \frac{A}{B}$ if $B \neq 0$

▷ **proof** : Skip !.

Continuous functions

Def : Continuous functions

X, Y : metric spaces, $E \subset X, f : E \rightarrow Y$. $P \in E$ f is continuous at p if for every $\epsilon > 0$, there exists a $\delta > 0$ such that, $d_Y(f(x), f(p)) < \epsilon$ for all $x \in E$ for which $d_X(x, p) < \delta$. The key here is the difference between limits and functions. The limit of a function deals with

whether the value of $f(x)$ converges to a point as x approaches a specific point p , even if the function is not defined at that point. In contrast, a continuous function must not only have a limit, but the limit value must also be the same as the function value at that point.

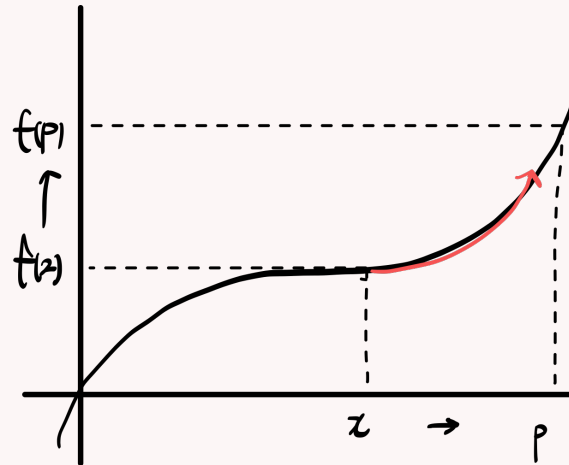


Figure 19: Continuous function

Thm : Characterization of Continuity at a Limit Point

In the situation given in Definition of continuous function, assume that p is a limit point of E . then f is continuous at $p \Leftrightarrow \lim_{x \rightarrow p} f(x) = f(p)$

▷ **proof :**

($\Rightarrow$) Suppose f is continuous at p and p is a limit point of E . then for every $\epsilon > 0$ there exists $\delta > 0$ such that $d_X(x, p) < \delta$ and $x \in E \setminus \{p\} \Rightarrow d_Y(f(x), f(p)) < \epsilon$ and this precisely the definition of $\lim_{x \rightarrow p} f(x) = f(p)$.

($\Leftarrow$) Suppose $\lim_{x \rightarrow p} f(x) = f(p)$ then this indicate $\forall \epsilon > 0, \exists \delta > 0$ such that $0 < d_X(x, p) < \delta \Rightarrow d_Y(f(x), f(p)) < \epsilon$ particular, since p is a limit point of E , we can always find $x \in E \setminus \{p\}$ satisfying $0 < d_X(x, p) < \delta$. Therefore, the condition $d_Y(f(x), f(p)) < \epsilon$ holds for all such x . This is exactly the definition of f being continuous at p . ■

Thm : Continuity of Composite Functions

Suppose X, Y, Z are metric spaces, $E \subset X$, f maps E into Y , g maps the range of f , $f(E)$, into Z and h is the mapping of E into Z defined by $h(x) = g(f(x))$, ($x \in E$) If f is continuous at a point $p \in E$ and if g is continuous at the point $f(p)$, then h is continuous at p . This function h is called the composition or the composite of f and g . The notation $h = g \cdot f$.

▷ **proof :** To show that h is continuous at p , we must prove: $\forall \epsilon > 0, \exists \delta > 0$ such that $d_Z(h(x), h(p)) < \epsilon$ for all $x \in E$ with $d_X(p, x) < \delta$. Since g is continuous at $q = f(p)$, for the given $\epsilon > 0$, there exists $\Gamma > 0$ such that $d_Y(y, q) < \Gamma \Rightarrow d_Z(g(y), g(q)) < \epsilon$. Similarly, since f is continuous at p , for this $\Gamma > 0$, there exists $\delta > 0$ such that $d_X(p, x) < \delta \Rightarrow d_Y(f(x), f(p)) < \Gamma$. Now, combining these two results If $d_X(p, x) < \delta$, then $d_Y(f(x), f(p)) < \Gamma$. By the continuity of g , this implies $d_Z(g(f(x)), g(f(p))) < \epsilon$. Thus $d_Z(h(x), h(p)) = d_Z(g(f(x)), g(f(p))) < \epsilon$ for all $d_X(p, x) < \delta$. This proves that $h = g \cdot f$ is continuous at p . ■

Thm : Continuity and Openness of Preimages ★

f map a metric space X into a metric space Y . Then f is continuous on $X \Leftrightarrow f^{-1}(V)$ is open in X for every open set V in Y .

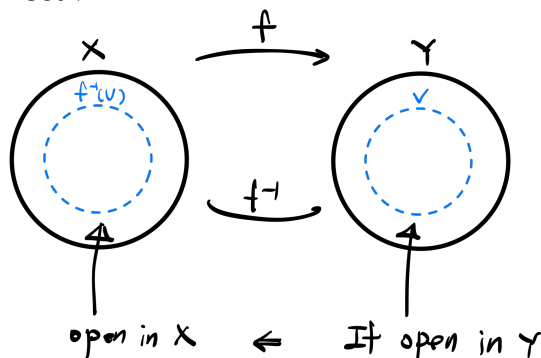


Figure 20: Openness of inverse function

▷ **proof :**

($\Rightarrow$) Let's assume f is continuous on X , and V is an open set in Y . then we wish to show that for this open set V in Y the following statement holds: $f^{-1}(V)$ is open in X . for this, we need to show that $p \in f^{-1}(V)$ and p is interior point of $f^{-1}(V)$. Suppose $p \in X$ and $f(p) \in V$. Then since f is continuous on X , we can know, $\forall \epsilon > 0, \exists \delta > 0$ such that $d_Y(f(x), f(p)) < \epsilon$ whenever $d_X(x, p) < \delta$. And this statement implies that $x \in N_\delta(p) \Rightarrow f(x) \in N_\epsilon(f(p)) = V$. So, this holds : $N_\delta(p) \subseteq f^{-1}(V)$ and this means p is interior point of $f^{-1}(V)$ thus, $f^{-1}(V)$ is open as V is open.

($\Leftarrow$) Assume that for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X . To show that f is continuous on X , we use the definition of continuity: for any $p \in X$ and $\epsilon > 0$, there exists $\delta > 0$ such that $d_Y(f(x), f(p)) < \epsilon$ whenever $d_X(x, p) < \delta$. Let $p \in X$, and consider the open ball $N_\epsilon(f(p)) \subseteq Y$, where $N_\epsilon(f(p)) = \{f(x) \in Y \mid d_Y(f(x), f(p)) < \epsilon\}$. Since $N_\epsilon(f(p))$ is open in Y , its preimage under f , $f^{-1}(N_\epsilon(f(p)))$, is open in X by assumption. Since $f^{-1}(N_\epsilon(f(p)))$ is open in X , there exists $\delta > 0$ such that the open ball $N_\delta(p) \subseteq f^{-1}(N_\epsilon(f(p)))$. That is: $x \in N_\delta(p) \Rightarrow f(x) \in N_\epsilon(f(p))$. By the definition of $N_\epsilon(f(p))$, this implies: $d_Y(f(x), f(p)) < \epsilon$ for all x with $d_X(x, p) < \delta$. Thus, f is continuous at p . Since $p \in X$ was arbitrary, f is continuous on X . ■

Cor : Continuity and Closeness of Preimages

A mapping f of a metric space X into a metric space Y is continuous $\Leftrightarrow f^{-1}(C)$ is closed in X for every closed set C in Y .

▷ **proof :**

($\Rightarrow$) Suppose f is continuous on X and $C \subseteq Y$ be the closed set in Y . Then ,by the Thm that compliment open is close, $Y \setminus C$ is open in Y . so, by upper Thm : Continuity and Openness of Preimages, $f^{-1}(Y \setminus C)$ is open in X . Thus , $\underbrace{f^{-1}(C)}_{\text{close}} = \underbrace{X \setminus f^{-1}(Y \setminus C)}_{\text{open}}$ is closed.

($\Leftarrow$) Suppose for $C \subseteq Y$ be closed set then $f^{-1}(C)$ be also closed set. Let $V \subseteq Y$ is open set and $Y \setminus V = C$ is closed set. then by assumption $f^{-1}(Y \setminus V = C)$ is also closed in X and same manner as previous one, we can know $\underbrace{f^{-1}(V)}_{\text{open}} = \underbrace{X \setminus f^{-1}(Y \setminus V)}_{\text{close}}$. Since,by

Thm : Continuity and Openness of Preimages, $f^{-1}(V)$ is open for V open in $Y \Rightarrow f$ is continuous. ■

Thm : Properties of Continuous Functions ★

Let f and g be complex continuous functions on a metric space X . Then $f + g$, $f \cdot g$, f/g are continuous on X . Note that $g(x) \neq 0$ for all $x \in X$

▷ **proof :**

- $f + g$

Let $p \in X$, since f, g are continuous at p for every $\epsilon > 0 \exists \delta_f, \delta_g > 0$ such that

$$\begin{aligned} d_X(x, p) < \delta_f &\Rightarrow |f(x) - f(p)| < \frac{\epsilon}{2} \\ d_X(x, p) < \delta_g &\Rightarrow |g(x) - g(p)| < \frac{\epsilon}{2} \end{aligned}$$

And let choose $\delta = \min(\delta_f, \delta_g)$ Then for all $x \in X$ with $d_X(x, p) < \delta$ implies

$$|f(x) + g(x) - (f(p) + g(p))| \leq |f(x) - f(p)| + |g(x) - g(p)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Thus, $f + g$ is continuous at p . Since $p \in X$ was arbitrary, $f + g$ is continuous on X . ■

- $f \cdot g$

Let $\epsilon > 0$. Since f and g are continuous at p , there exist $\delta_f > 0$ and $\delta_g > 0$ such that

$$\begin{aligned} d_X(x, p) < \delta_f &\Rightarrow |f(x) - f(p)| < \frac{\epsilon}{2(|g(p)| + 1)} \\ d_X(x, p) < \delta_g &\Rightarrow |g(x) - g(p)| < \frac{\epsilon}{2(|f(p)| + 1)}. \end{aligned}$$

Let $\delta = \min(\delta_f, \delta_g)$. For $d_X(x, p) < \delta$, we have:

$$|f(x)g(x) - f(p)g(p)| \leq |f(x)||g(x) - g(p)| + |g(p)||f(x) - f(p)|.$$

And we use the fact that $|f(x)| \leq |f(p)| + 1$ and $|g(x)| \leq |g(p)| + 1$ Substituting the bounds for $|f(x) - f(p)|$ and $|g(x) - g(p)|$, and noting that $f(x)$ and $g(x)$ are bounded near p , we get: $|f(x)g(x) - f(p)g(p)| < \epsilon$. Thus, $f \cdot g$ is continuous at p . Since $p \in X$ was arbitrary, $f \cdot g$ is continuous on X . ■

- f/g

By the continuity of g at p , there exists $\delta_g > 0$ such that:

$$d_X(x, p) < \delta_g \implies d_Y(g(x), g(p)) < |g(x) - g(p)| < \frac{\epsilon |g(p)|^2}{4|f(p)|}.$$

This ensures $g(x) \neq 0$ for all x with $d_X(x, p) < \delta_g$. Moreover, $|g(x)| > \frac{|g(p)|}{2}$. By the continuity of f at p , there exists $\delta_f > 0$ such that:

$$d_X(x, p) < \delta_f \implies d_Y(f(x), f(p)) < \frac{\epsilon |g(p)|}{4}.$$

Let $\delta = \min(\delta_f, \delta_g)$. Then for all $x \in X$ with $d_X(x, p) < \delta$, we have:

$$d_Y\left(\frac{f(x)}{g(x)}, \frac{f(p)}{g(p)}\right) \leq \frac{d_Y(f(x), f(p)) \cdot |g(p)| + |f(p)| \cdot d_Y(g(x), g(p))}{|g(x)g(p)|}.$$

Since $|g(x)| > \frac{|g(p)|}{2}$

$$\begin{aligned} d_Y \left(\frac{f(x)}{g(x)}, \frac{f(p)}{g(p)} \right) &< \frac{2}{|g(p)|^2} (d_Y(f(x), f(p)) \cdot |g(p)| + |f(p)| \cdot d_Y(g(x), g(p))) \\ \Rightarrow \frac{2}{|g(p)|^2} (|g(p)| |f(x) - f(p)| + |f(p)| |g(x) - g(p)|) \\ &< \frac{2}{|g(p)|^2} \left(|g(p)| \cdot \frac{\epsilon |g(p)|}{4} + |f(p)| \cdot \frac{\epsilon |g(p)|^2}{4 |f(p)|} \right) \\ &= \frac{2}{|g(p)|^2} \left(\frac{\epsilon |g(p)|^2}{4} + \frac{\epsilon |g(p)|^2}{4} \right) = \frac{2}{|g(p)|^2} \cdot \frac{\epsilon |g(p)|^2}{2} = \epsilon. \end{aligned}$$

Thus, We can know $\left| \frac{f(x)}{g(x)} - \frac{f(p)}{g(p)} \right| < \epsilon$, Which means f/g is continuous at p . Since $p \in X$ was arbitrary, f/g is continuous on X . ■

Thm : Continuity of Vector-Valued Functions

Let $f_1 \cdots f_k$ be real functions on metric space X , and let f be the mapping of X into $\mathbb{R}^k$ defined by $f(x) = (f_1(x), \dots, f_k(x))$ ($x \in X$) then f is continuous $\Leftrightarrow$ each of the functions $f_1 \cdots f_k$ is continuous.

▷ **proof** : ($\Rightarrow$) Assume $f : X \rightarrow \mathbb{R}^k$ is continuous. By definition, for every $p \in X$ and every $\epsilon > 0$, there exists $\delta > 0$ such that $d_{\mathbb{R}^k}(f(x), f(p)) < \epsilon$ whenever $d_X(x, p) < \delta$. for here $d_{\mathbb{R}^k}(f(x), f(p)) = \sqrt{\sum_{i=1}^k (f_i(x) - f_i(p))^2}$. And since, $d_{\mathbb{R}^k}(f(x), f(p)) < \epsilon$ we have $|f_i(x) - f_i(p)| \leq d_{\mathbb{R}^k}(f(x), f(p)) < \epsilon$ for all $i = 1, \dots, k$. Fix $i \in \{1, \dots, k\}$. For any $\epsilon > 0$, the above inequality ensures that $|f_i(x) - f_i(p)| < \epsilon$ whenever $d_X(x, p) < \delta$. Thus, f_i is continuous at p . Since $p \in X$ was arbitrary, f_i is continuous on X .

($\Leftarrow$) Assume each $f_i : X \rightarrow \mathbb{R}$ is continuous. Thus, for every $i = 1, \dots, k$, for any $p \in X$ and any $\epsilon > 0$, there exists $\delta_i > 0$ such that $d_X(x, p) < \delta_i \implies |f_i(x) - f_i(p)| < \frac{\epsilon}{\sqrt{k}}$. Let $\delta = \min(\delta_1, \dots, \delta_k)$. Then for all $x \in X$ with $d_X(x, p) < \delta$ we have $|f_i(x) - f_i(p)| < \frac{\epsilon}{\sqrt{k}}$ for all $i = 1, \dots, k$. Then with Euclidean norm in $\mathbb{R}^k$ we have $d_{\mathbb{R}^k}(f(x), f(p)) = \sqrt{\sum_{i=1}^k (f_i(x) - f_i(p))^2}$. Since $|f_i(x) - f_i(p)| < \frac{\epsilon}{\sqrt{k}}$ for all i , $\sum_{i=1}^k (f_i(x) - f_i(p))^2 < \sum_{i=1}^k \left(\frac{\epsilon}{\sqrt{k}} \right)^2 = \epsilon^2$. Thus, we can obtain $d_{\mathbb{R}^k}(f(x), f(p)) < \epsilon$. which indicate, f is continuous at p . Since $p \in X$ was arbitrary, f is continuous on X . ■

Continuity and Compactness

Def : Continuity and Compactness

A mapping f of a set E into $\mathbb{R}^k$ is said to be **bounded** if there is a real number M such that $|f(x)| \leq M$ for all $x \in E$.

For example, Suppose $f(x) = \sin x$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ then we easily know that f is bounded since $|f(x)| = |\sin x| \leq 1$ for here M is 1.

Thm : Continuity Preserves Compactness

Suppose f is a continuous mapping of a compact metric space X into a metric space Y . Then $f(X)$ is compact.

▷ **proof** : Let $\{V_\alpha\}$ be an open cover of $f(X)$. Since f is continuous, by Thm : Continuity and Openness of Preimages. $f^{-1}(V_\alpha)$ is open. Since X is compact, there are finitely many indices, which say $\alpha_1, \alpha_2, \dots, \alpha_n$ such that, $X \subset f^{-1}(V_{\alpha_1}) \cup f^{-1}(V_{\alpha_2}) \cup \dots \cup f^{-1}(V_{\alpha_n})$. Since $f(f^{-1}(E)) \subset E$ for every $E \subset Y$ upper statement can be written $f(X) \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$. Thus, We find finitely many open cover of $f(X)$. ■

Thm : Compactness Implies Closed and Bounded Image

If f is continuous mapping of a compact metric space X into $\mathbb{R}^k$, then $f(X)$ is closed and bounded thus, f is bounded.

▷ **Proof**: Let $f : X \rightarrow \mathbb{R}^k$ be a continuous mapping, where X is a compact metric space. Since f is continuous on X and X is compact, $f(X)$ is compact in $\mathbb{R}^k$ (by Thm : Continuity Preserves Compactness) we can verify in $\mathbb{R}^k$ that compact implies both closed and bounded (by the Heine-Borel theorem). Thus, $f(X)$ is closed and bounded. which means there exists $M > 0$ such that $y \in f(X)$ implies $|y| \leq M$ which means, for $y = f(x) \exists M > 0$ such that $|f(x)| \leq M$ for all $x \in X$ thus, f is bounded. ■

NOTE : f is bounded $\Leftrightarrow f(X)$ (where $f : X \rightarrow \mathbb{R}^k$) is a bounded set.

▷ **proof** : ($\Rightarrow$) By definition, f is bounded if there exists a constant $M > 0$ such that $\|f(x)\| \leq M$ for all $x \in X$. The set $f(X) = \{f(x) \mid x \in X\}$ consists of all function values of f on X . Since $\|f(x)\| \leq M$ for all x , we have $\|y\| \leq M$ for all $y \in f(X)$. Thus, $f(X)$ is a bounded set in $\mathbb{R}^k$.

($\Leftarrow$) Assume $f(X)$ is bounded. By definition, this means there exists a constant $M > 0$ such that: $\|y\| \leq M$ for all $y \in f(X)$. For all $x \in X$, $f(x) \in f(X)$, so we have $\|f(x)\| \leq M$ for all $x \in X$. This implies that f is bounded on X .

For example, suppose $f : (0, 1) \rightarrow \mathbb{R}$ and $f(x) = \frac{1}{x}$. First of all to check rather $\frac{1}{x}$ is continuous or isn't, for $|f(x) - f(p)| = \left| \frac{1}{x} - \frac{1}{p} \right| = \left| \frac{p-x}{xp} \right| = \frac{|x-p|}{|x||p|}$. If we choose $\delta = \min\left(\frac{\epsilon p^2}{2}, \frac{p}{2}\right)$, then for $|x - p| < \delta$ following statement holds $|f(x) - f(p)| = \frac{|x-p|}{|x||p|} < \frac{|x-p|}{\frac{p}{2} \cdot p} = \frac{2|x-p|}{p^2} < \epsilon$ thus, $\frac{1}{x}$ is continuous. But, f is **not bounded**. Because we cannot find $M > 0 \in \mathbb{R}$ such that $\left|\frac{1}{x}\right| \leq M$. As the domain is not bounded, $\frac{1}{x} \rightarrow 1$ for $x \rightarrow 1$ like $M = 0.9999 \dots$, we cannot find a certain M in real numbers. On the other hands, if we define $f : [\frac{1}{2}, 1] \rightarrow \mathbb{R}$. then f is bounded and M is $|f(x)| \leq 1$ ■

Thm : Max, Min in continuous function

Suppose f is a continuous real function on a compact metric space X , and $M = \sup_{p \in X} f(p)$, $m = \inf_{p \in X} f(p)$. Then there exists points $p, q \in X$ such that $f(p) = M$ and $f(q) = m$. Now on we call $M = \text{Max}$, $m = \text{min}$.

For example, $x \in X$, $\sup\{f(x) : 0 < x < 1\} = \sup(0, 1) = 1$. However, for here, there is no $\sup_{p \in X} f(p) = M = \text{Max}$.

▷ **proof** : Suppose $f : X \rightarrow \mathbb{R}$, f is continuous and X is compact. As X is compact and f is continuous, by Thm : Continuity Preserves Compactness, $f(X)$ closed and bounded (by Heine Borell theorem). Thus, $M = \sup f(X)$, $m = \inf f(X)$ are well-defined and finite. Upper Thm says that if $y = \sup E$ then $y \in \bar{E}$, which also indicate if E is closed $y \in E$. Therefore, $\sup f(X) = M = f(p) \in f(X)$ holds, also $\inf f(X) = m = f(q) \in f(X)$. Thus, there exist $p, q \in X$ such that $f(p) = M$ and $f(q) = m$. ■

Thm : Inverse of a Continuous Bijective Mapping ★

Suppose f is a continuous 1-1 mapping of a compact metric space X onto a metric space Y . Then the inverse mapping f^{-1} defined on Y by $f^{-1}(f(x)) = x$ where $x \in X$ is a continuous mapping of Y onto X .

▷ **proof** : We prove the continuity with the Thm: Continuity and Openness of Preimages. Thus, let $f^{-1} = g$ and for V is open in X if it is sufficient to show that $g^{-1}(V)$ is open in Y . ($g^{-1}(V) = f(V)$). Now, suppose f is a 1-1 and onto function. For here, we aim to prove that $f(V)^c = g^{-1}(V)^c$ is closed in Y . Since we assumed that V is open in X , then the complement of V , V^c is closed in X . And V^c is a closed subset of the compact set X , so V^c is compact as well. Which indicates $f(V^c)$ is closed (by Thm: Continuity Preserves Compactness). Since f is 1-1 and onto, $f(V)^c = f(V^c)$. Therefore, $f(V)^c$ is closed and $f(V) = g^{-1}(V)$ is open as V is open in X . ■

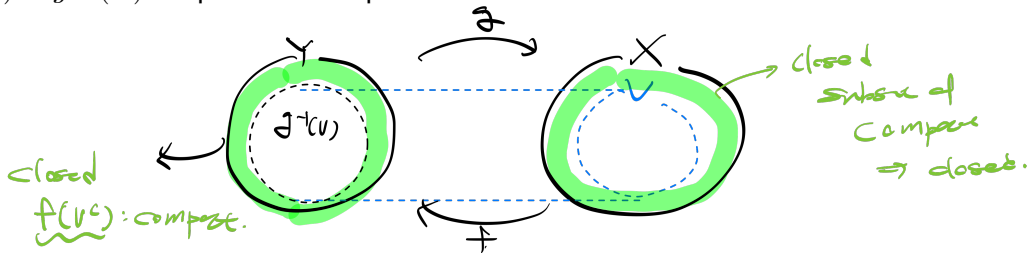


Figure 21: Inverse of Continuous bijective function

For example, suppose $f : [0, 1] \cup (2, 3] \rightarrow [0, 2]$ and

$$f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ x - 1 & 2 \leq x \leq 3 \end{cases}$$

Q1. Is f continuous on $[0, 1] \cup (2, 3]$? $\Rightarrow$ Yes.

For $x \in [0, 1]$, $f(x) = x$. Let $\epsilon > 0$ be given and set $\delta = \epsilon$. For all $x \in [0, 1]$ such that $|x - p| < \delta$, we have $|f(x) - f(p)| = |x - p| < \delta = \epsilon$. thus, f is continuous at p in $[0, 1]$. on the other hand, For $x \in (2, 3]$, $f(x) = x - 1$. Let $\epsilon > 0$ be given and set $\delta = \epsilon$. For all $x \in (2, 3]$ such that $|x - p| < \delta$, we have $|f(x) - f(p)| = |(x - 1) - (p - 1)| = |x - p| < \delta = \epsilon$. thus, f is continuous at p in $(2, 3]$.

Q2. Is f one to one and onto and then is there f^{-1} . If there exists f^{-1} , is it continuous? $\Rightarrow$ Yes.

Note that Def of 1-1 is $f : E \rightarrow Y$ is injective if $f(x_1) = f(x_2)$ implies $x_1 = x_2$ for all $x_1, x_2 \in E$ and onto is $f : E \rightarrow Y$ is surjective if for every $y \in Y$, there exists $x \in E$ such that $f(x) = y$.

To show one to one, let $x_1, x_2 \in [0, 1] \cup (2, 3]$ and suppose $f(x_1) = f(x_2)$. We consider three cases:

1. $x_1, x_2 \in [0, 1]$
for, $f(x_1) = x_1$, $f(x_2) = x_2$ if $f(x_1) = f(x_2)$, then $x_1 = x_2$
2. $x_1, x_2 \in (2, 3]$
for, $f(x_1) = x_1 - 1$, $f(x_2) = x_2 - 1$. if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

3. $x_1 \in [0, 1]$, $x_2 \in (2, 3]$ Vice versa for $f(x_1) \in [0, 1]$ and $f(x_2) \in (1, 2]$ if $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

Thus, f is **one to one** To show onto, Let $y \in [0, 2]$. We find $x \in [0, 1] \cup (2, 3]$ such that $f(x) = y$. Consider Two cases :

1. $y \in [0, 1]$

Choose $x = y \in [0, 1]$ then , Then $f(x) = x = y$.

2. $y \in (1, 2]$

Choose $x = y + 1 \in (2, 3]$ then, $f(x) = (y + 1) - 1 = y$.

Thus, f is **onto**

As we show that f is one-to-one and onto, it has an inverse, and f is continuous; thus, by the Thm: Inverse of a Continuous Bijective Mapping, the inverse function f^{-1} is continuous as well.

Def: Uniformly continuous

Let f be a mapping of a metric space X into a metric space Y . We say that f is **uniformly continuous** on X if for every $\epsilon > 0$ there exists $\delta > 0$ such that $d_Y(f(p), f(q)) < \epsilon$ whenever $d_X(x, p) < \delta$

The important point here is the distinction from continuous. In the case of continuous, when we define δ , it depends on the limit value p in the X space and the restriction of the function value ϵ in the Y space. That is, if $\delta(p, \epsilon) = \delta$, it is continuous, whereas **uniformly continuous** only depends on ϵ ; that is, we can write it as follows: $\delta(\epsilon) = \delta$.

For example, if $f : (0, 1) \rightarrow \mathbb{R}$ and $f(x) = \frac{1}{x}$, to prove that f is continuous, for $p \in (0, 1)$ and $\forall \epsilon > 0$, need to find $\delta > 0$. In this case, $\delta = \min(\frac{p}{4}, \frac{\epsilon p^2}{4})$. However, if $f : (\frac{1}{2}, 1) \rightarrow \mathbb{R}$ and $f(x) = \frac{1}{x}$, then $\delta = \min(\frac{1}{4}, \frac{\epsilon}{4})$; in this case, it does not depend on p , so it can be called **uniformly continuous**.

Thm : Compact continuous preserves uniformly continuous

$f : X \rightarrow Y$ and f is continuous in compact set X . Then, f becomes uniformly continuous.

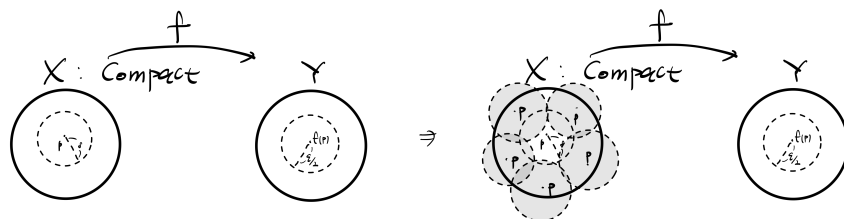


Figure 22: Compact continuous preserves uniformly continuous

▷ **proof** : Suppose $\epsilon > 0$ given, and wish to show that there exists $\delta > 0$ such that $d_Y(f(p), f(q)) < \delta$ whenever $d_X(p, q) < \delta$ and δ must be depend on choice of p, q . Now Suppose there exists open set which is $V_p = \{q \in X; d_X(p, q) < \frac{1}{2}\delta(\epsilon, p)\} = N_{\frac{\delta}{2}}(p)$, and $p \in X$. Then since X is compact set, by the definition of compact set we can know there exists finitely open cover of X . $\bigcup_{p \in X} V_p \supset X \Rightarrow V_{p_1} \cup V_{p_2} \cup V_{p_3} \cup \dots \cup V_{p_n} \supset X$ And

denote one of these open covers as : $V_{p_i} = \{q \in X; d_X(p, q) \leq \frac{1}{2}\delta(\delta, p_i)\} = N_{\frac{1}{2}\delta(\epsilon, p_i)}(p_i)$. And choose $\delta = \frac{1}{2} \min\{\delta(\epsilon, p_1), \delta(\epsilon, p_2), \dots, \delta(\epsilon, p_n)\}$. Then we wish to know that for this δ the following statement holds : Let $p, q \in X$ such that $d_X(p, q) < \delta$. We need to show that $d_Y(f(p), f(q)) < \epsilon$. One might easily know that $p \in V_{p_k}$ since the union of V_{p_i} covers the X . Then $d_X(p_k, p) < \frac{1}{2}\delta(\epsilon, p_k)$, using triangular inequality : $d_X(p_k, q) \leq d_X(p_k, p) + d_X(p, q) < \frac{1}{2}\delta(\epsilon, p_k) + \delta \leq \frac{1}{2}\delta(\epsilon, p_k) + \frac{1}{2}\delta(\epsilon, p_k) = \delta(\epsilon, p_k)$. Thus, $d_X(p_k, q) < \delta(\epsilon, p_k) \Rightarrow d_Y(f(p_k), f(q)) < \frac{\epsilon}{2}$. Similarly since $d_X(p_k, p) < \frac{1}{2}\delta(\epsilon, p_k)$ we also have $d_Y(f(p_k), f(p)) < \frac{\epsilon}{2}$. Using the triangle inequality in Y : $d_Y(f(p), f(q)) \leq d_Y(f(p), f(p_k)) + d_Y(f(p_k), f(q))$ Therefore, $d_Y(f(p), f(q)) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. ■

To summarize again, the property that f is continuous and X , where f is defined, is compact leads to many characteristics. First, the continuity of f and the compactness of X guarantee that $f(X)$ is compact, and this fact relates to $f(X)$ being closed and bounded. Since $f(X)$ is closed and bounded, minimum and maximum values exist. Furthermore, the property that f is continuous and X , where f is defined, is compact ensures uniform continuity. Below are a few related practice problems.

- $f : (0, 1) \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$

▷ **proof** : for $f(x)$, $|f(x) - f(y)| = \left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|y-x|}{xy}$.

Thm : Connected preserve Connected

Suppose $f : E \rightarrow Y$ and $E \subseteq X$, and f is continuous, X, Y is metric spaces. Then E is connected $\Rightarrow f(E)$ is connected.

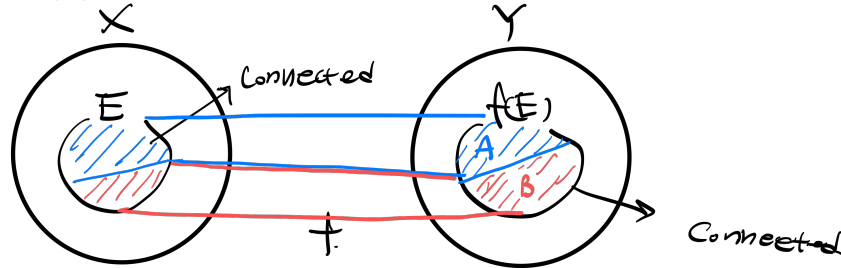


Figure 23: Connected preserve Connected

▷ **proof** : Assume $f(E)$ is not connected (to get contradiction) we wish to show E is not connected ($\sim B \Rightarrow \sim A \Leftrightarrow A \Rightarrow B$) Now since $f(E)$ is not connected $\Rightarrow A \cup B = f(E)$ and $\bar{A} \cap B = \bar{B} \cap A = \emptyset$. And let's denote that $G = E \cap f^{-1}(A), H = E \cap f^{-1}(B)$ then we are going to show $E = G \cup H$ this is trivial. For every $e \in E$, $f(e) \in f(E) = A \cup B$ Without loss of generality If $f(e) \in A$, then $e \in f^{-1}(A)$, and thus $e \in G$. Therefore, $E = G \cup H$. Now let's prove $\bar{G} \cap H = G \cap \bar{H} = \emptyset$. Note that $A \subseteq \bar{A}$ so $f(A) \subseteq f(\bar{A})$ indicate $G \subseteq f^{-1}(\bar{A})$ is closed set since f preserve closeness thus, $\bar{G} \subseteq f^{-1}(\bar{A})$. Then let $x \in \bar{G} \cap H$ then, $x \in \bar{G} \Rightarrow f(x) \in \bar{A}, x \in H \Rightarrow f(x) \in B$. This implies $f(x) \in \bar{A} \cap B$. However, by assumption, $\bar{A} \cap B = \emptyset$. Hence, such an x cannot exist $\bar{G} \cap H = \emptyset$. Similarly, we use the fact that $\bar{B} \cap A = \emptyset \Rightarrow H \subseteq f^{-1}(\bar{B}) \Rightarrow \bar{H} \subseteq f^{-1}(\bar{B})$. Now let $x \in G \cap \bar{H}$. Then, $x \in G \Rightarrow f(x) \in A$ and $x \in \bar{H} \Rightarrow f(x) \in \bar{B}$. This implies $f(x) \in A \cap \bar{B}$, which is empty by assumption. Thus: $G \cap \bar{H} = \emptyset$. Thus, G and H form a separation of E , meaning E is not connected. ■

Thm : Intermediate Value Theorem

Suppose $f : [a, b] \rightarrow \mathbb{R}$ be continuous and if $f(a) < f(b)$, c is number such that $f(a) < c < f(b)$ then there exists a point $x \in (a, b)$ such that $f(x) = c$

▷ **proof** : By the theorem that $E \subseteq \mathbb{R}$ is connected $\Leftrightarrow$ it has following properties ' if $x \in E, y \in E$ then $x < z < y$ and $z \in E$. Since $[a, b]$ is connected and let $E = [a, b]$. $f(E)$ is also connected by the Thm : Connected preserve Connected and $f(E) \subseteq \mathbb{R}$. So we can know that if $f(a) \in f(E)$, $f(b) \in f(E)$ and $f(a) < c < f(b)$ then $c \in f(E)$ by the property of connected set in real. This implies there has to be some $x \in E$ such that $f(x) = c$.

Discontinuities

Def : Left and Right limits

Let f be defined on (a, b) . Then we can write left hand limit and right hand limit as :

$$f(a-) = \lim_{x \rightarrow a, x < a} f(x), \text{ if } \forall \epsilon > 0, \exists \delta > 0$$

$$\text{such that } |f(x) - f(a-)| < \epsilon \text{ whenever } a - \delta < x < a$$

$$f(a+) = \lim_{x \rightarrow a, x > a} f(x), \text{ if } \forall \epsilon > 0, \exists \delta > 0$$

$$\text{such that } |f(x) - f(a+)| < \epsilon \text{ whenever } a < x < a + \delta$$

Thus, there are two type of discontinuities :

- $f(a+) \neq f(a-)$
- $f(a+) = f(a-) \neq f(a)$

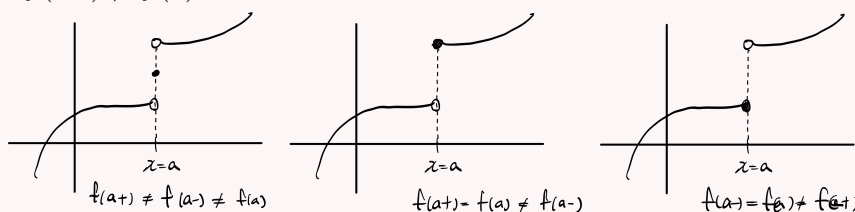


Figure 24: Discontinuity

For example, Consider this case :

$$f(x) = \begin{cases} x & \text{If } x \text{ is rational number} \\ 0 & \text{If } x \text{ is irrational number} \end{cases}$$

In such cases, we should consider various cases.

- when $a = 0$

$$f(0-) = \lim_{x \rightarrow 0, x \in \mathbb{Q}, x < 0} f(x) = 0, f(0+) = \lim_{x \rightarrow 0, x \in \mathbb{Q}, x > 0} f(x) = 0,$$

$$f(0-) = \lim_{x \rightarrow 0, x \notin \mathbb{Q}, x < 0} f(x) = 0, f(0+) = \lim_{x \rightarrow 0, x \notin \mathbb{Q}, x > 0} f(x) = 0,$$

$$\text{thus } f(0-) = f(0+) = \lim_{x \rightarrow 0} f(x)$$

- when $a \neq 0, a \in \mathbb{Q}$

$$f(a-) = \lim_{x \rightarrow a, x \in \mathbb{Q}, x < a} f(x) = a, f(a+) = \lim_{x \rightarrow a, x \in \mathbb{Q}, x > a} f(x) = a,$$

$$f(a-) = \lim_{x \rightarrow a, x \notin \mathbb{Q}, x < a} f(x) = 0, f(a+) = \lim_{x \rightarrow a, x \notin \mathbb{Q}, x > a} f(x) = 0,$$

$$\text{thus } f(a-) \neq f(a+) = \lim_{x \rightarrow 0} f(x)$$

- when $a \neq 0, a \notin \mathbb{Q}$

$$f(a-) = \lim_{x \rightarrow a, x \in \mathbb{Q}, x < a} f(x) = a, f(a+) = \lim_{x \rightarrow a, x \in \mathbb{Q}, x > a} f(x) = a,$$

$$f(0-) = \lim_{x \rightarrow 0, x \notin \mathbb{Q}, x < 0} f(x) = 0, f(0+) = \lim_{x \rightarrow 0, x \notin \mathbb{Q}, x > 0} f(x) = 0,$$

$$\text{thus } f(a-) \neq f(a+) = \lim_{x \rightarrow 0} f(x)$$

Monotonic function

suppose $f: (a, b) \rightarrow \mathbb{R}$ monotonic function following statements.

f is monotonically increasing on (a, b) if $a < x < y < b \Rightarrow f(x) \leq f(y)$

f is monotonically decreasing on (a, b) if $a < x < y < b \Rightarrow f(x) \geq f(y)$

Thm : ★

suppose $f : (a, b) \rightarrow \mathbb{R}$. f is monotonically increasing $\Rightarrow f(x+), f(x-)$ exists at every point. and x in (a, b) more precisely $\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t)$. Furthermore if $a < x < y < b$ then $f(x+) \leq f(y-)$.

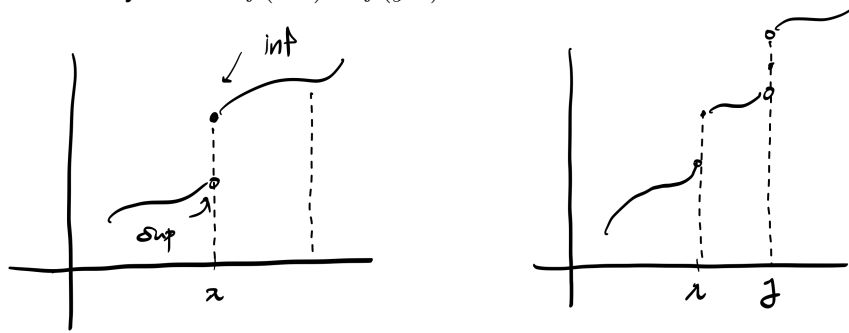


Figure 25: Monotonically increasing continuous

▷ **proof** : By hypothesis, the set of numbers $f(t)$, where $a < t < x$, is bounded above by the number $f(x)$, and therefore has a least upper bound which we shall denote by A . Evidently, $A \leq f(x)$. We have to show that $A = f(x-)$. For a given ϵ , we can choose an appropriate $\delta > 0$ such that $a < x - \delta < x$, and $A - \epsilon < f(x - \delta) \leq A$. Since f is a monotonic increasing function, for $x - \delta < t < x$, the following is true: $f(x - \delta) \leq f(t) \leq A$. With those two equations, we can get $A - \epsilon < f(t) \leq A$ ($x - \delta < t < x$) $\Rightarrow |f(t) - A| < \epsilon$. Thus, we can know that for $t \rightarrow x-$, it indicates $f(t) \rightarrow A$; thus, $f(x-) = A$. On the other hand, in the same manner, we can prove that $f(x+) = \inf_{x < t < b} f(t)$. Now let's prove that $f(x+) \leq f(y-)$. For $x < y$, we define that $f(x+) = \inf_{x < t < b} f(t)$, $f(y-) = \sup_{a < t < y} f(t)$. Since the definition of monotonic increasing, $x < t < y$ implies $f(x) \leq f(t) \leq f(y)$. Thus, $f(x+) \leq f(t) \leq f(y-)$, this implies $f(x+) \leq f(y-)$.

Thm : Countable of discontinuous

Let f be monotonic on (a, b) . Then the set of points of (a, b) at which f is discontinuous is at most countable.

▷ **proof** : At the discontinuity point $x \in E$, it holds that $f(x-) \neq f(x+)$. Since f is a monotonically increasing function, it follows that $f(x-) < f(x+)$. Also, due to the density of $\mathbb{Q}$, there always exists a rational number $r(x)$ satisfying $f(x-) < r(x) < f(x+)$. If we

collect such $r(x)$ for each discontinuity point x , then each element of E corresponds to $r(x)$, and if we consider this as a subset of E , since the set of rational numbers is countable, and E is a subset of the rational numbers, a subset of a countable set is trivially countable, thus E is also a countable set.

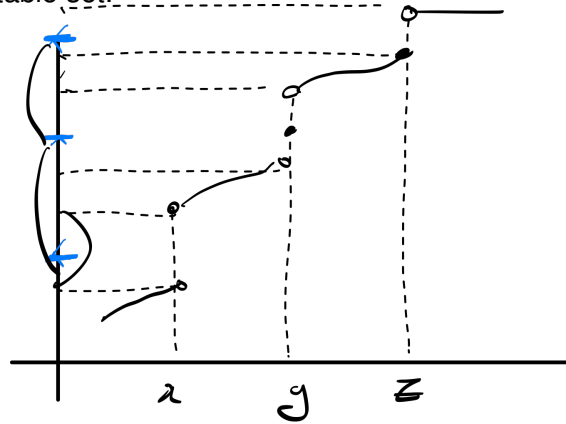


Figure 26: Countable of discontinuous

Infinite limits and limits at infinity

Def :Neighborhood of $+\infty$ and $-\infty$

For any real C , $(C, +\infty)$ denote neighborhood(=nhd) of $+\infty$ Similarly $(-\infty, C)$ denotes nhd of $-\infty$.

Def :Extended Real Limit Definition

Suppose $f : E \rightarrow \mathbb{R}$, $E \subseteq \mathbb{R}$ then we say that $f(t) \rightarrow A$ as $t \rightarrow x$, where A and x are in the extended real number system, if for every nhd U of A there is a nhd of V of x such that $V \cap E$ is not empty, and such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$. A moment's consideration will show that this coincides with upper definition when A and x are real.

Thm : Properties of Limits of Functions

Suppose $f, g : E \rightarrow \mathbb{R}$ and $f(t) \rightarrow A, g(t) \rightarrow B$ as $t \rightarrow x$ then following statements are hols.:

- $f(t) \rightarrow A' \Rightarrow A = A'$
 - ▷ **proof** : According to the limit definition, when $f(t) \rightarrow A$ as $t \rightarrow x$, for every neighborhood U of A , there is a neighborhood V of x such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$. Similarly, if $f(t) \rightarrow A'$ then for every neighborhood U' of A' , there exists a neighborhood V' of x such that $f(t) \in U'$ for all $t \in V' \cap E$, $t \neq x$. If $A \neq A'$, we can find neighborhoods U of A and U' of A' that do not intersect, which contradicts $f(t)$ being in both U and U' as $t \rightarrow x$. Thus, A must equal A' .
- $f(t + g)(t) \Rightarrow A + B$
 - ▷ **proof** : Let $\epsilon > 0$. Since $f(t) \rightarrow A$ and $g(t) \rightarrow B$ as $t \rightarrow x$, there exist neighborhoods V_f and V_g of x such that: $|f(t) - A| < \frac{\epsilon}{2}$ and $|g(t) - B| < \frac{\epsilon}{2}$ for all $t \in V_f \cap V_g$, $t \neq x$. Then, for $t \in V_f \cap V_g$, $t \neq x$, we have: $|(f + g)(t) - (A + B)| = |f(t) - A + g(t) - B| \leq$

$|f(t) - A| + |g(t) - B| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. Hence, $(f + g)(t) \rightarrow A + B$ as $t \rightarrow x$.

- $fg(t) \Rightarrow AB$

▷ **proof** : Let $\epsilon > 0$. Since $f(t) \rightarrow A$ and $g(t) \rightarrow B$, there exist neighborhoods V_f and V_g of x such that: $|f(t) - A| < \min\left(\frac{\epsilon}{2|B|+1}, 1\right)$ and $|g(t) - B| < \min\left(\frac{\epsilon}{2|A|+1}, 1\right)$ for all $t \in V_f \cap V_g$, $t \neq x$. For $t \in V_f \cap V_g$, $t \neq x$, we have: $|(fg)(t) - AB| = |f(t)g(t) - AB| = |f(t)g(t) - f(t)B + f(t)B - AB| \leq |f(t)||g(t) - B| + |B||f(t) - A|$. Using the bounds on $|f(t) - A|$ and $|g(t) - B|$, we can ensure that: $|(fg)(t) - AB| < \epsilon$. Hence, $(fg)(t) \rightarrow AB$ as $t \rightarrow x$.

- $f/g(t) \Rightarrow A/B$

▷ **proof** : Assume $B \neq 0$. Let $\epsilon > 0$. Since $g(t) \rightarrow B$, there exists a neighborhood V_g of x such that $|g(t) - B| < \frac{|B|}{2}$, implying $|g(t)| > \frac{|B|}{2}$, for all $t \in V_g$, $t \neq x$. Similarly, since $f(t) \rightarrow A$, there exists a neighborhood V_f of x such that $|f(t) - A| < \frac{\epsilon|B|}{2}$ for all $t \in V_f$, $t \neq x$. For $t \in V_f \cap V_g$, $t \neq x$, we have: $\left|\frac{f(t)}{g(t)} - \frac{A}{B}\right| = \left|\frac{f(t)B - Ag(t)}{g(t)B}\right| = \frac{|f(t)(B - g(t)) + g(t)(f(t) - A)|}{|g(t)B|}$. Using the bounds on $|f(t) - A|$ and $|g(t) - B|$, and the fact that $|g(t)| > \frac{|B|}{2}$, we can ensure that: $\left|\frac{f(t)}{g(t)} - \frac{A}{B}\right| < \epsilon$. Hence, $\frac{f}{g}(t) \rightarrow \frac{A}{B}$ as $t \rightarrow x$.

Part II — Analysis II

Differentiation, Integration, and Sequences of Functions

Differentiation

Def : Derivative

Let f be defined (and real-valued) on $[a, b]$. For any $x \in [a, b]$ from the quotient $\phi(t) = \frac{f(t) - f(x)}{t - x}$ ($a < t < b, t \neq x$) and define $f'(x) = \lim_{t \rightarrow x} \phi(t)$. This exists by the def: limits of functions implies that If X and Y be metric spaces and $E \subset X$, f maps E into Y . We write $f(x) \lim_{x \rightarrow p} f(x) = q$ if p is a limit point of E . And if f' is defined at a point x , we say that f is **differentiable** at x . And here we don't care derivative at a corner likes for $f'(a)$ and $f'(b)$ in $a < x < b$.

Another Def : Derivative

Suppose every setting is given same as above than we can also this

$$\exists \lim_{x \rightarrow a} \frac{f(a) - f(x)}{a - x} = f'(a) \iff \exists A \text{ s.t } f(x) = f(a) + A(x - a) + (x - a)\eta(x)$$
$$\lim_{x \rightarrow a} \eta(x) = 0 \text{ then } A = f'(x)$$

We can represent about differentiable at $x = a$ with both methods and they yield the same result when we consider that

$$\frac{f(a) - f(x)}{a - x} = A + \eta(x) \Rightarrow \lim_{x \rightarrow a} \frac{f(a) - f(x)}{a - x} = f'(a) = A$$

For here, it is easy to understand that the $\eta(x)$ indicates the error

Thm : Basic Differentiation Rules

Suppose f and g defined on $[a, b]$ and are differentiable at a point $x \in [a, b]$. Then followings are differentiable at x .

- $(f + g)'(x) = f'(x) + g'(x)$
- $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$
- $\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$
- $(f + g)'(x) = f'(x) + g'(x)$

▷ **proof** : by the another def, we can write $f(x) = f(a) + A(x - a) + (x - a)\eta_1(x)$, $\lim_{x \rightarrow a} \eta_1(x) = 0$ and $g(x) = g(a) + B(x - a) + (x - a)\eta_2(x)$ $\lim_{x \rightarrow a} \eta_2(x) = 0$ for here $A = f'(a)$ and $B = g'(a)$ since property of function $(f + g)(x) = f(x) + g(x)$

indicate that $f(x) + g(x) = [f(a) + g(a)] + [A + B](x - a) + (x - a)[\eta_1(x) + \eta_2(x)]$ and this can be rewritten like $(f + g)(x) = (f + g)(a) + (A + B)(x - a) + (x - a)[\eta_1(x) + \eta_2(x)]$ then by the def $\frac{(f+g)(x)-(f+g)(a)}{x-a} = A + B + \eta_1(x) + \eta_2(x)$ and if we go $x \rightarrow a$ then $\lim_{x \rightarrow a} \frac{(f+g)(x)-(f+g)(a)}{x-a} = A + B = f'(a) + g'(a)$, therefore $(f + g)'(a) = f'(a) + g'(a)$. ■

- $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$

▷ **proof** : With similar manner we can easily prove. by def and the property of function $f(x)g(x) = [f(a) + A(x - a) + (x - a)\eta_1(x)] \cdot [g(a) + B(x - a) + (x - a)\eta_2(x)]$ and this again can be $f(x)g(x) = f(a)g(a) + (x - a)[f(a)B + g(a)A] + (x - a)[f(a)\eta_2(x) + g(a)\eta_1(x)] + (x - a)^2\psi(x)$ for here $\psi(x)$ indicate term related to the $(x - a)^2$ and indicate $\frac{f(x)g(x)-f(a)g(a)}{x-a} = f(a)B + g(a)A + f(a)\eta_2(x) + g(a)\eta_1(x) + (x - a)\psi(x)$ and if we go $x \rightarrow a$ then $\lim_{x \rightarrow a} \frac{f(x)g(x)-f(a)g(a)}{x-a} = f(a)g'(a) + f'(a)g(a)$. ■

- $\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$

▷ **proof** : With same condition above we can consider equation little bit different way like. $\left(\frac{f}{g}\right)'(a) = \lim_{x \rightarrow a} \frac{\frac{f(x)}{g(x)} - \frac{f(a)}{g(a)}}{x - a} = \lim_{x \rightarrow a} \frac{f(x)g(a) - f(a)g(x)}{(x - a)g(x)g(a)}$ and we will use to fact that

$$\begin{aligned} f(x) &= f(a) + A(x - a) + (x - a)\eta_1(x) \\ \Rightarrow f(x)g(a) &= f(a)g(a) + A(x - a)g(a) + (x - a)\eta_1(x)g(a) \\ g(x) &= g(a) + B(x - a) + (x - a)\eta_2(x) \\ \Rightarrow f(a)g(x) &= f(a)g(a) + f(a)B(x - a) + f(a)(x - a)\eta_2(x) \\ \Rightarrow f(x)g(a) - f(a)g(x) &= [A(x - a)g(a) + (x - a)\eta_1(x)g(a)] - [f(a)B(x - a) + f(a)(x - a)\eta_2(x)] \\ &= (x - a)[Ag(a) - f(a)B + g(a)\eta_1(x) - f(a)\eta_2(x)] \end{aligned}$$

Now we can get another form of numerator. Let's replace the function with def

$$\begin{aligned} \frac{\frac{f(x)}{g(x)} - \frac{f(a)}{g(a)}}{x - a} &= \frac{(x - a)[Ag(a) - f(a)B + g(a)\eta_1(x) - f(a)\eta_2(x)]}{(x - a)g(x)g(a)} \\ &= \frac{Ag(a) - f(a)B + g(a)\eta_1(x) - f(a)\eta_2(x)}{g(x)g(a)} \end{aligned}$$

as $x \rightarrow a$

$$\begin{aligned} \lim_{x \rightarrow a} \frac{\frac{f(x)}{g(x)} - \frac{f(a)}{g(a)}}{x - a} &= \frac{Ag(a) - f(a)B}{g(a)^2} = \frac{f'(a)g(a) - f(a)g'(a)}{[g(a)]^2} \\ \Leftrightarrow \left(\frac{f}{g}\right)'(x) &= \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2} \end{aligned}$$

■

Thm : derivative of Chain Rule

Suppose f is continuous on $[a,b]$, $f'(x)$ exists at some point $x \in [a,b]$, g is defined well and then following is true

$$h(t) = g(f(t)) \quad (a \leq t \leq b) \Rightarrow h'(x) = g'(f(x))f'(x)$$

▷ **proof** : Since f is differentiable at x we can write $f(t) = f(x) + A(t-x) + (t-x)\eta_1(t)$, with $A = f'(x)$, $\lim_{t \rightarrow x} \eta_1(t) = 0$. On the other hand g is differentiable at $f(x)$ thus, $g(y) = g(f(x)) + B(y-f(x)) + (y-f(x))\eta_2(y)$ with $B = g'(f(x))$, $\lim_{y \rightarrow f(x)} \eta_2(y) = 0$.

And if we represent derivative of composition function according to the def. $g(f(t)) = g(f(x)) + B(f(t)-f(x)) + (f(t)-f(x))\eta_2(f(t))$. And for previous clue we can know what is the term $f(t) - f(x)$ and substitute it, then $g(f(t)) = g(f(x)) + B(t-x)[A + \eta_1(t)] + (t-x)[A + \eta_1(t)] \cdot \eta_2(f(t))$ holds and make it simply then $g(f(t)) = g(f(x)) + (t-x)[BA + B\eta_1(t) + (A + \eta_1(t))\eta_2(f(t))]$ $\Rightarrow \frac{g(f(t))-g(f(x))}{t-x} = BA + B\eta_1(t) + (A + \eta_1(t))\eta_2(f(t))$ and then let's $t \rightarrow x$ then $\lim_{t \rightarrow x} \frac{g(f(t))-g(f(x))}{t-x} = BA = g'(f(x)) \cdot f'(x)$. ■

Def : Local maximum

Let f be a real function on a metric space X . We say that f has a local maximum at a point $p \in X$. If $\exists \delta > 0$ such that $f(q) \leq f(p) \quad \forall q \in X$ with $d(p, q) < \delta$

Review : Supremum and infimum

The expression $\sup_{x \in E} f(x) = A$ means that there exists a real number A that satisfies the following two conditions simultaneously. First, there is the condition of an upper bound: $\forall x \in E, f(x) \leq A$. This means that the values of $f(x)$ do not exceed A for any x within the given range. The second condition is $\forall \varepsilon > 0, \exists x_\varepsilon \in E$ such that $f(x_\varepsilon) > A - \varepsilon$, which indicates that A is the least upper bound. Conversely, $\inf_{x \in E} f(x) = A$ is defined in the opposite manner.

Thm : Stationary Points of local maximum

Let f be defined on $[a,b]$; if f has a local maximum at a point $x \in (a,b)$, and if $\exists f'(x)$ then $f'(x) = 0$

▷ **proof** : First of all, we can change def of derivative like $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{(x+h)-x} = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$. If we let $f(x)$ is the local maximum then according to the def we can consider both side of h if $h > 0$ then $f(x+h) \leq f(x) \Rightarrow \frac{f(x+h)-f(x)}{h} \leq 0$ on the other hand if $h < 0$ indicate $f(x+h) \leq f(x) \Rightarrow \frac{f(x+h)-f(x)}{h} \geq 0$ since $f'(x)$ exists the both should be same. thus, $0 \leq \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} \leq 0$. Therefore, If f has a local max at $x \in (a,b)$ and $\exists f'(x)$, then $f'(x) = 0$. ■

★ Thm : Cauchy's Mean Value Theorem

Let f and g be continuous real functions on $[a, b]$, and differentiable on (a, b) . Then there exists some $x \in (a, b)$ such that: $[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x)$.

▷ **proof** : let $H(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t)$ then H is continuous in $[a, b]$ and also differentiable in (a, b) then make let's consider the $H(\cdot)$ at a, b then we can make $H(b) - H(a) = [f(b) - f(a)](g(b) - g(a)) - [g(b) - g(a)](f(b) - f(a))$ then we can know $H(a) = H(b)$ then by the Rolle's Theorem ($\exists x \in (a, b)$ such that $H'(x) = 0$). $H'(t) = [f(b) - f(a)]g'(t) - [g(b) - g(a)]f'(t)$ holds. Thus, $H'(x) = 0$ indicates $[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x)$. ■

★ Thm : Mean Value Theorem

Let f be a real continuous function on $[a, b]$, which is differentiable on (a, b) . Then there exists a point $x \in (a, b)$ such that $f(b) - f(a) = (b - a)f'(x) \iff \frac{f(b) - f(a)}{b - a} = f'(x)$.

▷ **proof** : Let's make auxiliary function, $\phi(t) = f(t) - \left(\frac{f(b) - f(a)}{b - a}\right)(t - a)$ f is continuous in $[a, b]$ and differentiable in (a, b) , ϕ also continuous and differentiable. Therefore, we can get

$$\begin{aligned}\phi(a) &= f(a) - \left(\frac{f(b) - f(a)}{b - a}\right)(a - a) = f(a) \\ \phi(b) &= f(b) - \left(\frac{f(b) - f(a)}{b - a}\right)(b - a) = f(b) - (f(b) - f(a)) = f(a) \\ \Rightarrow \phi(a) &= \phi(b)\end{aligned}$$

Now, we can use Rolle's Theorem as we did in previous theorem. ($\exists x \in (a, b)$ such that $\phi'(x) = 0$) and if we differentiate at t

$$\begin{aligned}\phi'(t) &= f'(t) - \left(\frac{f(b) - f(a)}{b - a}\right) = 0 \\ \Rightarrow f'(x) &= \frac{f(b) - f(a)}{b - a}.\end{aligned}$$

■

Thm : Intermediate Value Theorem for Derivatives

Suppose f is a real differentiable function on $[a, b]$, and suppose $f'(a) < \lambda < f'(b)$. Then there exists a point $x \in (a, b)$ such that: $f'(x) = \lambda$

▷ **proof** : We will use Mean Value Theorem (MVT) and let's make auxiliary function $F(x) = f(x) - \lambda x$. since F is the sum of differentiable two function ($f(\cdot), x$) thus, F also differentiable. Therefore, $F'(x) = f'(x) - \lambda$ can be defined. Since the assumption of theorem that

$$\begin{aligned}f'(a) < \lambda < f'(b) &\iff f'(a) - \lambda < 0 < f'(b) - \lambda \iff F'(a) < 0 < F'(b) \\ \Rightarrow F'(a) = f'(a) - \lambda &< 0, \quad F'(b) = f'(b) - \lambda > 0.\end{aligned}$$

We are enough to show that and It is obvious that there exists $F'(c) = 0$ for $c \in [a, b]$. Therefore, $\exists x \in (a, b)$ such that $F'(x) = 0 \Rightarrow f'(x) = \lambda$ ■

★ Thm : L'HOSPITAL's rule

Suppose f, g are differentiable in (a, b) and $g'(x) \neq 0 \quad \forall x \in (a, b) \subset (-\infty, \infty)$ and suppose $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = A$, $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ or $\lim_{x \rightarrow a} g(x) = +\infty$ then,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = A$$

1. Case $f(a) = g(a) = 0$

▷ **proof** : Since f, g are differentiable in (a, b) , $g'(x) \neq 0$ and $f(a) = g(a) = 0$. Let, $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$. We wish to show $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$. According to the Cauchy's Mean Value Theorem $\exists c \in (a, x)$ such that $\frac{f(x)-f(a)}{g(x)-g(a)} = \frac{f'(c)}{g'(c)}$. Since $f(a) = g(a) = 0$ $\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}$ for some $c \in (a, x)$ and let it go $x \rightarrow a^+$ then $c \in (a, x)$ lead $c \rightarrow a^+$ therefore,

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(c)}{g'(c)} = \lim_{c \rightarrow a^+} \frac{f'(c)}{g'(c)} = L$$

■

2. Case $f(a) = g(a) = \infty$

▷ **proof** : let $\lim_{x \rightarrow a^+} f(x) = \pm\infty$, $\lim_{x \rightarrow a^+} g(x) = \pm\infty$ and $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$. And as we did, we wish to show $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$. Let's review the def

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } a < x < a + \delta \Rightarrow L - \varepsilon < \frac{f'(x)}{g'(x)} < L + \varepsilon$$

and also by the Cauchy's Mean Value Theorem $\exists c \in (x, b)$

$$\begin{aligned} \frac{f(b) - f(x)}{g(b) - g(x)} &= \frac{f'(c)}{g'(c)} \\ \Rightarrow \frac{f'(c)}{g'(c)} &= \frac{f(x)}{g(x)} - \left(\frac{f(b)}{g(x)} - \frac{f'(c)g(b)}{g'(c)g(x)} \right) \\ \Rightarrow \frac{f'(c)}{g'(c)} &= \frac{f(x)}{g(x)} - r(x) \end{aligned}$$

For above equation, let $x = c$

$$\begin{aligned} L - \varepsilon &< \frac{f'(c)}{g'(c)} < L + \varepsilon \\ \Rightarrow L - \varepsilon + r(x) &< \frac{f(x)}{g(x)} < L + \varepsilon + r(x) \end{aligned}$$

And it's enough to show that $r(x) \rightarrow 0$ since $g(x) \rightarrow \infty \Rightarrow \frac{f(b)}{g(x)} \rightarrow 0, \frac{f'(c)g(b)}{g'(c)g(x)} \rightarrow 0$. Therefore $r(x) \rightarrow 0$ ■

3. Case $L = \pm\infty$

▷ **proof** : for Case 1 and Case 2 we wish to show that $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = \infty$ indicate $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \infty$. The fact that the limit of a derivative is infinity means the following:

$$\forall M > 0, \exists \delta > 0 \text{ such that } a < x < a + \delta \Rightarrow \frac{f'(x)}{g'(x)} > M$$

And we will use again Cauchy Mean Value Theorem

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)} + r(x) \quad \text{with } c \in (a, x), \quad r(x) \rightarrow 0$$

And if we plug this to the definition

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)} + r(x) > M + r(x)$$

And we've shown that $r(x) \rightarrow 0$ thus we can write that $\forall \varepsilon > 0, \exists \delta_1 > 0$ such that $a < x < a + \delta_1 \Rightarrow |r(x)| < \varepsilon$. Now, let $\varepsilon = \frac{1}{2}M$ then $\frac{f(x)}{g(x)} > M + r(x) > M - |r(x)| > M - \frac{1}{2}M = \frac{1}{2}M$. Therefore,

$$\begin{aligned} \frac{f(x)}{g(x)} &> \frac{M}{2} \quad \text{for all } x \in (a, a + \delta_1) \\ \Rightarrow \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} &= \infty \end{aligned}$$

■

4. Case $x \rightarrow \infty$

▷ **proof** : since the case $x \rightarrow \infty$ is tricky to handle. $t = \frac{1}{x} \Rightarrow x = \frac{1}{t}$ so $x \rightarrow \infty \iff t \rightarrow 0^+$. Now let's rewrite the limits like

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{t \rightarrow 0^+} \frac{f'(1/t)}{g'(1/t)}$$

since $t \rightarrow 0^+$ we can use L'Hopital's Rule. By the chain rule, if the function be $F(t) = f(1/t)$, $G(t) = g(1/t)$ then

$$\begin{aligned} F'(t) &= \frac{d}{dt} f(1/t) = -\frac{1}{t^2} f'(1/t) \\ G'(t) &= \frac{d}{dt} g(1/t) = -\frac{1}{t^2} g'(1/t) \end{aligned}$$

therefore,

$$\frac{F'(t)}{G'(t)} = \frac{-t^{-2} f'(1/t)}{-t^{-2} g'(1/t)} = \frac{f'(1/t)}{g'(1/t)}$$

According to the L'Hopital's Rule

$$\lim_{t \rightarrow 0^+} \frac{F(t)}{G(t)} = L \Rightarrow \lim_{t \rightarrow 0^+} \frac{f(1/t)}{g(1/t)} = L \Rightarrow \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

■

★ Thm : Taylor's Theorem

Suppose f is a real function on $[a, b]$, n is a positive integer, $f^{(n-1)}$ is continuous on $[a, b]$. $\exists f^{(n)}(t), \forall t \in (a, b)$. Let α, β be distinct points of $[a, b]$ and define

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k$$

Then there exists a point $x \in (\alpha, \beta)$ then,

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n$$

for here, $P(\beta)$ indicate α Polynomial approximation and $f(\beta)$ indicate real function Value and The error is proportional to $f^{(n)}$

▷ **proof** : if we let $f(\beta) = P(\beta) + M(\beta - \alpha)^n$ then we wish to show $M = \frac{f^{(n)}(x)}{n!}$

$$\begin{aligned} g(t) &= f(t) - P(t) - M(t - \alpha)^n \\ \Rightarrow g^{(n)}(t) &= f^{(n)}(t) - n!M \end{aligned}$$

Since, we defined that $P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k$, $P^{(k)}(\alpha) = f^{(k)}(\alpha) \quad \forall k = 0, \dots, n-1$

holds. Therefore, $g(\alpha) = g'(\alpha) = \dots = g^{(n-1)}(\alpha) = 0$ indicate that $g^{(k)}(\alpha) = f^{(k)}(\alpha) - P^{(k)}(\alpha) = 0$ Now, we can use Rolle's Theorem recursively to get inductive result.

$$\begin{aligned} g(\alpha) &= g(\beta) = 0 \Rightarrow \exists x_1 \in (\alpha, \beta) \text{ such that } g'(x_1) = 0 \\ g'(\alpha) &= g'(x_1) = 0 \Rightarrow \exists x_2 \in (\alpha, x_1) \text{ such that } g''(x_2) = 0 \\ g''(\alpha) &= g''(x_2) = 0 \Rightarrow \exists x_3 \in (\alpha, x_2) \text{ such that } g^{(3)}(x_3) = 0 \\ &\vdots \\ g^{(n)}(\alpha) &= g^{(n)}(x_n) = 0 \Rightarrow f^{(n)}(x_n) = n!M \Rightarrow M = \frac{f^{(n)}(x_n)}{n!} \end{aligned}$$

Therefore, we can conclude that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n \quad \text{for some } x \in (\alpha, \beta)$$

■

Thm : MVT in Vector space

Suppose f is open interval which in $[a, b]$, continuous function correspond to vector space $\mathbb{R}^k$ and differentiable in open set (a, b) then for $x \in (a, b)$ following statement hold :

$$|f(b) - f(a)| \leq (b - a) |f'(x)|$$

for here $|\cdot|$ indicate euclidean norm and $f'(x)$ is vector derivative.

▷ **proof** : We project vectors to scalars via inner products. By assumption, the Mean Value Theorem (MVT) holds for scalars, which we then extend back to the vector space. we can

define the scalar function

$$\phi(t) = \langle f(t), f(b) - f(a) \rangle$$

this indicate $f(t)$ project to $f(b) - f(a)$ since scalar ϕ is differentiable at (a, b) and continuous at $[a, b]$ and now use MVT

$$\phi(b) - \phi(a) = \phi'(x)(b - a) \quad \text{for some } x \in (a, b)$$

for here we can extend left-hand.

$$\begin{aligned} \phi(b) - \phi(a) &= \langle f(b), f(b) - f(a) \rangle - \langle f(a), f(b) - f(a) \rangle \\ &= \langle f(b) - f(a), f(b) - f(a) \rangle \\ &= |f(b) - f(a)|^2 = \phi'(x) = \langle f'(x), f(b) - f(a) \rangle \end{aligned}$$

Therefore, we can rewrite as

$$|f(b) - f(a)|^2 = (b - a) \langle f'(x), f(b) - f(a) \rangle$$

and if we using Cauchy–Schwarz Inequality which is $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq |\mathbf{u}| \cdot |\mathbf{v}|$ for two vector $\mathbf{u}, \mathbf{v} \in \mathbb{R}^k$ we can know

$$(b - a) \langle f'(x), f(b) - f(a) \rangle \leq (b - a) |f'(x)| \cdot |f(b) - f(a)|$$

This implies that

$$\begin{aligned} |f(b) - f(a)|^2 &\leq (b - a) |f'(x)| \cdot |f(b) - f(a)| \\ \Rightarrow |f(b) - f(a)| &\leq (b - a) |f'(x)| \end{aligned}$$

Integral

For this lecture we only consider the Riemann integral.

Def : Partition, lower and upper integrals

In this section we will consider various notation to define what is the integral

1. Partition : for given interval $[a, b]$ partition P indicate that

$$P = \{x_0, x_1, \dots, x_n\} \quad \text{with} \quad a = x_0 \leq x_1 \leq \dots \leq x_n = b$$

In other words, $[a, b]$ is divided into finite points. and also we notate that $\Delta x_i = x_i - x_{i-1}$ for $i = 1, \dots, n$

2. Upper Sum : Suppose f is bounded in $[a, b]$ and for each Δx_i we define

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}$$

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i$$

3. Lower Sum : Suppose f is bounded in $[a, b]$ and for each Δx_i we define

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$$

$$L(P, f) = \sum_{i=1}^n m_i \Delta x_i$$

4. Upper Integral : If we get upper sum from all possible partition then Upper integral is :

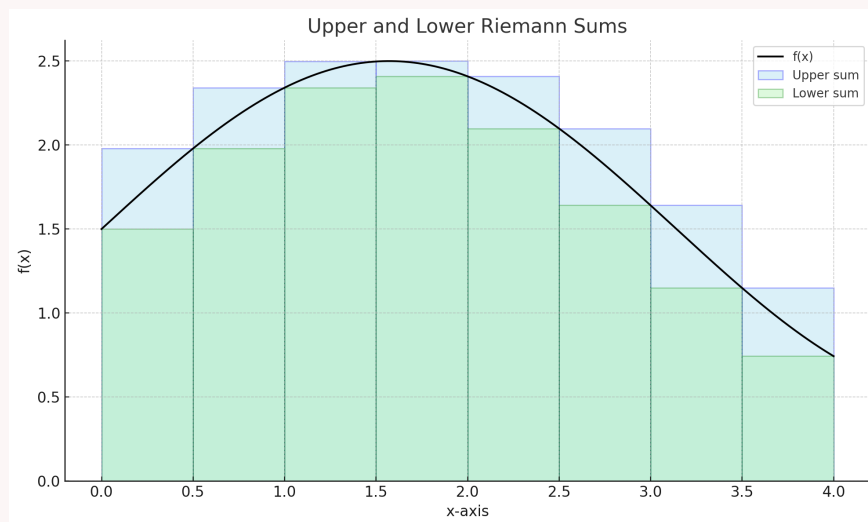
$$\int_a^b f(x) dx = \inf\{U(P, f)\}$$

5. Upper Integral : If we get lower sum from all possible partition then Lower integral is :

$$\int_a^b f(x) dx = \sup\{L(P, f)\}$$

And If following statement true then we call it Riemann Integrability or Integrability

$$\overline{\int_a^b f(x) dx} = \underline{\int_a^b f(x) dx}$$



Def : Riemann–Stieltjes Integral

It is a more general concept than the Riemann integral, allowing a function f to be integrated with respect to another function α . We usually cut the partition into rectangles of consistent size, but in fact, we don't have to. The size of the partitions can vary. let's assume that α is monotonically increasing function in $[a, b]$ and f is bounded real function then for partition $P = \{x_0, x_1, \dots, x_n\}$. For δx_i we can consider this $\Delta \alpha_i =$

$\alpha(x_i) - \alpha(x_{i-1}) \geq 0$ then all of above concepts can change according to the α like

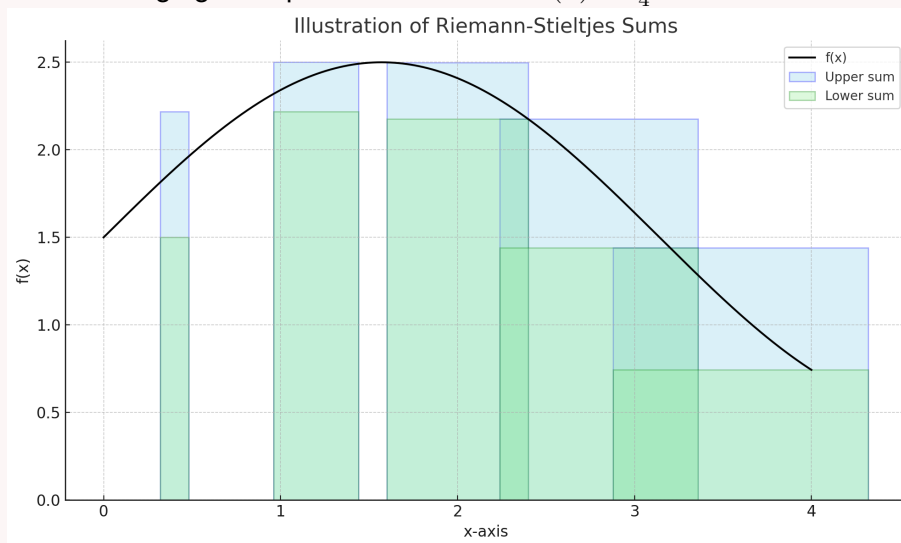
$$\text{Upper sum: } U(P, f, \alpha) = \sum_{i=1}^n M_i \Delta \alpha_i$$

$$\text{Lower sum: } L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta \alpha_i$$

$$\text{Upper integral: } \int_a^b f d\alpha = \inf U(P, f, \alpha)$$

$$\text{Lower integral: } \int_a^b f d\alpha = \sup L(P, f, \alpha)$$

If we consider $\alpha(x) = x$ then the Riemann–Stieltjes Integral equal to the Riemann Integral. and following figure represent the case if $\alpha(x) = \frac{x^2}{4}$



Def : Refinement

In Riemann integration, the approximation of the integral becomes more accurate as the partition becomes finer and finer, and to that end, we introduce the concept of refinement.

$$P^* \supseteq P$$

P^* is a more finely divided segment containing P . On the other hand we usually use the term common refinement which means

$$P^* = P_1 \cup P_2$$

Thm : refinement

If P^* is a refinement of P , then $L(P, f, \alpha) \leq L(P^*, f, \alpha)$, $U(P^*, f, \alpha) \leq U(P, f, \alpha)$

▷ **proof :** Case $L(P, f, \alpha) \leq L(P^*, f, \alpha)$

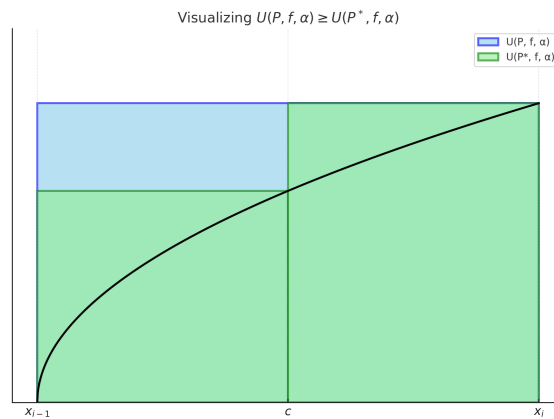
Consider the partition $[x_{i-1}, x_i]$ and then refinement P^* divide this more samller partition. Therefore, we can divide two $m_1^* := \inf_{x \in [x_{i-1}, c]} f(x)$ and $m_2^* := \inf_{x \in [c, x_i]} f(x)$ for $[x_{i-1}, c] \cup$

$[c, x_i]$. Thus, we can easily this $m_i \leq m_1^*$, $m_i \leq m_2^*$ and also we can derive this

$$\begin{aligned} m_i \Delta \alpha_i &= m_i(\alpha(x_i) - \alpha(x_{i-1})) = m_i(\alpha(x_i) - \alpha(c) + \alpha(c) - \alpha(x_{i-1})) \\ &= m_i \Delta \alpha_1 + m_i \Delta \alpha_2 \leq m_1^* \Delta \alpha_1 + m_2^* \Delta \alpha_2 \\ &\Rightarrow L(P, f, \alpha) \leq L(P^*, f, \alpha) \end{aligned}$$

On the other hand, for $P^* = P \cup \{c\}$ and $c \in (x_{i-1}, x_i)$ again, for $M_i \geq M_1^*$, $M_i \geq M_2^*$ we can derive.

$$\begin{aligned} M_i \Delta \alpha_i &= M_i(\alpha(x_i) - \alpha(x_{i-1})) = M_i(\alpha(x_i) - \alpha(c) + \alpha(c) - \alpha(x_{i-1})) \\ &= M_i \Delta \alpha_1 + M_i \Delta \alpha_2 \geq M_1^* \Delta \alpha_1 + M_2^* \Delta \alpha_2 \\ &\Rightarrow U(P, f, \alpha) \geq U(P^*, f, \alpha) \end{aligned}$$



Thm : relationship between lower and upper integrals

$$\int_a^b f d\alpha \leq \overline{\int_a^b f d\alpha}$$

▷ **proof** : By the Thm : refinement we proved that

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P, f, \alpha)$$

therefore,

$$\begin{aligned} L(P, f, \alpha) &\leq U(P, f, \alpha) \\ \Rightarrow \sup_P L(P, f, \alpha) &\leq \inf_P U(P, f, \alpha) \\ \Rightarrow \int_a^b f d\alpha &\leq \overline{\int_a^b f d\alpha} \end{aligned}$$

★ **Thm : equivalent representation of integrable**

It is equivalent that the function f is Riemann-Stieltjes integrable for α in $[a, b]$:

$$f \in \mathcal{R}(\alpha) \iff \forall \varepsilon > 0, \exists P \text{ such that } U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

▷ **proof :** ($\Rightarrow$)

$$f \in \mathcal{R}(\alpha) \iff \int_a^b f d\alpha = \overline{\int_a^b f d\alpha} \iff \int_a^b f d\alpha - \overline{\int_a^b f d\alpha} < \varepsilon \text{ And if we let } \varepsilon \text{ like that for}$$

$$L(P_1, f, \alpha) > \int_a^b f d\alpha - \frac{\varepsilon}{3}$$

$$U(P_2, f, \alpha) < \overline{\int_a^b f d\alpha} + \frac{\varepsilon}{3}$$

for, $P = P_1 \cup P_2$ by above thm we can know

$$L(P_1, f, \alpha) \leq L(P, f, \alpha)$$

$$U(P_2, f, \alpha) \geq U(P, f, \alpha)$$

$$\Rightarrow U(P, f, \alpha) - L(P, f, \alpha) \leq U(P_2, f, \alpha) - L(P_1, f, \alpha)$$

$$= \left[U(P_2, f, \alpha) - \overline{\int_a^b f d\alpha} \right] + \left[\int_a^b f d\alpha - L(P_1, f, \alpha) \right] < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \frac{2}{3}\varepsilon < \varepsilon$$

▷ **proof :** ($\Leftarrow$)

Suppose that $\forall \varepsilon > 0, \exists P_{1,2}$ such that $\overline{\int} f d\alpha - U(P_1, f, \alpha) < \varepsilon/3$, $L(P_2, f, \alpha) - \underline{\int} f d\alpha < \varepsilon/3$
Then with the assumption

$$\overline{\int} f d\alpha - U(P_1, f, \alpha) + U(P, f, \alpha) - L(P, f, \alpha) + \underline{\int} f d\alpha < \varepsilon$$

$$\Rightarrow U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

According to the above theorem, increasing the partition reduces U and increases L

$$U(P^*, f, \alpha) - L(P^*, f, \alpha) \leq \varepsilon$$

For here, $P^* = P_1 \cup P_2$

$$L(P^*, f, \alpha) \leq \int_a^b f d\alpha = \sup L(P, f, \alpha)$$

$$U(P^*, f, \alpha) \geq \overline{\int_a^b f d\alpha} = \inf U(P, f, \alpha)$$

$$\Rightarrow \overline{\int} f d\alpha - \underline{\int} f d\alpha \leq U - L < \varepsilon \Rightarrow \overline{\int} f d\alpha = \underline{\int} f d\alpha \Rightarrow f \in \mathcal{R}(\alpha)$$

■

Thm : Various properties of integrable

- If $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds for some partition P and some ε , then $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ also holds for every refinement of P .

▷ **proof** : We proved that

$$\begin{aligned} L(P, f, \alpha) &\leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P, f, \alpha) \\ \Rightarrow U(P^*, f, \alpha) - L(P^*, f, \alpha) &\leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon \\ \Rightarrow U(P^*, f, \alpha) - L(P^*, f, \alpha) &< \varepsilon \end{aligned}$$

■

- Let's assume that for $P = \{x_0, x_1, \dots, x_n\}$, the condition $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds. Additionally, for each i , if we arbitrarily choose two points $s_i, t_i \in [x_{i-1}, x_i]$, it follows that $\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon$.

▷ **proof** : For every partition, $[x_{i-1}, x_i]$. we can generally define $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$ therefore, this indicate that $M_i - m_i \geq |f(s_i) - f(t_i)|$ and by the definition of the Upper sum and Lower sum which is :

$$\begin{aligned} U(P, f, \alpha) &= \sum_{i=1}^n M_i \cdot \Delta\alpha_i \\ L(P, f, \alpha) &= \sum_{i=1}^n m_i \cdot \Delta\alpha_i \\ \Rightarrow U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \cdot \Delta\alpha_i < \varepsilon \end{aligned}$$

From above, by using the fact that $|f(s_i) - f(t_i)| \leq M_i - m_i$, we can derive $|f(s_i) - f(t_i)| \cdot \Delta\alpha_i \leq (M_i - m_i) \cdot \Delta\alpha_i$ by combine these two,

$$\begin{aligned} \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i &\leq \sum_{i=1}^n (M_i - m_i) \cdot \Delta\alpha_i = U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon \\ \Rightarrow \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i &< \varepsilon \end{aligned}$$

■

- If $f \in \mathcal{R}(\alpha)$ (that is, f is Riemann-Stieltjes integrable with respect to α), and if the assumptions in above are satisfied, then $\left| \sum_{i=1}^n f(t_i) \Delta\alpha_i - \int_a^b f d\alpha \right| < \varepsilon$

▷ **proof** : Continue the upper thm condition, $f(t_i) \in [m_i, M_i]$ still holds then we can easilly dreive that

$$\begin{aligned} \sum m_i \Delta\alpha_i &\leq \sum f(t_i) \Delta\alpha_i \leq \sum M_i \Delta\alpha_i \\ \Rightarrow L(P, f, \alpha) &\leq \sum f(t_i) \Delta\alpha_i \leq U(P, f, \alpha) \end{aligned}$$

and $f \in \mathcal{R}(\alpha)$ indicate that $\exists I = \int_a^b f d\alpha \quad \forall \varepsilon > 0, \exists P \text{ s.t } U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ thus, $L(P, f, \alpha) \leq I \leq U(P, f, \alpha)$ also holds. Fianlly we can get this

$$\begin{aligned} \left| \sum f(t_i) \Delta\alpha_i - I \right| &\leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon \\ \left| \sum_{i=1}^n f(t_i) \Delta\alpha_i - \int_a^b f d\alpha \right| &< \varepsilon \end{aligned}$$



Thm : Continuous and Integrable

If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

▷ **proof** : Since f is continuous on $[a, b]$, it is uniformly continuous. That is, for any $\varepsilon > 0$, there exists a $\delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{\alpha(b) - \alpha(a)}$. to set-off the term $\Delta\alpha_j$ it would better choose $\varepsilon \Rightarrow \frac{\varepsilon}{\alpha(b) - \alpha(a)}$

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \Delta\alpha_i \\ &< \frac{\varepsilon}{\alpha(b) - \alpha(a)} \cdot \sum_{i=1}^n \Delta\alpha_i = \frac{\varepsilon}{\alpha(b) - \alpha(a)} \cdot (\alpha(b) - \alpha(a)) = \varepsilon \end{aligned}$$

Thm : Continuous and Integrable

If f is monotonic on $[a, b]$, and α is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

▷ **proof** : Fix a positive number $\varepsilon > 0$. For a positive integer n , choose a partition $P = \{x_0, x_1, \dots, x_n\}$ with uniform α -length.

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \frac{\alpha(b) - \alpha(a)}{n} \quad (i = 1, \dots, n)$$

Since f is monotonically increasing, in each interval $[x_{i-1}, x_i]$

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \cdot \Delta\alpha_i \\ &= \sum_{i=1}^n (f(x_i) - f(x_{i-1})) \cdot \frac{\alpha(b) - \alpha(a)}{n} \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \end{aligned}$$

since $\sum_{i=1}^n [f(x_i) - f(x_{i-1})] = f(b) - f(a)$ therefore,

$$U(P, f, \alpha) - L(P, f, \alpha) = \frac{\alpha(b) - \alpha(a)}{n} \cdot [f(b) - f(a)]$$

for enough large n we can get

$$\frac{\alpha(b) - \alpha(a)}{n} \cdot [f(b) - f(a)] < \varepsilon$$



★Thm: Discontinuous and Integrable

f is bounded on $[a, b]$, and has only finitely many points of discontinuity on $[a, b]$, α is continuous at every point where f is discontinuous. Then $f \in \mathcal{R}(\alpha)$, i.e., f is Riemann–Stieltjes integrable with respect to α .

▷ **proof** : We need to show that $\forall \varepsilon > 0$, choose appropriate P such that $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$ then $f \in \mathcal{R}(\alpha)$. Therefore, we will divide the integrable interval into partitions of continuous and discontinuous parts. First, let's consider the discontinuous part.

- **Case Discontinuous**) Let $\varepsilon > 0$ be given. Since there exists finite discontinuity points that $D = \{d_1, \dots, d_N\}$. By the assumption each $d_k \in D$ α is continuous. Therefore we can define this :

$$\forall \varepsilon > 0, \exists \delta_k > 0 \text{ such that } |\alpha(x) - \alpha(d_k)| < \frac{\varepsilon}{4MN} \text{ for all } |x - d_k| < \delta_k$$

For each d_k following holds

$$\begin{aligned} \alpha(d_k + \delta_k) - \alpha(d_k - \delta_k) &= [\alpha(d_k + \delta_k) - \alpha(d_k)] + [\alpha(d_k) - \alpha(d_k - \delta_k)], \\ |\alpha(d_k + \delta_k) - \alpha(d_k)| &< \frac{\varepsilon}{4MN}, \quad |\alpha(d_k - \delta_k) - \alpha(d_k)| < \frac{\varepsilon}{4MN} \\ \Rightarrow \alpha(d_k + \delta_k) &< \alpha(d_k) + \frac{\varepsilon}{4MN}, \quad \alpha(d_k - \delta_k) > \alpha(d_k) - \frac{\varepsilon}{4MN} \end{aligned}$$

If subtract both inequality then we can get

$$\begin{aligned} \alpha(d_k + \delta_k) - \alpha(d_k - \delta_k) &< \left(\alpha(d_k) + \frac{\varepsilon}{4MN} \right) - \left(\alpha(d_k) - \frac{\varepsilon}{4MN} \right) = \frac{\varepsilon}{2MN} \\ \Rightarrow \alpha(d_k + \delta_k) - \alpha(d_k - \delta_k) &< \frac{\varepsilon}{2MN} \end{aligned}$$

We gather these intervals to define a small open set $U = \bigcup_{k=1}^N (d_k - \delta_k, d_k + \delta_k)$. Since f is bounded on each interval $[d_k - \delta_k, d_k + \delta_k]$, we have $M_k - m_k \leq 2M$, and the increase in α is $< \frac{\varepsilon}{2MN}$. Therefore, the difference between the upper and lower sums for each interval is:

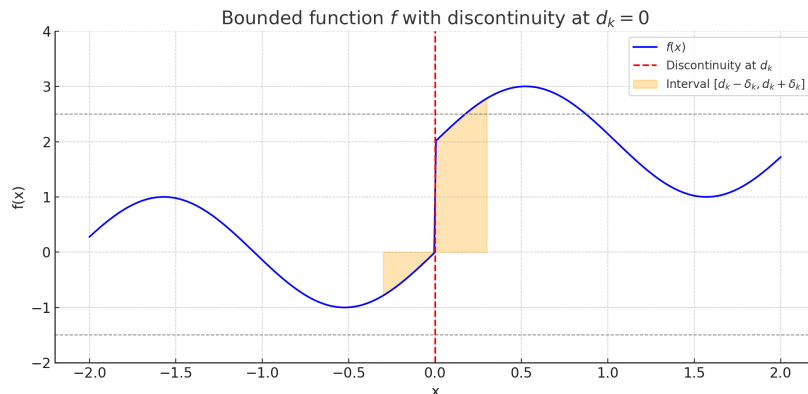
$$(M_k - m_k) \cdot (\alpha(d_k + \delta_k) - \alpha(d_k - \delta_k)) < 2M \cdot \frac{\varepsilon}{2MN} = \frac{\varepsilon}{N}$$

Here, the upper and lower sums can be expressed somewhat differently, with $M_k = \sup_{x \in [d_k - \delta_k, d_k + \delta_k]} f(x)$ and $m_k = \inf_{x \in [d_k - \delta_k, d_k + \delta_k]} f(x)$. Assuming that $f : [a, b] \rightarrow \mathbb{R}$ is bounded, we can write as follows.

$$\exists M > 0 \text{ such that } |f(x)| \leq M \text{ for all } x \in [a, b] \Rightarrow -M \leq f(x) \leq M \text{ for all } x \in [a, b]$$

Therefore, for any sub-interval $[d_k - \delta_k, d_k + \delta_k] \subset [a, b]$, $f(x) \leq M \Rightarrow \sup f(x) = M_k \leq M$ and $f(x) \geq -M \Rightarrow \inf f(x) = m_k \geq -M$ hold true. Thus,

$$M_k - m_k \leq M - (-M) = 2M$$



It can be said that, when combined with the results above,

$$(M_k - m_k)(\alpha(d_k + \delta_k) - \alpha(d_k - \delta_k)) \leq 2M \cdot \frac{\varepsilon}{2MN} = \frac{\varepsilon}{N}$$

$$\Rightarrow \sum_{k=1}^N \underbrace{(M_k - m_k)}_{\leq 2M} \cdot \underbrace{(\alpha(d_k + \delta_k) - \alpha(d_k - \delta_k))}_{< \frac{\varepsilon}{2MN}} < N \cdot \frac{\varepsilon}{N} = \varepsilon$$

Through this, we can conclude that the upper and lower sums for the discontinuous function are sufficiently small, and the interval is ultimately integrable.

- **Case Continuous)** Since f is continuous on $[a, b] \setminus U$, by the classical theorem, $f \in \mathcal{R}(\alpha)$ on $[a, b] \setminus U$. Therefore, there exists an appropriate partition P_{cont} such that $U(f, P_{\text{cont}}, \alpha) - L(f, P_{\text{cont}}, \alpha) < \frac{\varepsilon}{2}$
- **Sum of Cases)** Now, by combining the two partitions to form $P = P_{\text{cont}} \cup P_{\text{discont}}$,

$$U(f, P, \alpha) - L(f, P, \alpha) = [\text{Continuities}] + [\text{Discontinuities}] < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

$$\therefore \forall \varepsilon > 0, \exists P \text{ such that } U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon \Rightarrow f \in \mathcal{R}(\alpha)$$

■

★Thm : Composite functions' integrability

Suppose $f \in \mathcal{R}(\alpha)$ on $[a, b]$ $m \leq f \leq M$, ϕ is continuous on m, M , and $h(x) = \phi(f(x))$ on $[a, b]$. Then $h \in \mathcal{R}(\alpha)$ on $[a, b]$

▷ **proof:** we wish to show that for $h = \phi \circ f \Rightarrow h \in \mathcal{R}(\alpha)$. This indicate that

$$U(P, h, \alpha) - L(P, h, \alpha) = \sum_j (M_j^* - m_j^*) \Delta \alpha_j$$

$$\text{where : } M_j^* = \sup_{x \in [x_{j-1}, x_j]} h(x)$$

$$, m_j^* = \inf_{x \in [x_{j-1}, x_j]} h(x)$$

Since ϕ is uniformly continuous ϕ can be written as :

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \varepsilon$$

Here we choose $\varepsilon > 0$, and pick $\delta > 0$ such that $\delta < \varepsilon$, and we can express it as follows using the definition of integrability of f .

$$f \in \mathcal{R}(\alpha) \Rightarrow \exists P = \{x_0, \dots, x_n\} \text{ such that } \sum_{j=1}^n (M_j - m_j) \Delta \alpha_j < \delta^2$$

To use above definition let's divide the Upper sum minus lower sum with two separate parts such as :

$$j \in A : M_i - m_i < \delta$$

$$j \in B : M_i - m_i \geq \delta \quad A = B^c$$

- For $j \in A$ indicate that $M_i - m_i < \delta$ and since ϕ 's definition $M_i - m_i < \delta \Rightarrow \phi(M_j) - \phi(m_j) = M_j^* - m_j^* < \varepsilon$ $\sum_{j \in A} (M_j^* - m_j^*) \Delta \alpha_j \leq \sum_{j \in A} \varepsilon \Delta \alpha_j \leq \varepsilon(\alpha(b) - \alpha(a))$

- For $i \in B$ $(M_j - m_j) \geq \delta \Rightarrow \delta \leq (M_j - m_j) \Rightarrow \Delta\alpha_j \leq \frac{(M_j - m_j)}{\delta} \Delta\alpha_j$. Therefore, with this fact we can derive that. $\sum_{j \in B} \Delta\alpha_j \leq \frac{1}{\delta} \sum_{j \in B} (M_j - m_j) \Delta\alpha_j$ And the sum of the two sets clearly results in the following. $\sum_{j \in B} (M_j - m_j) \Delta\alpha_j \leq \sum_{j \in A \cup B} (M_j - m_j) \Delta\alpha_j$

$$\begin{aligned} \sum_{j \in B} \Delta\alpha_j &\leq \frac{1}{\delta} \sum_{j \in A} (M_j - m_j) \Delta\alpha_j \leq \frac{1}{\delta} \sum_{j \in A \cup B} (M_j - m_j) \Delta\alpha_j \leq \frac{\delta^2}{\delta} = \delta < \varepsilon \\ \Rightarrow \sum_{j \in B} \Delta\alpha_j &< \varepsilon \end{aligned}$$

And we can see that ϕ is bounded on the closed interval through continuity, and ultimately we can conclude that its composite function h satisfies the following condition.
 $|h(x) = \phi(f(x))| \leq K \Rightarrow h(x) = \phi(f(x)) \in [-K, K]$

$$\begin{aligned} M_j^* &= \sup \phi(f(x)) \leq K, \quad m_j^* = \inf \phi(f(x)) \geq -K \\ \Rightarrow M_j^* - m_j^* &\leq K - (-K) = 2K \\ \Rightarrow M_j^* - m_j^* &\leq 2K \end{aligned}$$

By this result we can derive :

$$\begin{aligned} \sum_{j \in B} (M_j^* - m_j^*) \Delta\alpha_j &\leq 2K \cdot \sum_{j \in B} \Delta\alpha_j < 2K \cdot \varepsilon < \varepsilon \\ \Rightarrow \sum_{j \in B} (M_j^* - m_j^*) \Delta\alpha_j &\leq \varepsilon \end{aligned}$$

If we combine with these two cases then, we can know that :

$$\begin{aligned} U(P, h, \alpha) - L(P, h, \alpha) &= \sum_{i \in A} (M_i^* - m_i^*) \Delta\alpha_i + \sum_{i \in B} (M_i^* - m_i^*) \Delta\alpha_i < \varepsilon(\alpha(b) - \alpha(a)) + 2K\varepsilon < \varepsilon \end{aligned}$$

■

Properties of Integral

Basic Properties of Integral

Thm : linearity

If $f_1, f_2 \in \mathcal{R}(\alpha)$ on $[a, b]$, then their sum $f_1 + f_2$ also belongs to $\mathcal{R}(\alpha)$, and for any constant $c \in \mathbb{R}$, the function cf belongs to $\mathcal{R}(\alpha)$ as well. Furthermore, the Riemann-Stieltjes integral is linear, meaning that $\int_a^b (f_1 + f_2) d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha$, and $\int_a^b cf d\alpha = c \int_a^b f d\alpha$.

1. $f_1 + f_2 \in \mathcal{R}(\alpha)$

▷ **proof** : Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. then $[x_{i-1}, x_i]$ $M_i^{(j)} := \sup_{x \in [x_{i-1}, x_i]} f_j(x)$ and $m_i^{(j)} := \inf_{x \in [x_{i-1}, x_i]} f_j(x)$, $j = 1, 2$ holds and by the definition of

Supremum and infimum following inequalities hold :

$$\begin{aligned}
\sup(f_1 + f_2) &\leq \sup f_1 + \sup f_2 = M_i^{(1)} + M_i^{(2)} \\
\inf(f_1 + f_2) &\geq \inf f_1 + \inf f_2 = m_i^{(1)} + m_i^{(2)} \\
\Rightarrow U(P, f_1 + f_2, \alpha) &\leq U(P, f_1, \alpha) + U(P, f_2, \alpha), \\
L(P, f_1 + f_2, \alpha) &\geq L(P, f_1, \alpha) + L(P, f_2, \alpha) \\
\Rightarrow U(P, f_1 + f_2, \alpha) - L(P, f_1 + f_2, \alpha) \\
&\leq [U(P, f_1, \alpha) - L(P, f_1, \alpha)] + [U(P, f_2, \alpha) - L(P, f_2, \alpha)] \\
\Rightarrow f_1 + f_2 &\in \mathcal{R}(\alpha)
\end{aligned}$$

■

2. $cf \in \mathcal{R}(\alpha)$

▷ **proof** : let $f \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$ then for the partition $P = \{x_0, \dots, x_n\}$ and interval $[x_{i-1}, x_i]$ following can be changed $M_i := \sup_{x \in [x_{i-1}, x_i]} f(x)$, $m_i := \inf_{x \in [x_{i-1}, x_i]} f(x) \Rightarrow M_i^c := \sup cf(x)$, $m_i^c := \inf cf(x)$ Therefore, for the subinterval Upper sum and Lower sum can be : $M_i^c = cM_i$, $m_i^c = cm_i \Rightarrow M_i^c - m_i^c = c(M_i - m_i)$ And this term also change this

$$\begin{aligned}
U(P, cf, \alpha) &= \sum_{i=1}^n M_i^{(c)} \Delta\alpha_i = c \sum_{i=1}^n M_i \Delta\alpha_i = cU(P, f, \alpha), \\
L(P, cf, \alpha) &= \sum_{i=1}^n m_i^{(c)} \Delta\alpha_i = c \sum_{i=1}^n m_i \Delta\alpha_i = cL(P, f, \alpha) \\
\Rightarrow U(P, cf, \alpha) - L(P, cf, \alpha) &= c[U(P, f, \alpha) - L(P, f, \alpha)]
\end{aligned}$$

since $f \in \mathcal{R}(\alpha)$, $\forall \varepsilon > 0$,

$$\begin{aligned}
\exists P \text{ such that } U(P, f, \alpha) - L(P, f, \alpha) &< \frac{\varepsilon}{|c|} \Rightarrow U(P, cf, \alpha) - L(P, cf, \alpha) < \varepsilon \\
\Rightarrow cf &\in \mathcal{R}(\alpha)
\end{aligned}$$

■

Thm : monotonicity

If $f_1(x) \leq f_2(x)$ for all $x \in [a, b]$, and both $f_1, f_2 \in \mathcal{R}(\alpha)$, then the Riemann-Stieltjes integrals satisfy the inequality $\int_a^b f_1 d\alpha \leq \int_a^b f_2 d\alpha$. That is, if one function is pointwise smaller than the other, then its integral with respect to the same integrator function α is also less than or equal to the other's.

▷ **proof** : If $f_1(x) \leq f_2(x)$ for all $x \in [a, b]$, and $f_1, f_2 \in \mathcal{R}(\alpha)$, let $g(x) := f_2(x) - f_1(x) \geq 0$ then $g(x)$ can be defined in $[a, b]$ and $g(x) \geq 0$. And above with linearity of integral we did show that $f_1, f_2 \in \mathcal{R}(\alpha) \Rightarrow g = f_2 - f_1 \in \mathcal{R}(\alpha)$ Now we wish to show that for $g(x) \geq 0$ and $g \in \mathcal{R}(\alpha)$, $\int_a^b g(x) d\alpha(x) \geq 0$ under that $U(P, g, \alpha) - L(P, g, \alpha) < \varepsilon$. No matter M or m we can show this, $L(P, g, \alpha) = \sum_{i=1}^n m_i \cdot (\alpha(x_i) - \alpha(x_{i-1})) \geq 0$ where, $m_i := \inf\{g(x) \mid x \in [x_{i-1}, x_i]\} \geq 0$ therefore, $\int_a^b g(x) d\alpha(x) \geq 0 \Rightarrow \int_a^b f_1(x) d\alpha(x) \leq \int_a^b f_2(x) d\alpha(x)$. holds for $f_1(x) \leq f_2(x) \quad \forall x \in [a, b]$. ■

Thm : Segmentation

If a function $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $a < c < b$, then $f \in \mathcal{R}(\alpha)$ on both $[a, c]$ and $[c, b]$, and the integral can be split as $\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$.

▷ **proof** : $f \in \mathcal{R}(\alpha)$ on $[a, b]$ indicate that $\varepsilon > 0$ $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ holds and lets divide the partition to match the assumption for $a < c < b$ we have $P = P_1 \cup P_2$ for here, $P_1 = [a, c]$ and $P_2 = [c, b]$. And by definition we can derive that

$$\begin{aligned} U(P, f, \alpha) &= U(P_1, f, \alpha) + U(P_2, f, \alpha), \quad L(P, f, \alpha) = L(P_1, f, \alpha) + L(P_2, f, \alpha) \\ \Rightarrow U(P, f, \alpha) - L(P, f, \alpha) &= (U(P_1, f, \alpha) - L(P_1, f, \alpha)) + (U(P_2, f, \alpha) - L(P_2, f, \alpha)) < \varepsilon \\ \Rightarrow U(P_i, f, \alpha) - L(P_i, f, \alpha) &< \varepsilon \quad i = 1, 2 \end{aligned}$$

since we actually divided the area of assumption we again consider this

$$\begin{aligned} \sum_{\forall x_i}^n f(x_i) \Delta \alpha_i &= \sum_{x_i \in [a, c]} f(x_i) \Delta \alpha_i + \sum_{x_i \in [c, b]} f(x_i) \Delta \alpha_i \\ \Rightarrow \int_a^b f d\alpha &= \int_a^c f d\alpha + \int_c^b f d\alpha \end{aligned}$$

■

Thm : Boundedness

If $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $|f(x)| \leq M$ for all $x \in [a, b]$, then $\left| \int_a^b f d\alpha \right| \leq M[\alpha(b) - \alpha(a)]$.

▷ **proof** : For all $x \in [a, b]$, $|f(x)| \leq M$ implies, $-M \leq f(x) \leq M \quad \forall x \in [a, b]$. Using the properties of linearity and monotonic, we can get this:

$$-M \leq f(x) \leq M \Rightarrow \int_a^b -M d\alpha \leq \int_a^b f(x) d\alpha \leq \int_a^b M d\alpha$$

for here, each integral indicate that

$$\int_a^b M d\alpha = M[\alpha(b) - \alpha(a)], \quad \int_a^b (-M) d\alpha = -M[\alpha(b) - \alpha(a)]$$

therefore,

$$\begin{aligned} \int_a^b -M d\alpha &\leq \int_a^b f(x) d\alpha \leq \int_a^b M d\alpha \\ \Rightarrow -M[\alpha(b) - \alpha(a)] &\leq \int_a^b f(x) d\alpha \leq M[\alpha(b) - \alpha(a)] \\ \Rightarrow \left| \int_a^b f(x) d\alpha \right| &\leq M[\alpha(b) - \alpha(a)] \end{aligned}$$

■

Thm : Linearity in function

If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and $\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2$.
Also, if $f \in \mathcal{R}(\alpha)$ and $c > 0$, then $f \in \mathcal{R}(c\alpha)$ and $\int_a^b f d(c\alpha) = c \int_a^b f d\alpha$.

1. $\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2$
▷ **proof** : Consider $\alpha := \alpha_1 + \alpha_2$ then

$$\begin{aligned}\alpha(x_i) - \alpha(x_{i-1}) &= (\alpha_1 + \alpha_2)(x_i) - (\alpha_1 + \alpha_2)(x_{i-1}) \\ &= [\alpha_1(x_i) - \alpha_1(x_{i-1})] + [\alpha_2(x_i) - \alpha_2(x_{i-1})]\end{aligned}$$

applying this to the above definition of intergral then we can get

$$\begin{aligned}U(P, f, \alpha) &= \sum_{i=1}^n M_i[\alpha(x_i) - \alpha(x_{i-1})] = \sum_{i=1}^n M_i \{[\alpha_1(x_i) - \alpha_1(x_{i-1})] + [\alpha_2(x_i) - \alpha_2(x_{i-1})]\} \\ &= \sum_{i=1}^n M_i[\alpha_1(x_i) - \alpha_1(x_{i-1})] + \sum_{i=1}^n M_i[\alpha_2(x_i) - \alpha_2(x_{i-1})] \\ &= U(P, f, \alpha_1) + U(P, f, \alpha_2)\end{aligned}$$

It is sufficient to show that $= \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2$

2. $\int_a^b f d(c\alpha) = c \int_a^b f d\alpha, \quad c > 0$
▷ **proof** : This process so analogues with previous one. Consider agian this:

$$U(P, f, \alpha) = \sum_{i=1}^n M_i[\alpha(x_i) - \alpha(x_{i-1})], \quad L(P, f, \alpha) = \sum_{i=1}^n m_i[\alpha(x_i) - \alpha(x_{i-1})]$$

for $c\alpha$ we have

$$(c\alpha)(x_i) - (c\alpha)(x_{i-1}) = c[\alpha(x_i) - \alpha(x_{i-1})]$$

With this we can transform the Upper sum of Lower sum, It also sufficient to consider one side.

$$\begin{aligned}U(P, f, c\alpha) &= \sum_{i=1}^n M_i[c(\alpha(x_i) - \alpha(x_{i-1}))] \\ &= c \sum_{i=1}^n M_i[\alpha(x_i) - \alpha(x_{i-1})] \\ &= c \cdot U(P, f, \alpha)\end{aligned}$$

■

Thm : Unit step function integral (X)

Let $\{c_n\}$ be a sequence of non-negative real numbers such that $\sum_{n=1}^{\infty} c_n < \infty$, and let $\{s_n\}$ be a sequence of distinct points in (a, b) . Define the integrator function as $\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$, where $I(x - s_n)$ is the unit step function: 0 if $x < s_n$, and 1 if $x \geq s_n$. If f is continuous on $[a, b]$, then $\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

▷ **proof** : Each term $c_n I(x - s_n)$ has a jump at $x = s_n$, and the size of that jump is c_n . The sum $\sum c_n$ converges, so by the comparison test, $c_n I(x - s_n) \leq c_n$, which means $\sum_{n=1}^{\infty} c_n I(x - s_n)$ converges. In the end, we can obtain a monotonically increasing function as follows.

$$\alpha(a) = 0, \quad \alpha(b) = \sum_{n=1}^{\infty} c_n < \infty$$

$$\Rightarrow \exists N \text{ s.t. } \sum_{n=N+1}^{\infty} c_n < \varepsilon$$

Therefore, we divide the given integral function into parts that satisfy the above conditions and parts that do not.

$$\alpha(x) = \underbrace{\sum_{n=1}^N c_n I(x - s_n)}_{\alpha_1(x)} + \underbrace{\sum_{n=N+1}^{\infty} c_n I(x - s_n)}_{\alpha_2(x)}$$

α_1 increases only at a finite number of jump points $s_1, s_2, \dots, s_N$, with each jump size being $c_1, c_2, \dots, c_N$. Therefore, the expression can be represented as follows.

$$\int_a^b f d\alpha_1 = \sum_{n=1}^N f(s_n) \cdot (\alpha_1(s_n^+) - \alpha_1(s_n^-)) = \sum_{n=1}^N c_n f(s_n)$$

Here, $\alpha(s_n^-) := \lim_{x \rightarrow s_n^-} \alpha(x)$ and $\alpha(s_n^+) := \lim_{x \rightarrow s_n^+} \alpha(x)$ are defined. Considering this as a linear combination, we can draw the following conclusion.

$$\int_a^b f d(c_n I(x - s_n)) = c_n f(s_n)$$

$$\Rightarrow \int_a^b f d\left(\sum_{n=1}^N c_n I(x - s_n)\right) = \sum_{n=1}^N c_n f(s_n)$$

Next, considering $\alpha_2(b) - \alpha_2(a) = \sum_{n=N+1}^{\infty} c_n < \varepsilon$, since f is continuous, it is a bounded function, and let $M := \sup_{x \in [a, b]} |f(x)|$.

$$\left| \int_a^b f d\alpha_2 \right| \leq M \cdot (\alpha_2(b) - \alpha_2(a)) < M\varepsilon$$

Let's combine the two separate subsets using the linear combination property of integrals that we proved earlier.

$$\left| \int_a^b f d\alpha - \sum_{n=1}^N c_n f(s_n) \right| < M\varepsilon$$

And as $N \rightarrow \infty$

$$\sum_{n=1}^N c_n f(s_n) \rightarrow \sum_{n=1}^{\infty} c_n f(s_n)$$

$$\Rightarrow \int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$$

■

Thm : Special case of Stieltjes

Let α be a monotonically increasing function on $[a, b]$, and suppose $\alpha' \in \mathcal{R}$ (i.e., α' is Riemann integrable on $[a, b]$). Let f be a bounded real-valued function on $[a, b]$.

▷ **proof** : If bounded, α is monotonically increasing, $\alpha' \in \mathcal{R}$, $f\alpha' \in \mathcal{R}$. Therefore, we can say that $U(P, \alpha') - L(P, \alpha') < \varepsilon$. To show that the Riemann-Stieltjes integral and the Riemann integral are equal, it is sufficient to demonstrate that $|f(s_j)\Delta\alpha_j - f(s_j)\alpha'(s_j)\Delta x_j|$ are equal. Here, $\Delta\alpha_j = \alpha(x_j) - \alpha(x_{j-1})$ represents the integrative function, and by using the mean value theorem, we can derive the following result.

$$\frac{\alpha(x_j) - \alpha(x_{j-1})}{x_j - x_{j-1}} = \alpha'(t_j) \Rightarrow \alpha'(t_j)\Delta x_j \quad \text{for some } t_j \in (x_{j-1}, x_j)$$

As a result, the following relational expressions can be derived.

$$\begin{aligned} |f(s_j)\Delta\alpha_j - f(s_j)\alpha'(s_j)\Delta x_j| &= |f(s_j)\alpha'(t_j)\Delta x_j - f(s_j)\alpha'(s_j)\Delta x_j| \\ &= |f(s_j)(\alpha'(t_j) - \alpha'(s_j))| \cdot \Delta x_j \leq |f(s_j)| \cdot |\alpha'(t_j) - \alpha'(s_j)| \cdot \Delta x_j \end{aligned}$$

since $|f(s_j)| \leq M$ bounded

$$|f(s_j)| \cdot |\alpha'(t_j) - \alpha'(s_j)| \cdot \Delta x_j \leq M \cdot |\alpha'(t_j) - \alpha'(s_j)| \cdot \Delta x_j < M\varepsilon$$

Here, since $t_j \approx s_j \in [x_j, x_{j-1}]$, we can obtain a very small ε such that $< M\varepsilon$. Ultimately, by moving some parts of the left-hand side to the right-hand side, we obtain the following result.

$$\begin{aligned} \sum f(s_j)\Delta\alpha_j &\leq M\varepsilon + \sum f(s_j)\alpha'(s_j)\Delta x_j = M\varepsilon + U(P, f\alpha') \\ \Rightarrow U(P, f, \alpha) &\leq M\varepsilon + U(P, f\alpha') \end{aligned}$$

In the same way, you can handle the Lower sum part. $L(P, f\alpha') - M\varepsilon \leq L(P, f, \alpha)$ It's easy to get that. After all, the difference between the upper and lower ends is

$$U(P, f, \alpha) - L(P, f, \alpha) \leq U(P, f\alpha') - L(P, f\alpha') + 2M\varepsilon$$

From the above assumption, $U - L < \varepsilon$.

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon + 2M\varepsilon$$

in the end, it is clear that

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx$$

■

Change of Variable in Riemann–Stieltjes Integral

Let $\varphi : [A, B] \rightarrow [a, b]$ be a strictly increasing, continuous function such that $\varphi(A) = a, \varphi(B) = b$ and α is monotonically increasing on $[a, b]$, $f \in \mathcal{R}(\alpha)$ on $[a, b]$. Define $\beta(y) := \alpha(\varphi(y))$, $g(y) := f(\varphi(y))$ for $y \in [A, B]$. Then $g \in \mathcal{R}(\beta)$, and $\int_A^B g d\beta = \int_a^b f d\alpha$. Furthermore, we can express the partition in two forms as follows (utilizing the monotonicity property): $P = \{x_0, x_1, \dots, x_n\}$ can be transposed that $Q = \{y_0, y_1, \dots, y_n\} := \{\varphi^{-1}(x_0), \varphi^{-1}(x_1), \dots, \varphi^{-1}(x_n)\}$, where $Q = \varphi^{-1}(P)$. Thus, $x_j = \varphi(y_j)$, which allows us to see that $\Delta\alpha_j = \alpha(x_j) - \alpha(x_{j-1}) = \beta(y_j) - \beta(y_{j-1})$.

Through these conditions, we can discern the following fact about the upper sum.

$$\begin{aligned} M_j &= \sup_{x \in [x_{j-1}, x_j]} f(x) \\ \tilde{M}_j &= \sup_{y \in [y_{j-1}, y_j]} g(y) = \sup_{y \in [y_{j-1}, y_j]} f(\varphi(y)) \\ \therefore \sup_{y \in [y_{j-1}, y_j]} f(\varphi(y)) &= \sup_{x \in [x_{j-1}, x_j]} f(x) = M_j \end{aligned}$$

Now, by the definition of a superposition, let's check the following.

$$\begin{aligned} U(P, f, \alpha) &= \sum_{j=1}^n M_j \cdot (\alpha(x_j) - \alpha(x_{j-1})) \\ U(Q, g, \beta) &= \sum_{j=1}^n \tilde{M}_j \cdot (\beta(y_j) - \beta(y_{j-1})) \end{aligned}$$

For here, $\beta(y_j) = \alpha(\varphi(y_j)) = \alpha(x_j)$ we can easily notice that

$$U(P, f, \alpha) = U(Q, g, \beta)$$

The fundamental theorem of calculus

Thm : Fundamental Theorem of Calculus 1

Let $f \in \mathcal{R}$ on $[a, b]$. For $a \leq x \leq b$, define $F(x) = \int_a^x f(t) dt$. Then F is continuous on $[a, b]$.
If f is continuous at $x_0 \in [a, b]$,

1. F is differentiable at x_0

▷ **proof** : $f \in \mathcal{R}$ implies that f is bounded on $[a, b]$ then Let $|f(t)| \leq M$ then for any $a \leq x < y \leq b$

$$\begin{aligned} |F(y) - F(x)| &= \left| \int_a^y f(t) dt - \int_a^x f(t) dt \right| = \left| \int_x^y f(t) dt \right| \\ &\Rightarrow \left| \int_x^y f(t) dt \right| \leq \int_x^y |f(t)| dt \leq M(y - x) \end{aligned}$$

Therefore, for given $\varepsilon > 0$ following holds.

$$|y - x| < \varepsilon/M \Rightarrow |F(y) - F(x)| < \varepsilon$$

This result indicate that for all point in F is continuous. ■

2. $F'(x_0) = f(x_0)$

▷ **proof** : What we want to show is this :

$$F'(x_0) := \lim_{h \rightarrow 0} \frac{F(x_0 + h) - F(x_0)}{h} = f(x_0)$$

let $x_0 - \delta < s \leq x_0 \leq t < x_0 + \delta$ and we are going to consider the $\frac{F(t) - F(s)}{t - s}$. According to the definition of integral we can write this :

$$\frac{F(t) - F(s)}{t - s} = \frac{1}{t - s} \int_s^t f(u) du$$

And let's subtract $f(x_0)$ to reach out the result we are aiming.

$$\begin{aligned}
\left| \frac{F(t) - F(s)}{t - s} - f(x_0) \right| &= \left| \frac{F(t) - F(s)}{t - s} - \frac{f(x_0)(t - s)}{t - s} \right| \\
&= \left| \frac{1}{t - s} \int_s^t (f(u) - f(x_0)) du \right| \\
&\leq \frac{1}{t - s} \int_s^t |f(u) - f(x_0)| du
\end{aligned}$$

For here, we can use the continuous definition that $|f(u) - f(x_0)| < \varepsilon$ then ,

$$\begin{aligned}
\frac{1}{t - s} \int_s^t |f(u) - f(x_0)| du &\leq \frac{1}{t - s} \cdot \int_s^t \varepsilon du = \frac{1}{t - s} \cdot \varepsilon(t - s) = \varepsilon \\
\therefore \left| \frac{F(t) - F(s)}{t - s} - f(x_0) \right| &< \varepsilon \Rightarrow \lim_{t \rightarrow x_0} \frac{F(t) - F(x_0)}{t - x_0} = f(x_0)
\end{aligned}$$

■

Thm : Fundamental Theorem of Calculus 2

Let $f \in \mathcal{R}$ on $[a, b]$. If there exists a differentiable function F on $[a, b]$ such that $F'(x) = f(x)$, Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

▷ **proof** : Let's consider the partition such as $P = \{x_0 = a < x_1 < \dots < x_n = b\}$ and then use the mean value theorem which gives :

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1}) \quad \text{for some } c_i \in (x_{i-1}, x_i)$$

And if we add this up for all sections:

$$\sum_{i=1}^n f(c_i)(x_i - x_{i-1}) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = F(b) - F(a)$$

Left side indicate the Riemann sum.

$$\int_a^b f(x) dx = F(b) - F(a)$$

■

Rectifiable curves

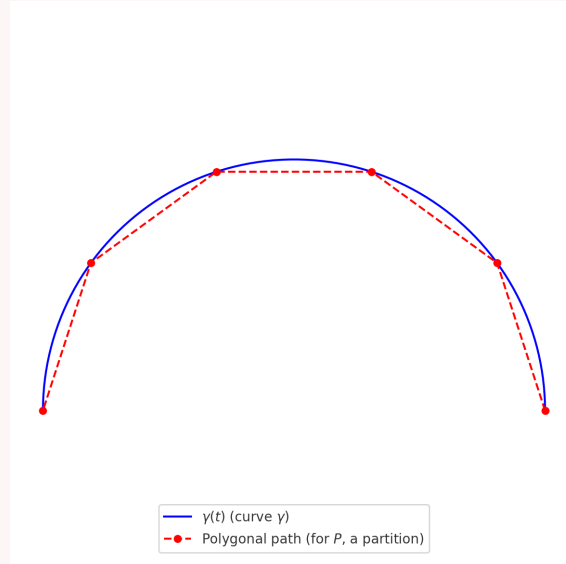
Def : Curve

let $\gamma : [a, b] \rightarrow \mathbb{R}^k$ A continuous function from the real interval $[a, b]$ to the space $\mathbb{R}^k$ is defined. This is a curve. And if γ is one-to-one then we call it arc and if $\gamma(a) = \gamma(b)$ end

point and start point are equivalent then we call it closed curve.

$$\Lambda(P, \gamma) = \sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})|$$

For here, Partition is $P = \{x_0, x_1, \dots, x_n\} \subset [a, b]$ and $\gamma(x_i) - \gamma(x_{i-1})$ indicate distance between to point in vector.



For every possible partition P , we take the ****supremum**** of $\Lambda(P, \gamma)$. This is because the finer the partition, the more accurately it follows γ . This supremum is defined as the length of the curve γ .

$$\Lambda(\gamma) = \sup_P \Lambda(P, \gamma)$$

And we call this rectifiable curve if the curve defined in finite interval $\gamma < \infty$

Thm : Curve is rectifiable and integrable (might be important?)

if $\gamma'(t)$ is continuous on $[a, b]$ then rectifiable and satisfy this :

$$\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$$

▷ **proof :**

($\Leftarrow$) suppose that $\gamma : [a, b] \rightarrow \mathbb{R}^k$, and γ' is continuous on $[a, b]$ this :

$$\exists \delta \text{ such that } |\gamma'(s) - \gamma'(t)| < \varepsilon, \text{ if } |s - t| < \delta$$

then let's consider the partition $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ and $|\gamma(x_i) - \gamma(x_{i-1})|$ by fundamental theorem calculus

$$\gamma(x_i) - \gamma(x_{i-1}) = \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| \leq \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt$$

Therefore, for the sum of the partition we can get :

$$\begin{aligned}\Lambda(P, \gamma) &= \sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt = \int_a^b |\gamma'(t)| dt \\ \Rightarrow \therefore \Lambda(\gamma) &= \sup_P \Lambda(P, \gamma) \leq \int_a^b |\gamma'(t)| dt\end{aligned}$$

($\Leftarrow$) Since γ' is uniformly continuous

$$\exists \delta > 0 \text{ such that } |s - t| < \delta \Rightarrow |\gamma'(s) - \gamma'(t)| < \varepsilon$$

And for the partition $P = \{x_0, x_1, \dots, x_n\}$, choose δ such that $\Delta x_i = x_i - x_{i-1} < \delta$ then we can apply the definition like this :

$$\forall t \in [x_{i-1}, x_i], \quad |t - x_i| < \delta \Rightarrow |\gamma'(t) - \gamma'(x_i)| < \varepsilon \Rightarrow |\gamma'(t)| < |\gamma'(x_i)| + \varepsilon$$

This implies that in $[x_{i-1}, x_i]$ following is always holds $|\gamma'(t)| \leq |\gamma'(x_i)| + \varepsilon$ Take integral both side then

$$\begin{aligned}\int_{x_{i-1}}^{x_i} |\gamma'(t)| dt &\leq \int_{x_{i-1}}^{x_i} (|\gamma'(x_i)| + \varepsilon) dt \\ &= (|\gamma'(x_i)| + \varepsilon)(x_i - x_{i-1}) \\ &= |\gamma'(x_i)| \Delta x_i + \varepsilon \Delta x_i\end{aligned}$$

And if we substitute this

$$\begin{aligned}|\gamma'(t)| &= |\gamma'(t) + \gamma'(x_i) - \gamma'(x_i)| \\ \Rightarrow \left| \int_{x_{i-1}}^{x_i} [\gamma'(t) + \gamma'(x_i) - \gamma'(x_i)] dt \right| + \varepsilon \Delta x_i &\leq \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| + \left| \int_{x_{i-1}}^{x_i} [\gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i\end{aligned}$$

And using fundamental theorem of Calculus

$$\left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| = |\gamma(x_i) - \gamma(x_{i-1})|$$

and since γ' is continuous

$$\left| \int_{x_{i-1}}^{x_i} [\gamma'(x_i) - \gamma'(t)] dt \right| < \left| \int_{x_{i-1}}^{x_i} \varepsilon dt \right| = \varepsilon \Delta x_i$$

Combine those then we can get

$$\int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \leq |\gamma(x_i) - \gamma(x_{i-1})| + 2\varepsilon \Delta x_i$$

Finally if we add up to all partition $[a, b]$

$$\begin{aligned}\int_a^b |\gamma'(t)| dt &= \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \\ &\leq \sum |\gamma(x_i) - \gamma(x_{i-1})| + 2\varepsilon(b - a) \\ &= \Lambda(P, \gamma) + 2\varepsilon(b - a)\end{aligned}$$

as $\varepsilon \rightarrow 0$ then

$$\int_a^b |\gamma'(t)| dt \leq \Lambda(\gamma)$$

From above we get that $\Lambda(\gamma) \leq \int_a^b |\gamma'(t)| dt$ and then $\Lambda(\gamma) \geq \int_a^b |\gamma'(t)| dt$. Thus, we can know that $\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$ ■

Sequences and series of Functions

Def : Uniform Convergence

A sequence $\{f_n\}$ uniformly converges to a function f on a set E if the following holds.

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, \forall x \in E, |f_n(x) - f(x)| \leq \varepsilon.$$

The key point here is that N depends only on ε and does not depend on x . In other words, uniform convergence requires that the rate of convergence can be controlled simultaneously for all $x \in E$. The difference from pointwise convergence is that for pointwise convergence, $N = N(\varepsilon, x)$, whereas for uniform convergence, the property of convergence is independent of x .

★Thm : Cauchy condition in uniform convergence

A sequence $\{f_n\}$ uniformly converges to a function f on a set $E \iff$:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N, \forall x \in E, |f_n(x) - f_m(x)| \leq \varepsilon.$$

▷ **proof :**

($\Rightarrow$)

By the assumption we can choose ε satisfy this :

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, \forall x \in E, |f_n(x) - f(x)| < \frac{\varepsilon}{2}$$

We wish to show that for this N the condition of this $|f_n(x) - f_m(x)| < \varepsilon$ holds. Therefore, we will change the given expression slightly :

$$\begin{aligned} |f_n(x) - f_m(x)| &= |f_n(x) - f(x) + f(x) - f_m(x)| \\ &\leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \\ &\therefore |f_n(x) - f_m(x)| < \varepsilon \end{aligned}$$

($\Leftarrow$)

With similar manner we can consume the ε such as :

$$\forall \varepsilon > 0, \exists N \text{ such that } \forall m, n \geq N, \forall x \in E, |f_n(x) - f_m(x)| \leq \frac{\varepsilon}{2}$$

Since $\mathbb{R}$ is a complete space, Cauchy sequences must converge. Thus, $\{f_n(x)\}$ converges in $\mathbb{R}$, and we define its limit as $f(x)$. Now, to summarize again,

$$|f_n(x) - f(x)| \leq |f_n(x) - f_m(x)| + |f_m(x) - f(x)|$$

By using this, taking $m \rightarrow \infty$ results in

$$\begin{aligned} |f_n(x) - f_m(x)| + |f_m(x) - f(x)| &< \frac{\varepsilon}{2} + \delta, \forall \delta > 0 \\ \therefore |f_n(x) - f(x)| &< \varepsilon \end{aligned}$$

Since we can choose $\delta > 0$ (δ Comes from the $\{f_m(x)\} \rightarrow f(x)$) to be small, the entire thing can be made less than ε . ■

Thm : Uniform consequence another criteria (X)

Let $\{f_n\}$ be a sequence of functions defined on a set E , and suppose $\forall x \in E, \lim_{n \rightarrow \infty} f_n(x) = f(x)$. And define $M_n := \sup_{x \in E} |f_n(x) - f(x)|$. Then

$$f_n \rightarrow f \text{ uniformly on } E \iff \lim_{n \rightarrow \infty} M_n = 0.$$

▷ **proof :**

($\Rightarrow$)

By definition, we know that

$$\forall \varepsilon > 0, \exists N \text{ such that } \forall n \geq N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$$

and assumed that $M_n = \sup_{x \in E} |f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ therefore, $\lim_{n \rightarrow \infty} M_n = 0$

($\Leftarrow$)

Suppose $\lim_{n \rightarrow \infty} M_n = 0$ then by the definition,

$$\forall \varepsilon > 0, \exists N \text{ such that } \forall n \geq N, M_n = \sup_{x \in E} |f_n(x) - f(x)| < \varepsilon$$

then, $\forall x \in E, |f_n(x) - f(x)| \leq M_n < \varepsilon$ for all $n \geq N$ therefore, $f_n \rightarrow f$ uniformly on E ■

★ Thm : interchangeable Limit

Let $\{f_n\}$ be a sequence of functions on a set E in a metric space, and suppose $f_n \rightarrow f$ uniformly on E , x is a limit point of E and For each n , the limit $A_n := \lim_{t \rightarrow x} f_n(t)$ exists. Then,

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$$

▷ **proof :**

We will first show that A_n converges by demonstrating that the space of the given operation is real and that A_n is a Cauchy sequence. We will then prove that $\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n = A$, thereby deriving the fact that $\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$.

1. let $\varepsilon > 0$ and there exists N such that $n \leq N$ and wish to show that $|A_n - A_m| < \varepsilon$ to reach out the definition of Cauchy sequence. To use triangular inequality change the expression slightly

$$\begin{aligned} |A_n - A_m| &= |A - f_n(x) + f_n(x) - f_m(x) + f_m(x) - A_m| \\ &\leq |A - f_n(x)| + |f_n(x) - f_m(x)| + |f_m(x) - A_m| \end{aligned}$$

we can take limits on x since f is uniformly consequence and A also independent to x therefore,

$$\begin{aligned} \lim_{x \rightarrow a} |A_n - A_m| &\leq \lim_{x \rightarrow a} (|A - f_n(x)| + |f_n(x) - f_m(x)| + |f_m(x) - A_m|) \\ &< 0 + \varepsilon + 0 = \varepsilon \\ \therefore \lim_{x \rightarrow a} |A_n - A_m| < \varepsilon &\Rightarrow |A_n - A_m| < \varepsilon \Rightarrow |A_n - A| < \varepsilon \end{aligned}$$

Above process ultimately indicate this $A_n \rightarrow A$. Next, we are going to show that $\lim_{t \rightarrow x} f(t) = A$

2. let $\varepsilon > 0$, and with similar manner we can get this :

$$\begin{aligned} |f(x) - A| &= |f(x) - f_n(x) + f_n(x) - A_n + A_n - A| \\ &\leq |f(x) - f_n(x)| + |f_n(x) - A_n| + |A_n - A| \end{aligned}$$

Since $f_n \rightarrow f$ uniformly, $\exists N_1$ such that $\forall n \geq N_1, \forall x \in E, |f_n(t) - f(t)| < \varepsilon/3$. And we've show that $|A_n - A| < \varepsilon \approx \frac{\varepsilon}{3}$. By the given condition that $\lim_{x \rightarrow a} f_n(x) \Rightarrow \exists \delta_n > 0$ such that $0 < d(x, a) < \delta_n \Rightarrow |f_n(x) - A_n| < \varepsilon/3$. Finally as above $A_n \rightarrow A \Rightarrow \exists N_2$ such that $\forall n \geq N_2, |A_n - A| < \varepsilon/3$. Therefore If we choose $n \geq \max(N_1, N_2)$ and take $x \in E$ such that $0 < d(x, a) < \delta_n$ then

$$\begin{aligned} |f(t) - A| &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \\ \therefore \lim_{t \rightarrow x} f(t) = A &= \lim_{n \rightarrow \infty} A_n \Rightarrow \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t) \end{aligned}$$

■

★ Thm : Dini's Theorem

Let K be a compact set. Suppose that $\{f_n\}$ is a sequence of continuous functions on K $f_n \rightarrow f$ pointwise on K , and f is continuous and $f_n(x) \geq f_{n+1}(x), \forall x \in K, \forall n$ which means The sequence is monotonically decreasing Then $f_n \rightarrow f$ **uniformly** on K . The above theorem states that if a sequence of functions satisfies pointwise convergence, it can be strengthened to uniform convergence under additional conditions.

▷ **proof** : We prove that if the intersection of finitely many decreasing sequences of compact sets is not empty, then the whole set of compact sets is not empty. That is, we develop the proof by assuming the negation of uniform convergence to derive a contradiction. First, if it is not uniformly convergent, the following holds: There exists $\varepsilon > 0$ such that for all $n \in \mathbb{N}$, there exists $x_n \in K$ such that $f_n(x_n) - f(x_n) \geq \varepsilon$. In other words, this means that the following sets are not empty:

$$K_n := \{x \in K \mid f_n(x) - f(x) = g(x) \geq \varepsilon\} = K \cap g_n^{-1}([\varepsilon, \infty)) \neq \emptyset$$

Here, K_n is a compact set because K is both a compact set and a closed set, and g_n^{-1} is also a closed set, so the intersection of the two sets is a compact set. Moreover, according to the given conditions, K_n is a decreasing sequence of sets, and based on the given decreasing function condition,

$$\begin{aligned} f_n(x) \geq f_{n+1}(x) &\Rightarrow f_n(x) - f(x) \geq f_{n+1}(x) - f(x) \\ &\Rightarrow K_{n+1} \subseteq K_n \end{aligned}$$

Pointwise convergence implies $\bigcap_{n=1}^{\infty} K_n = \emptyset$ because $f_n(x) \rightarrow f(x)$, so for each $x \in K$

$$f_n(x) - f(x) \rightarrow 0 \Rightarrow \exists N_x \text{ such that } f_n(x) - f(x) < \varepsilon, \forall n \geq N_x$$

In other words, each x eventually leaves some K_n . Since $x \notin \bigcap_n K_n$, the finite intersection of this set ultimately becomes empty. $\bigcap_{n=1}^{\infty} K_n = \emptyset$. If the intersection of the decreasing closed sets K_n is empty, it must be empty beyond a certain stage. $\exists N \in \mathbb{N}$ such that $K_N = \emptyset$

$$\begin{aligned} \forall x \in K, f_N(x) - f(x) &< \varepsilon \\ \Rightarrow f_n \leq f_N &\Rightarrow f_n(x) - f(x) \leq f_N(x) - f(x) < \varepsilon \\ \Rightarrow \forall n \geq N, \forall x \in K, &|f_n(x) - f(x)| < \varepsilon \end{aligned}$$

Def : supremum norm and metric

When X is a metric space, $\mathcal{C}(X)$ is the set of functions that satisfy the following condition.

$$\mathcal{C}(X) = \{f : X \rightarrow \mathbb{R} \mid f : \text{Continuous and bounded}\}$$

For each function $f \in \mathcal{C}(X)$, we define the norm as follows:

$$\|f\| := \sup_{x \in X} |f(x)|$$

We define the distance between two functions $f, g \in \mathcal{C}(X)$ as follows:

$$d(f, g) := \|f - g\| = \sup_{x \in X} |f(x) - g(x)|$$

This definition satisfies the axioms of a metric space (3 properties: symmetry, identity, and triangle inequality).

Thm : complete metric space

The metric defined by the supremum norm makes $\mathcal{C}(X)$ into a complete metric space.

▷ **proof** : Assume that $\{f_n\} \subset \mathcal{C}(X)$ is a Cauchy sequence with respect to the sup-norm.

$$\forall \varepsilon > 0, \exists N \text{ such that } \forall m, n \geq N, d(f_n, f_m) = \sup_{x \in X} |f_n(x) - f_m(x)| < \varepsilon$$

For a fixed $x \in X$, $\{f_n(x)\} \subset \mathbb{C}$ is a Cauchy sequence, and since $\mathbb{C}$ is a complete space, it converges to $f(x) \in \mathbb{C}$. We actually need to show that $\|f_n - f\| = \sup_{x \in X} |f_n(x) - f(x)| \rightarrow 0$. As is often the case, we will achieve the desired result by slightly modifying the equation.

$$\begin{aligned} |f_n(x) - f(x)| &\leq |f_n(x) - f_m(x)| + |f_n(x) - f_m(x)| < \varepsilon + |f_m(x) - f(x)| \\ |f_n(x) - f(x)| &< \varepsilon \end{aligned}$$

Now, as $m \rightarrow \infty$, we have $f_m(x) \rightarrow f(x)$ by the definition of pointwise limit. We have shown that $f_n \rightarrow f$ uniformly, but finally we need to verify that the limit function f is indeed continuous, meaning $f \in \mathcal{C}(X)$. Therefore, the completeness of the space $\mathcal{C}(X)$ is established. If we list the conditions we have, it can be easily shown.

$$\begin{aligned} f_n &\Rightarrow f \Rightarrow \\ \forall \varepsilon > 0, \exists N \text{ such that } \forall x, |f_n(x) - f(x)| &< \varepsilon/3 \quad \forall n \geq N, \\ f_n \in \mathcal{C}(X) &\rightarrow \exists \delta > 0 \text{ such that } |x - y| < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon/3 \end{aligned}$$

Now, for any $x, y \in X$, if $|x - y| < \delta$:

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon \end{aligned}$$

■

★Thm : interchangeable integral

Let α be monotonically increasing in $[a, b]$. Suppose $f_n \in \mathcal{R}(\alpha)$ on $[a, b]$, for $n = 1, 2, 3, \dots$, and suppose $f_n \rightarrow f$ uniformly on $[a, b]$. Then $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and

$$\int_a^b \lim_{n \rightarrow \infty} f_n(x) d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_n(x) d\alpha$$

▷ **proof** : To prove this we need to show that $\lim_{n \rightarrow \infty} f_n \rightarrow f \in \mathcal{R}(\alpha)$

1. $f \in \mathcal{R}(\alpha)$

what we know is that $f_n \rightarrow f$ and $f_n \in \mathcal{R}(\alpha)$ this implies : let $\varepsilon > 0$, $\exists N$ s.t $\|f_n - f\| < \varepsilon$ if $n \geq N$ and also $\exists P(\text{Partition})$ s.t $U(P, f_n, \alpha) - L(P, f_n, \alpha) < \varepsilon$. Now let's consider this :

$$U(P, f_n, \alpha) - U(P, f, \alpha) = \sum \left| \sup_{[x_{j-1}, x_j]} f_n(x) - \sup_{[x_{j-1}, x_j]} f(x) \right| \Delta \alpha_j$$

For here, note that $f_n(x) = f_n(x) + f(x) - f_n(x) \leq f(x) + \varepsilon$ since $f_n \rightarrow f$. Therefore we can extend upper inequality as following :

$$\sum \left| \sup_{[x_{j-1}, x_j]} f_n(x) - \sup_{[x_{j-1}, x_j]} f(x) \right| \Delta \alpha_j \leq \sum \varepsilon \Delta \alpha_j = \varepsilon = \varepsilon(\alpha(b) - \alpha(a))$$

Where, $\sum_{i=1}^n \Delta\alpha_i = (\alpha(x_1) - \alpha(x_0 = a) + \alpha(x_2) - \alpha(x_1) + \cdots + \alpha(x_n = b) - \alpha(x_{n-1}))$.

And with this fact we can derive that

$$\begin{aligned} & U(P, f, \alpha) - L(P, f, \alpha) \\ &= U(P, f, \alpha) - U(P, f_n, \alpha) (< \varepsilon) \\ &+ U(P, f_n, \alpha) - L(P, f_n, \alpha) (< \varepsilon) \\ &+ L(P, f_n, \alpha) - L(P, f, \alpha) (< \varepsilon) < 3\varepsilon \approx \varepsilon \Rightarrow f \in \mathcal{R}(\alpha) \end{aligned}$$

Lastly, we need to show that $\int_a^b \lim_{n \rightarrow \infty} f_n(x) d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_n(x) d\alpha \Rightarrow \int_a^b \lim_{n \rightarrow \infty} f_n(x) d\alpha - \lim_{n \rightarrow \infty} \int_a^b f_n(x) d\alpha < \varepsilon$

$$2. \int_a^b \lim_{n \rightarrow \infty} f_n(x) d\alpha - \lim_{n \rightarrow \infty} \int_a^b f_n(x) d\alpha < \varepsilon$$

Since $f \in \mathcal{R}(\alpha)$ we can express this :

$$\begin{aligned} \exists P \text{ s.t } L(P, f, \alpha) &\leq \int_a^b f(x) d\alpha \leq U(P, f, \alpha), \\ U(P, f, \alpha) - L(P, f, \alpha) &< \varepsilon \end{aligned}$$

For the same Partition we got in number 1. We derived that :

$$\begin{aligned} U(P, f_n, \alpha) - U(P, f, \alpha) &< \varepsilon(\alpha(b) - \alpha(a)) \\ L(P, f_n, \alpha) - L(P, f, \alpha) &< \varepsilon(\alpha(b) - \alpha(a)) \end{aligned}$$

And according to the assumption that $f_n \in \mathcal{R}(\alpha)$

$$L(P, f_n, \alpha) \leq \int_a^b f_n(x) d\alpha \leq U(P, f_n, \alpha)$$

If we subtract two inequality then we get

$$\begin{aligned} L(P, f_n, \alpha) - L(P, f, \alpha) &\leq \int_a^b f_n(x) d\alpha - \int_a^b f(x) d\alpha \leq U(P, f_n, \alpha) - U(P, f, \alpha) \\ \Rightarrow L(P, f_n, \alpha) - \varepsilon(\alpha(b) - \alpha(a)) &\leq L(P, f, \alpha) \leq \int_a^b f_n(x) d\alpha - \int_a^b f(x) d\alpha \\ &\leq U(P, f_n, \alpha) \leq U(P, f, \alpha) + \varepsilon(\alpha(b) - \alpha(a)) \\ \Rightarrow U(P, f, \alpha) - L(P, f, \alpha) + 2\varepsilon(\alpha(b) - \alpha(a)) &< \varepsilon(\alpha(b) - \alpha(a)) + \varepsilon \\ \Rightarrow \int_a^b f_n(x) d\alpha - \int_a^b f(x) d\alpha &< \varepsilon \end{aligned}$$

■

★Thm : interchangeable derivative

Suppose $\{f_n\}$ is a sequence of functions, differentiable on $[a, b]$ and such that $\{f_n(x_0)\}$ converges for some point x_0 on $[a, b]$. If $\{f'_n\}$ converges uniformly on $[a, b]$, then $\{f_n\}$ converges uniformly on $[a, b]$, to a function f , and

$$\left(\lim_{n \rightarrow \infty} f_n(x) \right)' = \lim_{n \rightarrow \infty} f'_n(x) \quad (a \leq x \leq b)$$

▷ **proof** : As we did we first of all prove that $f_n(x) \rightarrow f(x)$

1. $f_n(x) \rightarrow f(x)$

It's enough to show that it is Cauchy. Therefore, consider this:

$$|f_n(x) - f_m(x)| = |f_n(x) - f_n(x_0) + f_n(x_0) - f_m(x_0) + f_m(x_0) - f_m(x)|$$

And let's use the assumption that $f_n(x_0)$ converges which indicate that $f_n(x_0) - f_m(x_0) < \varepsilon$. And let's define this $h_{n,m}(x) = f_n(x) - f_m(x)$. Then above equation can be change :

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq \varepsilon + |h_{n,m}(x) - h_{n,m}(x_0)| \leq \varepsilon + |h'_{n,m}(t)|(x - x_0), \quad t \in (x_0, x) \\ \therefore |f_n(x) - f_m(x)| &\leq \varepsilon + |f'_n(x) - f'_m(x)|(x - x_0) < \varepsilon + \varepsilon|x_0 + x| \approx \varepsilon \quad \text{if } n \geq N \\ \therefore f_n(x) &\rightarrow f \quad \forall x \in [a, b] \end{aligned}$$

And Now we will get result with the fact that $\lim_{n \rightarrow \infty} \lim_{t \rightarrow \infty} \phi_n(t) = \lim_{t \rightarrow \infty} \lim_{n \rightarrow \infty} \phi_n(t)$ we've proved. For here we define $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$ for here $(x \neq t)$ and $\lim_{t \rightarrow x} \phi_n(t) = f'_n(x)$

2. $\phi_n \Rightarrow \phi$

To show the ϕ_n converges it is enough to show that it is Cauchy

$$\begin{aligned} |\phi_n(t) - \phi_m(t)| &= \left| \frac{(f_n(t) - f_n(x)) - (f_m(t) - f_m(x))}{t - x} \right| = \left| \frac{h_{n,m}(t) - h_{n,m}(x)}{t - x} \right| \\ &\leq h'_{n,m}(c) \text{ (by MVT) } = f'_n(c) - f'_m(c) < \varepsilon \\ \therefore f'_n(c) &\rightarrow g \\ \therefore \phi_n(t) &\rightarrow \phi \end{aligned}$$

■

AFTER MIDTERM

Theorem 7.18 : continuous but not differentiable

There exists a real continuous function on the real line which is nowhere differentiable.

▷ **proof** : First of all, let's define

$$\varphi(x) = |x| \quad \text{for } -1 \leq x \leq 1$$

Now let us extend the above function to a periodic function with period 2.

$$\varphi(x+2) = \varphi(x)$$

That is, φ equals $|x|$ on $[-1, 1]$, extended periodically with period 2. Now, to construct a function that is continuous but nowhere differentiable, consider the following function.

$$\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$$

To check whether this series converges, let us first apply the Comparison Test. To verify convergence, consider a suitable sequence that dominates the given one but still converges.

$$\left| \left(\frac{3}{4}\right)^n \varphi(4^n x) \right| = \left(\frac{3}{4}\right)^n \cdot |\varphi(4^n x)| \leq \left(\frac{3}{4}\right)^n \cdot 1 = \left(\frac{3}{4}\right)^n$$

$\because \varphi(4^n x) \in [0, 1]$

Since $\left(\frac{3}{4}\right)^n$ converges to 0 as $n \rightarrow \infty$ (as a geometric sequence), and since $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n$ converges, the Comparison Test tells us that $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$ also converges. Moreover, since the terms are always defined on $[0, 1]$ regardless of x , this convergence is uniform. By the theorem "if $f_n \rightarrow f$ uniformly, then f is continuous," the function defined by this series is continuous. Now let us prove that this function is differentiable at no point. To do so, it suffices to show the following.

$$\nexists \lim_{\delta \rightarrow 0} \frac{f(x+\delta) - f(x)}{\delta}$$

Since the function $f(x) = \sum_{n=0}^{\infty} a_n \varphi(b_n x)$ has a complicated structure as an infinite sum, it is essentially impossible to consider every possible way $\delta \rightarrow 0$. Instead, we construct a specific sequence $\delta_n \rightarrow 0$ and carry the proof forward using it. What we ultimately need to show is the following.

$$\lim_{m \rightarrow \infty} \frac{f(x+\delta_m) - f(x)}{\delta_m} = \infty$$

Since the φ we constructed is non-differentiable at the integers, we must consider two conditions when choosing δ_m : first, it must converge to 0, and second, it must never land on an integer. The reason is as follows.

$$\frac{f(x+\delta_m) - f(x)}{\delta_m} = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \cdot \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$$

In the expression above, if an integer lies between $4^m x$ and $4^m(x+\delta_m)$, the difference quotient could vanish, creating a point that looks differentiable even though it is not. We must choose the sequence carefully to avoid this.

$$\delta_m = \pm \frac{1}{2} \cdot 4^{-m} \Rightarrow 4^m \delta_m = \pm \frac{1}{2}$$

With the sequence constructed this way, the difference between $4^n x$ and $4^n(x + \delta_m)$ is exactly $\frac{1}{2}$.

$$\gamma_n := \frac{\varphi(4^n(x + \delta_m)) - \varphi(4^n x)}{\delta_m} \quad \text{for each } n \in \mathbb{N}$$

Define this, and consider the case $n > m$ as follows.

$$\begin{aligned} \varphi(4^n(x + \delta_m)) - \varphi(4^n x) &= \varphi(4^n x + 4^n \delta_m) - \varphi(4^n x) \\ &\Rightarrow \varphi(4^n x + 4^n \delta_m) = \varphi(4^n x) \\ (\because 4^n \delta_m &= \pm \frac{1}{2} \cdot 4^n \cdot 4^{-m} = \pm \frac{1}{2} \cdot 4^{n-m}, n > m \Rightarrow 4^{n-m} \in \mathbb{Z}) \\ \varphi(4^n(x + \delta_m)) - \varphi(4^n x) &= 0 \Rightarrow \gamma_n = \frac{0}{\delta_m} = 0 \end{aligned}$$

So we do not need to consider the case $n \geq m$; we only need to consider $n < m$. Also note that our function has the following property.

$$|\varphi(s) - \varphi(t)| \leq |s - t| \quad \text{for all } s, t \in \mathbb{R}$$

Therefore,

$$\begin{aligned} |\gamma_n| &= \left| \frac{\varphi(4^n(x + \delta_m)) - \varphi(4^n x)}{\delta_m} \right| \leq \frac{|\varphi(4^n(x + \delta_m)) - \varphi(4^n x)|}{|\delta_m|} \\ &\leq \frac{|4^n(x + \delta_m) - 4^n x|}{|\delta_m|} = \frac{4^n |\delta_m|}{|\delta_m|} = 4^n \\ |\gamma_n| &\leq 4^n \end{aligned}$$

Using the above facts, we can derive the following.

$$\begin{aligned} \left| \frac{f(x + \delta_m) - f(x)}{\delta_m} \right| &= \left| \sum_{n=0}^m \left(\frac{3}{4} \right)^n \gamma_n \right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1) \quad (\because |\gamma_n| \leq 4^n) \\ \therefore m \rightarrow \infty &\Rightarrow \frac{1}{2}(3^m + 1) \rightarrow \infty \Rightarrow \left| \frac{f(x + \delta_m) - f(x)}{\delta_m} \right| \rightarrow \infty \end{aligned}$$

■

Definition 7.19,22 : Pointwise Bounded, Uniformly Bounded and equicontinuous

Review that

- **f : continuous** at $x = a \iff \forall \varepsilon > 0, \exists \delta > 0$ such that $d(f(x), f(a)) < \varepsilon$ if $d(x, a) < \delta$
- **f : uniformly continuous** at on $E \iff \forall \varepsilon > 0, \exists \delta > 0$ for $d(f(x), f(y)) < \varepsilon \forall x, y \in E$ if $d(x, y) < \delta$

From here on, we introduce the following new definitions.

- **$\{f_n\}$ point bounded** on $E \iff \forall x \in E \exists M_x$ such that $|f_n(x)| < M_x \quad \forall n$
- **$\{f_n\}$ Uniformly bounded** on $E \iff \exists M$ such that $|f_n(x)| < M \quad \forall n \in \mathbb{N}, \forall x \in E$
- **$\{f_\alpha\}_{\alpha \in A}$ equicontinuous** on $E \iff \exists \delta$ such that $d(f_\alpha(x), f_\alpha(y)) < \varepsilon \forall \alpha \in A, \quad \forall x, y \in E$ if $d(x, y) < \delta$

Each of the above definitions carries the following meaning.

1. Pointwise bounded $\rightarrow$ for each point $x \in E$, the sequence $\{f_n(x)\}$ is bounded individ-

ually.

2. Uniformly bounded $\rightarrow$ the entire sequence of functions is bounded at every point simultaneously by a single common constant M .
3. Equicontinuous $\rightarrow$ continuity holds simultaneously for every function f_α with the same δ (i.e., they share the same "strength" of continuity).

7.23 Theorem

If $\{f_n\}$ is a pointwise bounded sequence of complex functions on a countable set E , then $\{f_n\}$ has a subsequence $\{f_{n_k}\}$ such that $\{f_{n_k}(x)\}$ converges for every $x \in E$.

▷ **proof** : We prove this theorem using the Cantor diagonal process. First, $E = \{x_1, x_2, x_3, \dots\}$ is a countable set. Since the sequence $\{f_n(x_i)\} \subset \mathbb{C}$ is bounded, the Bolzano–Weierstrass theorem guarantees a convergent subsequence for each $\{x_i\}$. For example,

$$\begin{aligned} f_{1,1}(x_1), f_{1,2}(x_1), f_{1,3}(x_1), \dots &\rightarrow f(x_1) \\ f_{2,1}(x_2), f_{2,2}(x_2), f_{2,3}(x_2), \dots &\rightarrow f(x_2) \\ f_{3,1}(x_3), f_{3,2}(x_3), f_{3,3}(x_3), \dots &\rightarrow f(x_3) \\ &\vdots \end{aligned}$$

Here, each $\{f_{i,j}\}$ is a subsequence chosen from f_n satisfying $\{f_{i,j}(x_i)\}_{j=1}^\infty \rightarrow f(x_i)$. We can now construct a nested sequence of subsequences, each one a subsequence of the previous, as follows. $\{f_{1,j}\} \supset \{f_{2,j}\} \supset \{f_{3,j}\} \supset \dots$ with $f_{n,j}(x_k)$ convergent for $1 \leq k \leq n$

$$\begin{aligned} S_1 : f_{1,1}(x), f_{1,2}(x), f_{1,3}(x), f_{1,4}(x), \dots \{x_1\} = x \\ S_2 : f_{2,1}(x), f_{2,2}(x), f_{2,3}(x), f_{2,4}(x), \dots \{x_1, x_2\} = x \\ S_3 : f_{3,1}(x), f_{3,2}(x), f_{3,3}(x), f_{3,4}(x), \dots \{x_1, x_2, x_3\} = x \\ \vdots \end{aligned}$$

Here S_n is a subsequence of functions chosen from f_n , constructed so that S_n converges at $x_1, x_2, \dots, x_n$. That is, S_1 converges at x_1 , and S_2 converges once x_1, x_2 are included, and so on. Taking only the diagonal entries produces a new subsequence that converges at every x . That subsequence is as follows.

$$S = f_{1,1}(x), f_{2,2}(x), \dots, f_{n,n}(x)$$

■

7.24 Theorem

If K is a compact metric space, if $f_n \in \mathcal{C}(K)$ for $n = 1, 2, 3, \dots$, and if $\{f_n\}$ converges uniformly on K , then $\{f_n\}$ is equicontinuous on K .

▷ **proof** : Let us first separate what we are given from what we need to show. What we need to show is the following.

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x, y \in K, d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon \text{ for all } n$$

And what we are given is the following.

1. K is a compact metric space: a finite subcover exists,
2. $f_n \in \mathcal{C}(K) \Rightarrow \forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x \in K, d(x, x_0) < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon/3$
3. $\{f_n\}$ converges uniformly on $K \Rightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N, \forall x \in K, |f_n(x) - f(x)| < \varepsilon/3$

Let us first confirm equicontinuity for $n \geq N$, and then explain equicontinuity in terms of δ . To gain one more factor of $\varepsilon/3$ defined via N : since $f \in \mathcal{C}(K)$ and K is compact $\Rightarrow f$ is uniformly continuous, therefore

$$\exists \delta > 0 \text{ such that } \forall x, y \in K, d(x, y) < \delta \Rightarrow |f(x) - f(y)| < \varepsilon/3$$

Then, for the N_1 and δ above, the following holds for fixed $x, y \in K$.

$$\begin{aligned} |f_n(x) - f_n(y)| &\leq |f_n(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_n(y)| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon \end{aligned}$$

From this we see that equicontinuity holds for all $n \geq N$. On the other hand, for $n < N$ the above does not directly apply, but since K is still compact, $f_n \in \mathcal{C}(K) \Rightarrow f_n$ is uniformly continuous. So for each such n , the following holds.

$$\forall n < N, \exists \delta_n > 0 \text{ such that } d(x, y) < \delta_n \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$$

Hence, choosing the smallest such δ_0 gives equicontinuity that holds throughout. So if we set $\delta_0 := \min\{\delta, \delta_1, \delta_2, \dots, \delta_{N-1}\}$, the following holds for all $n \in \mathbb{N}$.

$$d(x, y) < \delta_0 \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$$

The conclusion is as follows.

$$\forall \varepsilon > 0, \exists \delta_0 > 0 \text{ such that } \forall x, y \in K, d(x, y) < \delta_0 \Rightarrow |f_n(x) - f_n(y)| < \varepsilon, \forall n$$

Note that equicontinuity places no restriction on n , so we obtained δ separately for the cases $n \geq N$ and $n < N$, and then took the smallest value as our final δ . ■

Note :

Let K be a compact metric space. Then there exists a countable subset $E \subset K$ such that $\overline{E} = K$.

▷ **proof** : Since K is compact, it can be covered by finitely many open sets; using this, let us cover K with balls of radius $1/n$ as follows.

$$K \subset \bigcup_{i=1}^{k_n} B(x_{n,i}, 1/n)$$

For each fixed n , collect all the centers $i = 1, 2, \dots, k_n$, namely $x_{n,1}, \dots, x_{n,k_n} \in K$, and call this collection E .

$$E := \bigcup_{\forall n} \{x_{n,1}, \dots, x_{n,k_n}\}$$

Recalling our goal, it suffices to show that near any point $x \in K$ there always exists a point belonging to E . So we just need to show the following.

$$\forall x \in K, \forall \varepsilon > 0, \exists e \in E \text{ such that } d(x, e) < \varepsilon \Rightarrow \overline{E} = K$$

So choose n with $\exists n \in \mathbb{N}$ such that $1/n < \varepsilon$. Then, since $x \in K \subset \bigcup_{i=1}^{k_n} B(x_{n,i}, 1/n)$, it automatically follows that $\exists i$ such that $d(x, x_{n,i}) < 1/n < \varepsilon$. Hence, whatever x we choose, there is always some $x_{n,i} \in E$ nearby, and since $x_{n,i}$ is an element of E , E is dense. ■

7.25 Theorem

Let K be a compact metric space, and let $\{f_n\} \subset \mathcal{C}(K)$. Suppose that $\{f_n\}$ is pointwise bounded: $\forall x \in K, \sup_n |f_n(x)| < \infty$ and equicontinuous on K then :

- $\{f_n\}$ is uniformly bounded on K
- $\{f_n\}$ has a uniformly convergent subsequence

▷ **proof** : Applying equicontinuity with $\varepsilon = 1$

$$\exists \delta > 0 \text{ such that } \forall x, y \in K, d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < 1$$

gives this. And since K is compact, a finite open cover exists.

$$K \subset \bigcup_{i=1}^k B(x_i, \delta)$$

Here $x_1, \dots, x_k \in K$ are finitely many points such that the open balls centered at them cover K . Since $\{f_n\}$ is pointwise bounded, let us take the maximum of the pointwise bounds over each x_i .

$$M := \max_{1 \leq i \leq k} \sup_n |f_n(x_i)| < \infty$$

Then for any $x \in K$ there is some x_i with $d(x, x_i) < \delta$, and then,

$$|f_n(x)| \leq |f_n(x_i)| + |f_n(x) - f_n(x_i)| < M + 1$$

this holds. That is, $\sup_{x \in K, n} |f_n(x)| \leq M + 1 < \infty$ holds.

Next, let $\{f_{n_k}\} := \{g_k\}$ be a subsequence of f_n ; then by Theorem 7.23, g_k converges. Expressed in Cauchy form, this can be stated as follows.

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N, |g_m(x) - g_n(x)| < \varepsilon, \forall x \in K$$

Next, using the equicontinuity property, we can write the following.

$$\begin{aligned} |g_n(x) - g_m(x)| &\leq |g_n(x) - g_n(p)| + |g_n(p) - g_m(p)| + |g_m(p) - g_m(x)| \\ \Rightarrow |g_n(x) - g_m(x)| &< 2\varepsilon + |g_n(p) - g_m(p)| \end{aligned}$$

This is because, since K is compact, if we choose N so that $\frac{1}{N} < \delta$,

$$K \subset \bigcup_{j=1}^{M_N} B(X_{N,j}, \frac{1}{N}) \quad \text{with} \quad \frac{1}{N} < \delta$$

we obtain an open cover as above that automatically gives $|g_k(x) - g_k(X_{N,j})| < \varepsilon$, where the number of sets in the cover depends on the radius $1/N$ (a smaller radius requires

more balls to cover K). So we may take $p \in \{X_{N,1}, X_{N,2}, \dots, X_{N,MN}\}$. As shown in the Note above, a dense countable subset $E \subset K$ exists. Since f_n is pointwise bounded, for $p \in E$ the real sequence $\{f_n(p)\}$ is bounded, so by Bolzano–Weierstrass we can extract a convergent subsequence $\{g_k\} := \{f_{n_k}\}$. This only guarantees convergence at $p \in E$, but since f_n has the strong property of equicontinuity, the same conclusion extends to any point $x \in K$ near $p \in E$:

$$|g_k(x) - g_k(p)| < \varepsilon \quad \text{for all } k$$

this holds. Hence, for $N < n, m$ the following holds.

$$\begin{aligned} |g_n(x) - g_m(x)| &\leq |g_n(x) - g_n(p)| + |g_n(p) - g_m(p)| + |g_m(p) - g_m(x)| \\ &\Rightarrow |g_n(x) - g_m(x)| < 3\varepsilon \approx \varepsilon \end{aligned}$$

■

7.26 Theorem : (Just memorize this one — better for your sanity)

Let f be a continuous complex-valued function on the closed interval $[a, b]$. Then there exists a sequence of polynomials $\{P_n\}$ such that

$$\lim_{n \rightarrow \infty} P_n(x) = f(x)$$

uniformly on $[a, b]$. Moreover, if f is real-valued, then the polynomials P_n may be taken to be real.

▷ **proof** : Without Loss of Generality, let $[a, b]$ be $[0, 1]$. This is because if we set $g(x) = f(a + (b - a)x)$, then whatever is defined on $[0, 1]$ is also defined on $[a, b]$. Furthermore, let us assume $f(0) = f(1) = 0$, and that f is defined to be 0 outside $[0, 1]$. This is valid because f can be compressed onto the interval $[0, 1]$ via a suitable linear transformation, and the resulting function is uniformly continuous on the whole real line. Now consider the following sequence of functions

$$Q_n(x) = c_n(1 - x^2)^n$$

Here c_n is chosen so that $\int_{-1}^1 Q_n(x) dx = 1$. Let us now examine the size of c_n .

$$\int_{-1}^1 (1 - x^2)^n dx \geq 2 \int_0^1 (1 - x^2)^n dx \geq 2 \int_0^{\frac{1}{\sqrt{n}}} (1 - x^2)^n dx \geq 2 \int_0^{\frac{1}{\sqrt{n}}} (1 - nx^2) dx$$

This is because $(1 - x^2)^n \geq (1 - nx^2)$ always holds for x between 0 and 1. Let us now evaluate the integral.

$$\begin{aligned} 2 \int_0^{\frac{1}{\sqrt{n}}} (1 - nx^2) dx &= 2 \left[x + \frac{n}{3} x^3 \right]_0^{\frac{1}{\sqrt{n}}} = \frac{4}{3\sqrt{n}} \\ \therefore \int_{-1}^1 (1 - x^2)^n dx &\geq \frac{4}{3\sqrt{n}} \geq \frac{1}{\sqrt{n}} \\ \Rightarrow \int_{-1}^1 c_n(1 - x^2)^n dx &= 1 \geq \frac{c_n}{\sqrt{n}} \Rightarrow \sqrt{n} \geq c_n \end{aligned}$$

For any $\delta > 0$, on $\delta \leq |x| \leq 1$ we have $Q(x) \leq \sqrt{n}(1 - \delta^2)^n$, which converges to 0 uniformly as $n \rightarrow \infty$. This fact will be used to show that P_n converges uniformly to f . Now let us

examine P_n , defined as follows.

$$P_n(x) := \int_{-1}^1 f(x+t)Q_n(t) dt$$

With a suitable substitution, we obtain the following.

$$P_n(x) = \int_{x-1}^{x+1} f(y)Q_n(y-x) dy$$

Setting $y := x+t$ changes the interval of integration to $[x-1, x+1]$; the resulting expression is well defined as a polynomial in x , since y is integrated out, and if f is real-valued, the resulting polynomial is also real. All that remains is to show $|P_n(x) - f(x)| < \varepsilon$. Before doing so, we need a couple of tools: first, since f is uniformly continuous on the closed interval, for the given ε there exists $\delta > 0$ such that $|y - x| < \delta \Rightarrow |f(y) - f(x)| < \varepsilon/2$. Below this will be used in the form $|t| < \delta \Rightarrow |f(x+t) - f(x)| < \varepsilon/2$. Let $M := \sup |f(x)|$.

$$\begin{aligned} |P_n(x) - f(x)| &= \left| \int_{-1}^1 f(x+t)Q(t)dt - f(x) \cdot \left(1 = \int_{-1}^1 Q(t)dt\right) \right| \\ &= \left| \int_{-1}^1 [f(x+t) - f(x)]Q(t)dt \right| \\ &\leq \int_{-1}^1 |f(x+t) - f(x)|Q(t)dt \end{aligned}$$

Splitting this into the range $|t| \leq \delta$ and $|t| > \delta$: on $|t| \leq \delta$, uniform continuity gives $|f(x+t) - f(x)| < \varepsilon/2$, while on $|t| > \delta$ we have $|f(x+t) - f(x)| \leq 2M$. Hence the expression above becomes

$$\leq 2M \int_{-1}^{-\delta} Q_n(t)dt + \frac{\varepsilon}{2} \int_{-\delta}^{\delta} Q_n(t)dt + 2M \int_{\delta}^1 Q_n(t)dt$$

And here the integral term can be simplified as follows,

$$\int_{|t|>\delta} Q_n(t)dt = c_n \int_{|t|>\delta} (1-t^2)^n dt \leq c_n(1-\delta^2)^n \cdot 2 < 2\sqrt{n}(1-\delta^2)^n.$$

and also

$$\int_{-1}^1 Q_n(t) dt = 1 \Rightarrow \int_{-\delta}^{\delta} Q_n(t) \leq \int_{-1}^1 Q_n(t) = 1$$

so we can finally conclude the following.

$$|P_n(x) - f(x)| < 4M\sqrt{n}(1-\delta^2)^n + \frac{\varepsilon}{2} < \varepsilon$$

Some special functions

8.1 Theorem :

Let the power series

$$\sum_{n=0}^{\infty} c_n x^n$$

have a radius of convergence $R > 0$, and define the function

$$f(x) := \sum_{n=0}^{\infty} c_n x^n \quad \text{for } |x| < R.$$

Then:

1. The series converges uniformly on every closed interval $[-R + \varepsilon, R - \varepsilon]$ for any $\varepsilon > 0$.
2. The function f is continuous and differentiable on the open interval $(-R, R)$.
3. The derivative of f is given by $f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ for $|x| < R$.

▷ **proof** : Suppose ε is given and $f(x)$ converges for $|x| < R$. Then the root test gives the following relation.

$$\begin{aligned} \limsup |c_n x^n|^{\frac{1}{n}} &= \limsup |c_n|^{\frac{1}{n}} |x|^{\frac{1}{n}} = \limsup |c_n|^{\frac{1}{n}} |x| < 1 \\ \Rightarrow \limsup |c_n|^{\frac{1}{n}} &= \frac{1}{R} \end{aligned}$$

Here, if we restrict to $|x| \leq R - \varepsilon$, i.e. shrink the radius, we get $|c_n x^n| \leq |c_n (R - \varepsilon)^n|$. Combining this with the convergence result above,

$$\begin{aligned} \limsup |c_n x^n|^{\frac{1}{n}} &= \limsup |c_n (R - \varepsilon)^n|^{\frac{1}{n}} \leq \limsup |c_n|^{\frac{1}{n}} (R - \varepsilon) = \frac{1}{R} (R - \varepsilon) \\ \Rightarrow |1 - \frac{\varepsilon}{R}| &< 1 \end{aligned}$$

this still falls within the convergence range of the root test, so the series still converges on the shrunken radius; since this convergence is independent of x , it is uniform. Since the series converges uniformly no matter how we choose the subinterval between $-R$ and R , we conclude it converges uniformly, and uniform convergence implies continuity. Now, if the derivative series is also uniformly continuous, parts 2 and 3 follow automatically. Since $\frac{\partial}{\partial x} c_n x^n = n c_n x^{n-1}$, let us apply the root test once more to the $n c_n$ term.

$$\limsup (n |c_n|)^{\frac{1}{n}} = \limsup |c_n|^{\frac{1}{n}}$$

This is because $n^{\frac{1}{n}}$ converges to 1^a, and this result shows that f and f' share the same radius of convergence. From this we obtain

$$f'(x) = \sum_{n=0}^{\infty} n c_n x^{n-1}$$

that this also converges uniformly. Now, using the fact that differentiating the series and then summing equals summing the series and then differentiating, we complete the proof of part 3 as follows.

$$\left(\lim_{k \rightarrow \infty} \sum_{n=0}^k c_n x^n \right)' = \lim_{k \rightarrow \infty} \left(\sum_{n=0}^k c_n x^n \right)' = \lim_{k \rightarrow \infty} \sum_{n=0}^k n c_n x^{n-1} = \sum_{n=0}^{\infty} n c_n x^{n-1}$$

^aFrom $\lim_{n \rightarrow \infty} n^{1/n} = e^{\lim_{n \rightarrow \infty} n / n^2} = e^0 = 1$, applying L'Hopital's rule easily shows this equals $e^0 = 1$.

8.3 Theorem

Given a double sequence $\{a_{ij}\}$, $i = 1, 2, 3, \dots$, $j = 1, 2, 3, \dots$, suppose that

$$\sum_{j=1}^{\infty} |a_{ij}| = b_i \quad (i = 1, 2, 3, \dots)$$

and $\sum b_i$ converges. Then

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

▷ **proof** : First, let $E = \{x_0, x_1, x_2, \dots\}$ be a countable set, with $x_n \rightarrow x_0$ pointwise. Define the following.

$$f_i(x_0) = \sum_{j=1}^{\infty} a_{ij}, \quad f_i(x_n) = \sum_{j=1}^n a_{ij}$$

We may define it this way because we can choose a different f value depending on n ; but since what we need to reach is the limit point,

$$g(x) = \sum_{i=1}^{\infty} f_i(x), \quad x \in E$$

Here, since $f_i(x_n) \rightarrow f_i(x_0)$, f_i is continuous at x_0 . And

$$|f_i(x_n)| = \left| \sum_{j=1}^n a_{ij} \right| \leq \sum_{j=1}^n |a_{ij}| = b_i$$

so applying the Weierstrass M-test shows that $g(x)$ converges uniformly. It converges uniformly because the bound above holds regardless of x . Using this uniform convergence, we can now interchange the order of the limit,

$$\begin{aligned} \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij} &= \sum_{i=1}^{\infty} f_i(x_0) = g(x_0) = \lim_{n \rightarrow \infty} g(x_n) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} f_i(x_n) = \lim_{n \rightarrow \infty} \sum_{j=1}^{\infty} \sum_{i=1}^n a_{ij} = \lim_{n \rightarrow \infty} \sum_{j=1}^n \sum_{i=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij} \end{aligned}$$

8.4 Theorem

Suppose

$$f(x) = \sum_{n=0}^{\infty} c_n x^n,$$

the series converging in $|x| < R$. If $-R < a < R$, then f can be expanded in a power series about the point $x = a$, which converges in $|x - a| < R - |a|$, and

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \quad (|x - a| < R - |a|)$$

This is an extension of Theorem 5.15 and is also known as *Taylor's theorem*. ▷ **proof** : First, let $f(x)$ be given as a power series with radius of convergence R . Let us shift the center and then expand back, as follows.

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n ((x-a) + a)^n$$

Here, using the binomial theorem on the exponent n , this can be rewritten as follows.

$$= \sum_{n=0}^{\infty} c_n \sum_{m=0}^n \binom{n}{m} (x-a)^m a^{n-m} = \sum_{n=0}^{\infty} \sum_{m=0}^n c_n \binom{n}{m} a^{n-m} (x-a)^m$$

Since what we want is the expression corresponding to $(x-a)^m$, we would like to interchange the order of summation. To swap the sum over n with the sum over m , we must first check that the resulting series still lies within the radius of convergence. The condition for the new radius of convergence is as follows.

$$|x| < R \Rightarrow |x-a| + |a| < R \Rightarrow |x-a| < R - |a|$$

We now need to derive the above condition from the given expression. To do so, let us take absolute values inside the series

$$\sum_{n=0}^{\infty} \sum_{m=0}^n \left| c_n \binom{n}{m} a^{n-m} (x-a)^m \right| \leq \sum_{n=0}^{\infty} \sum_{m=0}^n |c_n| \binom{n}{m} |a|^{n-m} |x-a|^m = \sum_{n=0}^{\infty} |c_n| (|a| + |x-a|)^n$$

By the definition of radius of convergence, this series converges when $|x-a| + |a| < R$, which fortunately is exactly the condition derived above. That is, this slightly modified power series converges. Now, interchanging the order of summation,

$$\sum_{n=0}^{\infty} \sum_{m=0}^n c_n \binom{n}{m} a^{n-m} (x-a)^m = \sum_{m=0}^{\infty} \left(\sum_{n=m}^{\infty} c_n \binom{n}{m} a^{n-m} \right) (x-a)^m$$

we obtain this rewriting. The lower limit $n = m$ follows from the properties of the binomial coefficient. Now set $A_m := \sum_{n=m}^{\infty} c_n \binom{n}{m} a^{n-m}$, and it remains only to show this equals $\frac{f^{(m)}(a)}{m!}$.

$$\begin{aligned} f^{(m)}(a) &= \sum_{n=m}^{\infty} c_n \frac{n!}{(n-m)!} a^{n-m} \\ \Rightarrow \frac{f^{(m)}(a)}{m!} &= \sum_{n=m}^{\infty} c_n \binom{n}{m} a^{n-m} = A_m \end{aligned}$$

8.5 Theorem

Suppose the series $\sum a_n x^n$ and $\sum b_n x^n$ converge in the segment $S = (-R, R)$. Let E be the set of all $x \in S$ at which

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n \quad (20)$$

If E has a limit point in S , then $a_n = b_n$ for $n = 0, 1, 2, \dots$. Hence (20) holds for all $x \in S$.

▷ **proof** : Define the difference of the two sequences by $c_n = a_n - b_n$, and let the power series $f(x) = \sum_{n=0}^{\infty} c_n x^n$ satisfy $f(x) = 0$ on E by hypothesis. Add a few more definitions: let $A :=$ the set of limit points of E , and $B := S \setminus A$. Then $S = A \cup B$, and what we want to show is $B = \emptyset$. It suffices to show A is open, because if A is open and nonempty, and S is connected, then $B = \emptyset$ is forced. Let us follow this outline. Consider $x_0 \in A$; to show A is open, it suffices to show the following.

$$\exists \delta > 0 \text{ s.t. } B(x_0, \delta) \subset A$$

That is, it suffices to show x_0 is an interior point. Expanding the series near x_0 gives the following power series.

$$f(x) = \sum_{n=0}^{\infty} d_n (x - x_0)^n, \quad |x - x_0| < R - |x_0|$$

This value is 0 on E , but for x_0 to be interior we need $x - x_0 \neq 0$. What we ultimately need to show is that d_n is always 0. Let us prove this by contradiction.

$$\text{for some } d_k = 0, d_0 = d_1 = d_2 = \cdots = d_{k-1} = 0$$

Assume this holds. Then since the terms of $f(x)$ before k vanish, we can factor $f(x)$ as follows.

$$f(x) = (x - x_0)^k g(x) \quad \text{where } g(x) = \sum_{m=0}^{\infty} d_{k+m} (x - x_0)^m$$

Here $g(x_0) = \sum_{m=0}^{\infty} d_{k+m} (x_0 - x_0)^m = d_k \neq 0$. Since $g(x)$ is continuous and $g(x_0) \neq 0$, by definition the following holds.

$$\exists \delta > 0 \text{ s.t. } |x - x_0| < \delta \Rightarrow g(x) \neq 0$$

Since $g(x)$ is continuous, if $g(x_0) \neq 0$ then g is nonzero throughout a neighborhood of x_0 . This forces $f(x) \neq 0$, a contradiction. Hence d_n vanishes everywhere, so x_0 is an interior point, and hence A is open. Since $A \neq \emptyset$, we get $B = \emptyset$, meaning $S = A$.

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = 0 \quad \text{on } S \Rightarrow c_n = 0 \Rightarrow a_n = b_n \quad \forall n$$

8.8 Theorem

Suppose $a_0, \dots, a_n$ are complex numbers, $n \geq 1$, $a_n \neq 0$,

$$P(z) = \sum_{k=0}^n a_k z^k.$$

Then $P(z) = 0$ for some complex number z .

▷ **proof** : WLOG assume $a_n := 1$. This is valid because normalizing by a_n only produces a rescaled polynomial $\tilde{P}(Z)$, which does not affect whether a complex root exists.

$$\text{where } \tilde{P}(Z) := \frac{1}{a_n} P(Z) = \frac{a_0}{a_n} + \frac{a_1}{a_n} z_1 + \cdots + z^n$$

In any case, what we want to show is that there exists $z_0 \in \mathbb{C}$ such that $P(z_0) = 0$. To this end, let us first define the infimum.

$$\mu := \inf_{z \in \mathbb{C}} |P(z)|$$

With this defined, it suffices to show $\mu = 0$.

$$|P(z)| = |z^n + a_{n-1}z^{n-1} + \dots + a_0 + a_1z| \geq |z^n| \left(1 - \frac{|a_{n-1}|}{|z|} - \dots - \frac{|a_0|}{|z|^n} \right)$$

Note that $|A + B| \geq |A| - |B|$; this shows that as $|z| \rightarrow \infty$, $|p(z)| \rightarrow \infty$, meaning $|p(z)|$ diverges as we move farther out. This guarantees that the infimum μ is actually attained at some point, i.e. there exists z_0 with $|p(z_0)| = \mu = 0$. Let us prove this by contradiction. Suppose $|p(z_0)| = \mu > 0$. Define a new function as follows

$$Q(w) := \frac{P(z_0 + w)}{P(z_0)}$$

$Q(w = 0) = 1$, and $|Q(w)| \geq 1$, because

$$Q(w) = 1 + b_k w^k + b_{k+1} w^{k+1} + \dots + b_n w^n, \quad b_k \neq 0$$

here $|p(z)| \geq |p(z_0)| = \mu$, so the expansion on the right always keeps $Q(w)$ at least 1. Now, converting to the exponential form $w = re^{i\theta}$,

$$\begin{aligned} Q(w) &= Q(re^{i\theta}) = 1 + b_k (re^{i\theta})^k + b_{k+1} (re^{i\theta})^{k+1} + \dots + b_n (re^{i\theta})^n \\ \Rightarrow |Q(re^{i\theta})| &\leq 1 + |b_k| (re^{i\theta})^k + |b_{k+1}| (re^{i\theta})^{k+1} + \dots + |b_n| (re^{i\theta})^n \\ &\leq 1 + |b_k| r^k e^{i\theta k} + |b_{k+1}| r^{k+1} + \dots + |b_n| r^n \end{aligned}$$

here we can choose a suitable argument so that $|b_k| e^{i\theta k} = -|b_k|$. Hence

$$\begin{aligned} |Q(re^{i\theta})| &\leq 1 - |b_k| r^k + |b_{k+1}| r^{k+1} + \dots + |b_n| r^n \\ &= r^k |b_k| \left[1 - r \frac{|b_{k+1}|}{|b_k|} - \dots - r^{n-k} \frac{|b_n|}{|b_k|} \right] \end{aligned}$$

here, a sufficiently small r makes $|Q(re^{i\theta})| < 1$, but our assumption $|P(z_0)| > 0 \Rightarrow |Q(w)| \geq 1$ gives a contradiction. Hence there exists z_0 with $|P(z_0)| = 0$.

8.11 Theorem

Let $\{\phi_n\}$ be orthonormal on $[a, b]$. Let

$$s_n(x) = \sum_{m=1}^n c_m \phi_m(x)$$

be the n th partial sum of the Fourier series of f , and suppose

$$t_n(x) = \sum_{m=1}^n \gamma_m \phi_m(x).$$

Then

$$\int_a^b |f(x) - s_n(x)|^2 dx \leq \int_a^b |f(x) - t_n(x)|^2 dx,$$

and equality holds if and only if

$$\gamma_m = c_m \quad (m = 1, \dots, n).$$

That is to say, among all functions t_n , s_n gives the best possible mean square approximation to f .

▷ **proof** : First recall that $(c_n = \int_a^b f(t) \overline{\phi_n(t)} dt)$ are the Fourier coefficients. Our goal is to minimize $\int_a^b |f(x) - t_n(x)|^2 dx$. To do this we need a few preliminary tools; first, let us examine the inner product of f with the general series t_n .

$$\begin{aligned} \langle f, t_n \rangle &= \int_a^b f \overline{t_n} = \int_a^b f \overline{\left(\sum_{m=1}^n \gamma_m \phi_m \right)} = \int_a^b f \sum_{m=1}^n \overline{\gamma_m \phi_m} \\ &= \sum_m \overline{\gamma_m} \int_a^b f \overline{\phi_m} = \sum_m \overline{\gamma_m} c_m. \end{aligned}$$

Next, let us examine the norm of t_n . For convenience, we omit the integration domain from now on.

$$\int |t_n|^2 = \int t_n \overline{t_n} = \int \left(\sum_m \gamma_m \phi_m \right) \left(\sum_k \overline{\gamma_k \phi_k} \right) = \sum_m \sum_k \gamma_m \overline{\gamma_k} \int \phi_m \overline{\phi_k}.$$

Here, since we assumed $\{\phi_n\}$ is orthonormal,

$$\int \phi_m \overline{\phi_k} = \delta_{mk} \Rightarrow \sum_m \gamma_m \overline{\gamma_m} = \sum_m |\gamma_m|^2,$$

where δ_{mk} equals 1 when $m = k$, and 0 when $m \neq k$.

$$\int |t_n|^2 = \sum_m |\gamma_m|^2.$$

So the norm of t_n can be expressed as above. Now, using these tools, let us prove that the difference is minimized when $t_n = s_n$.

$$\begin{aligned} \int |f - t_n|^2 &= \int (f - t_n) \overline{(f - t_n)} = \int f \overline{f} - f \overline{t_n} - \overline{f} t_n + t_n \overline{t_n} \\ &= \int |f|^2 - \int f \overline{t_n} - \int \overline{f} t_n + \int |t_n|^2. \end{aligned}$$

The distance between f and the general t_n can be written as above,

$$\begin{aligned} \int f \overline{t_n} &= \sum_m \overline{\gamma_m} c_m, \\ \int \overline{f} t_n &= \int \overline{f} \sum_m \gamma_m \phi_m = \sum_m \gamma_m \int \overline{f} \phi_m = \sum_m \gamma_m \overline{\int f \overline{\phi_m}} = \sum_m \gamma_m \overline{c_m}, \\ \int |t_n|^2 &= \sum_m |\gamma_m|^2. \end{aligned}$$

having derived the three results above, substituting each in gives the following,

$$\begin{aligned}
& \int |f|^2 - \sum_m \overline{\gamma_m} c_m - \sum_m \gamma_m \overline{c_m} + \sum_m |\gamma_m|^2 \\
&= \int |f|^2 - \sum_m |c_m|^2 + \sum_m (|\gamma_m|^2 - \overline{\gamma_m} c_m - \gamma_m \overline{c_m} + |c_m|^2) \\
&= \int |f|^2 - \sum_m |c_m|^2 + \sum_m |\gamma_m - c_m|^2.
\end{aligned}$$

To complete the square, we add and subtract $\sum_m |c_m|^2$. This expression is minimized when $\gamma_m = c_m$, and for any other choice of γ_m the second term on the right is always positive. Hence

$$\int |f - s_n|^2 \leq \int |f - t_n|^2.$$

8.12 Theorem

If $\{\phi_n\}$ is orthonormal on $[a, b]$, and if

$$f(x) \sim \sum_{n=1}^{\infty} c_n \phi_n(x),$$

then

$$\sum_{n=1}^{\infty} |c_n|^2 \leq \int_a^b |f(x)|^2 dx.$$

In particular,

$$\lim_{n \rightarrow \infty} c_n = 0.$$

▷ **proof** : In the proof above we obtained the following Bessel inequality.

$$\sum_{m=1}^n |c_m|^2 \leq \int_b^a |f|^2 dx$$

This holds for every finite n , and the left side is monotonically increasing, while the integral on the right is a constant independent of n . Hence the left side is bounded above, so $\sum_{n=1}^{\infty} |c_n|^2$ converges. By the term test for divergence, we can then derive the following.

$$\lim_{n \rightarrow \infty} |c_n|^2 = 0 \Rightarrow \lim_{n \rightarrow \infty} c_n = 0$$

From this fact we obtain the Riemann–Lebesgue lemma, which can be summarized as follows.

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \langle f, e^{int} \rangle = \lim_{n \rightarrow \infty} \int f \cos(nt) + i \int f \sin(nt) = 0$$

8.14 Theorem

If, for some x , there are constants $\delta > 0$ and $M < \infty$ such that

$$|f(x+t) - f(x)| \leq M|t| \quad (1)$$

for all $t \in (-\delta, \delta)$, then

$$\lim_{N \rightarrow \infty} s_N(f; x) = f(x). \quad (2)$$

▷ **proof** : First, let us define the following function, which will be useful later when computing the difference between S_N and f .

$$g(t) = \begin{cases} \frac{f(x-t) - f(x)}{\sin(t/2)} & \text{for } 0 < |t| \leq \tau, \\ 0 & \text{if } t = 0 \end{cases}$$

Let us first identify a few properties of g ,

$$|g(t)| = \left| \frac{f(x-t) - f(x)}{\sin(t/2)} \right| = \left| \frac{f(x-t) - f(x)}{-t} \cdot \frac{-t}{\sin(t/2)} \right|.$$

Here, keep in mind the hypothesis given.

$$|f(x+t) - f(x)| \leq M|t| \Rightarrow |f(x-t) - f(x)| \leq M|t| = M|t|.$$

$$\lim_{t \rightarrow 0} \frac{\sin(t/2)}{t/2} = 1 \Rightarrow \lim_{t \rightarrow 0} \frac{t}{\sin(t/2)} = 2,$$

Using the trigonometric identity above, we obtain the following.

$$\lim_{t \rightarrow 0} |g(t)| = \lim_{t \rightarrow 0} \left| \frac{f(x-t) - f(x)}{-t} \cdot \frac{-t}{\sin(t/2)} \right| \leq M \cdot 2 = 2M.$$

so we conclude that g is bounded near 0, which is one of the necessary conditions for Riemann integrability and will be used later. Now let us examine the difference between S_N and f directly. The partial sum of the Fourier series can be written via the Dirichlet kernel,

$$S_N(f; x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t) dt,$$

the Dirichlet kernel has the following property.

$$D_N(t) = \sum_{n=-N}^N e^{int}, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} dt = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}.$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(t) dt = 1.$$

This identity follows from the property of the Dirichlet kernel: it equals 1 when $n = 0$, and $\frac{1}{\pi} \int_{-\pi}^{\pi} 1 dx = \frac{1}{2\pi} 2\pi = 1$. Using this, we can express f in terms of the Dirichlet kernel as follows.

$$f(x) = f(x) \cdot 1 = f(x) \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) D_N(t) dt.$$

Now let us look at the difference between the Fourier partial sum and f .

$$\begin{aligned} S_N(f; x) - f(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(x-t) - f(x)] D_N(t) dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(x-t) - f(x)] \cdot \frac{\sin((N + 1/2)t)}{\sin(t/2)} dt. \end{aligned}$$

Here, let us decompose the sine term using a trigonometric identity.

$$\sin\left(\left(N + \frac{1}{2}\right)t\right) = \sin(Nt) \cos(t/2) + \cos(Nt) \sin(t/2),$$

Hence, the difference between the Fourier series and the function can be written as follows.

$$S_N(f; x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(t) \cos\left(\frac{t}{2}\right) \sin(Nt) dt + \frac{1}{2\pi} \int_{-\pi}^{\pi} g(t) \sin\left(\frac{t}{2}\right) \cos(Nt) dt.$$

Since $g(t)$ is bounded and integrable, both integrals $\rightarrow 0$ as $N \rightarrow \infty$ by the Riemann–Lebesgue lemma.

$$\Rightarrow S_N(f; x) \rightarrow f(x) \quad \text{as } N \rightarrow \infty.$$

References

- [1] Walter Rudin et al. *Principles of mathematical analysis*, volume 3. McGraw-hill New York, 1964.